

Market Risk Management

Short-Term Monitoring · Medium- and Long-Term Testing

市场风险管理：短期监测 · 中期测试 · 长期测试

VaR & ES · Backtesting · Stress testing · Drawdown · Scenario analysis

In-depth theory · Practitioner detail with reproducible calculations · AIFMD, UCITS and MiFID II requirements

VaR 与 ES · 回溯 · 压力测试 · 回撤 · 情景分析 · 理论深度解析 · 实操细节与可复现计算 · 附 AIFMD、UCITS、MiFID II 监管要求
英文为主，附中文翻译

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Regulatory content as of 30 September 2026. Every numerical example has been independently recomputed. Study notes only, not investment or regulatory-compliance advice.
监管内容截至 2026 年 9 月 30 日。所有数值示例均已独立重算。本手册为学习笔记，不构成投资或监管合规建议。

How to Use This Guide & Notation

导读与符号约定

The handbook is organised by risk horizon. **Part I, short-term monitoring** (1 day to 1 month): probability language, VaR and ES, volatility models, historical simulation, non-Normal corrections and backtesting, with reproducible S&P 500 case studies. **Part II, medium-term testing** (weeks to a year): stress testing, including sensitivity, historical, hypothetical, reverse and liquidity stress tests. **Part III, long-term testing** (beyond a year): drawdown analysis and multi-period scenario analysis. Each part sets out the corresponding AIFMD, UCITS and MiFID II requirements.

Layout: English text on the left, Chinese translation on the right, paragraph by paragraph; formulas span both columns. Five coloured boxes recur:

- **Deep dive**: theory beginners can compute but struggle to explain;
- **In practice**: details the source material leaves open but a real bank or fund must decide;
- **Pitfall**: frequent computational or conceptual errors, including typos in the sources;
- **Worked example**: fully numerical examples;
- **Regulatory mapping**: related AIFMD, UCITS and MiFID II requirements.

本手册按风险期限分为三部分。**第一部分 短期风险监测**（1日至1个月）：概率语言、VaR与ES、波动率模型、历史模拟、非正态修正与回测，并用真实的标普500数据完成可复现的案例。**第二部分 中期风险测试**（数周至1年）：压力测试，包括敏感性、历史、假设、反向与流动性压力测试。**第三部分 长期风险测试**（1年以上）：回撤分析与多期情景分析。每一部分都给出AIFMD、UCITS与MiFID II的对应监管要求。

版式：左栏为英文正文，右栏为中文翻译，逐段对应；公式跨栏居中。五种彩框贯穿全书：

- **深度解析**：初学者最容易“会算不会讲”的理论；
- **实操要点**：原材料中没有写明、但在银行或基金实际搭建模型时必须决定的细节；
- **常见陷阱**：计算或概念上的高频错误，以及原材料中的笔误；
- **例题**：带完整数字的算例；
- **监管对应**：AIFMD、UCITS与MiFID II的相关要求。

Cut-off date • Regulatory content as of 30 September 2026 / 截止日期 • 监管内容截至 2026 年 9 月 30 日

Regulatory content reflects rules and public information as of 30 September 2026. The following are still moving; check the latest texts before use: (1) AIFMD II Level 2 measures and the new supervisory reporting template (expected after October 2027); (2) the Central Bank of Ireland's revised UCITS rules and AIF Rulebook; (3) EU FRTB application from 1 January 2027 and transitional relief to end-2029; (4) ESMA guidelines and technical standards on liquidity management tools and the integrated reporting framework.

本手册的监管内容反映截至 2026 年 9 月 30 日的规则与公开信息。以下事项仍在变化，使用前请核对最新文本：(1) AIFMD II 的 Level 2 细则与新版监管报告模板（预计 2027 年 10 月以后）；(2) Central Bank of Ireland 修订后的 UCITS 规则与 AIF Rulebook；(3) 欧盟 FRTB 自 2027 年 1 月 1 日起适用及至 2029 年底的过渡安排；(4) ESMA 关于流动性管理工具与统一报告框架的指引和技术标准。

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The three-horizon framework 三期限风险框架

	Short-term monitoring 短期风险监测	Medium-term testing 中期风险测试	Long-term testing 长期风险测试
期限 Horizon	1 日 – 1 个月	数周 – 1 年	1 – 10 年
核心问题 Question	How much can we lose in normal markets? 正常市场下，明天/本月最多亏多少？	Can we survive a severe but plausible shock? 严重但合理的冲击下，能否承受？	How deep and long can losses run? 亏损会有多深、持续多久、多久恢复？
主要工具 Tools	VaR, ES, volatility, backtesting VaR、ES、波动率、回测	Sensitivity, historical, hypothetical, reverse, liquidity stress tests 敏感性、历史、假设、反向、流动性压力测试	Drawdown, multi-period scenarios, Monte Carlo 回撤、多期情景分析、蒙特卡洛
统计基础 Basis	Distributions, quantiles 概率分布、分位数	Conditional scenarios, sensitivities 条件情景、敏感度映射	Path dependence, narrative, simulation 路径依赖、叙事与模拟
典型频率 Frequency	Daily; monthly backtest 每日计算，每月回测	Monthly to quarterly 每月至每季	Quarterly to annually 每季至每年
关键监管 Key rules	UCITS global exposure; AIFMD limits and leverage; PRIIPs MRM; MiFID 10% loss report UCITS 全球风险暴露 VaR; AIFMD 风险限额与杠杆; PRIIPs MRM; MiFID 10% 损失通知	UCITS/AIFMD stress tests; ESMA LST guidelines; IFD internal capital UCITS/AIFMD 压力测试与 ESMA 流动性压力测试指引; IFD 内部资本评估	MiFID product scenario analysis; PRIIPs performance scenarios; HWM; sustainability risk MiFID 产品治理情景分析; PRIIPs 业绩情景; 业绩费高水位; 可持续风险整合

Notation 符号约定

符号 Symbol	Meaning	中文含义
X	Portfolio P&L or return, a random variable; losses negative	组合盈亏 (P&L) 或收益率，随机变量；亏损为负
$L = -X$	Loss expressed as a positive number	以正数表示的损失
$c, \alpha = 1 - c$	Confidence level (e.g. 99%) and tail probability (e.g. 1%)	置信水平 (如 99%) 与尾部概率 (如 1%)
$F_X(x) = \Pr(X \leq x)$	Cumulative distribution function	累积分布函数 (CDF)
Φ, φ, Φ^{-1}	Standard Normal CDF, PDF and inverse CDF (quantile function)	标准正态 CDF、密度函数、逆 CDF (分位数函数)
μ, σ	Mean and standard deviation of returns (daily unless stated)	收益率的均值与标准差 (默认日频)
u_t, r_t	Day- t log-return $\ln(S_t/S_{t-1})$	第 t 日对数收益率 $\ln(S_t/S_{t-1})$
h (或 T)	Holding period / risk horizon in days, e.g. 1D, 10D	持有期 / 风险期限 (天)，如 1 日、10 日
n	Estimation window length (observations), unrelated to h	估计窗口长度 (观测数)，与 h 无关
S, K	Cornish–Fisher skewness and excess-kurtosis parameters	Cornish–Fisher 偏度参数与超额峰度参数
γ_1, γ_2	Sample skewness and sample excess kurtosis	样本偏度与样本超额峰度
T, x (回测)	Number of eligible comparisons and number of breaches	有效比较次数与突破次数

Pitfall • Sign conventions / 常见陷阱 • 符号方向

Following the course, formulas write VaR as a **negative return percentile** (e.g. -2.33σ); a breach occurs when the realised return is below VaR. Reports and regulators usually quote VaR and ES as **positive loss amounts**. The two differ only by a sign, yet the comparison operator in Excel (< vs >) is a classic source of errors: use one convention per sheet and state it in the header.

本手册与课程一致，公式层面把 VaR 写成**负的收益率分位数**（例如 -2.33σ ），"突破"定义为实现收益低于 VaR。报告与监管口径通常把 VaR 与 ES 写成**正的损失金额**。两种写法只差一个负号，但在 Excel 中比较大小时 (< 还是 >) 时极易出错——每张表只用一种约定，并在表头写明。

Short-Term Risk Monitoring · 1 day to 1 month

第一部分 短期风险监测

Short-term monitoring answers "how much can we lose in normal markets?". It rests on distributions and quantiles, runs daily with monthly backtests, and underpins UCITS global exposure and bank market-risk capital.

短期监测回答"正常市场下最多亏多少"。它依赖概率分布和分位数，每日计算、每月回溯，是 UCITS 全球风险暴露和银行市场风险资本的计量基础。

1 Foundations: Returns, Volatility and Time Scaling

预备知识：收益率、波动率与时间缩放

1.1 Why log-returns 为什么用对数收益率

Log-returns $u_t = \ln(S_t/S_{t-1})$ are additive over time: the 10-day return is the sum of ten daily returns, $r_{10D,t} = \ln(S_{t+10}/S_t) = \sum_{i=1}^{10} u_{t+i}$. That is exactly what makes variances add and volatility scale with $\sqrt{h}$. For daily data log and simple returns are close, but the gap becomes material at 10-day horizons and in large sell-offs.

对数收益率 $u_t = \ln(S_t/S_{t-1})$ 在时间上可加：10 日收益等于 10 个日收益之和， $r_{10D,t} = \ln(S_{t+10}/S_t) = \sum_{i=1}^{10} u_{t+i}$ 。这正是"方差可加、波动率按 $\sqrt{h}$ 放大"的前提。对于日度数据，对数收益与简单收益差异很小，但在 10 日及以上期限、或在大幅下跌时差异会变得可观。

1.2 Variance is the average squared return 方差就是平方收益的平均

$$\hat{\sigma}_{1D}^2 = \frac{1}{n-1} \sum_{i=1}^n (u_i - \bar{u})^2 \xrightarrow{\bar{u} \approx 0} \hat{\sigma}_{1D}^2 \approx \frac{1}{n} \sum_{i=1}^n u_i^2$$

The daily mean ($\approx 0.03\%$) is more than an order of magnitude smaller than the daily standard deviation ($\approx 1\%$), and its estimation error is huge (s.e. = $\sigma/\sqrt{n}$). Risk managers therefore set $\mu \rightarrow 0$, and variance reduces to the plain average of squared returns. This simplification is the key to EWMA, which merely replaces the simple average with an exponentially weighted one.

日收益率的均值（约 0.03%）比标准差（约 1%）小一个数量级以上，而且均值的估计误差巨大（s.e. = $\sigma/\sqrt{n}$ ）。因此风险管理中通常令 $\mu \rightarrow 0$ ，方差退化为平方收益的简单平均。这一简化是理解 EWMA 的钥匙：EWMA 只是把"简单平均"换成"指数加权平均"。

Whether the window holds 21, 42 or 66 observations, the resulting σ is always a **daily** quantity. The window length n controls responsiveness and noise; the horizon h controls scaling. They are different concepts.

无论窗口取 21、42 还是 66 个观测，得到的 σ 永远是日频的。窗口长度 n 决定估计的灵敏度和噪声，风险期限 h 决定缩放倍数，两者是完全不同的概念。

1.3 Square-root-of-time rule and annualisation 平方根法则与年化

$$\sigma_{hD} = \sqrt{\sigma_{1D}^2 + \dots + \sigma_{1D}^2} = \sigma_{1D}\sqrt{h}, \quad \mu_{hD} = h\mu_{1D}, \quad \sigma_{1Y} = \sigma_{1D}\sqrt{252}$$

深度解析 · $\sqrt{h}$ 是 $\ln$ 的结果吗？ / Deep dive · Is $\sqrt{h}$ a consequence of the log?

Not entirely. Two layers do different jobs:

- **The log makes returns additive:** the multi-day log-return is the sum of daily ones, $r_{hD} = u_1 + \dots + u_h$. It only turns the problem into a sum.
- **$\sqrt{h}$ is statistical:** how the variance of that sum grows depends on how the terms relate to each other.

Whether the multiplier is $\sqrt{h}$ or something else is decided by the second layer. With simple returns the result holds approximately for small daily moves; the log merely makes the summation exact.

不完全是。这里有两层，分工不同：

- **$\ln$ 让收益可加：**多日对数收益等于日收益之和， $r_{hD} = u_1 + \dots + u_h$ 。它只负责把问题变成"求和"。
- **$\sqrt{h}$ 来自统计：**和的方差如何增长，取决于各项之间的关系。

放大倍数是 $\sqrt{h}$ 还是别的，由后者决定。换成简单收益，只要日收益足够小，结论近似成立； $\ln$ 只是让相加这一步变得精确。

$$\text{Var}\left(\sum_{i=1}^h u_i\right) = \sum_{i=1}^h \text{Var}(u_i) + 2 \sum_{i < j} \text{Cov}(u_i, u_j) \xrightarrow{\text{Cov}=0, \text{Var}(u_i)=\sigma^2} h\sigma^2 \Rightarrow \sigma_{hD} = \sigma\sqrt{h}$$

So $\sqrt{h}$ needs two conditions: (1) daily returns are **uncorrelated** (zero covariance terms; this is the key, and Normality is **not** required); (2) daily variance is **constant** (otherwise $\sqrt{\sigma_1^2 + \dots + \sigma_h^2}$). In one line: **the log makes returns additive; "uncorrelated, equal variance" makes variances additive; $\sqrt{h}$ follows from the latter.**

因此 $\sqrt{h}$ 需要两个条件：(1) 各日收益**不相关**（协方差项为 0，这是关键，不要求正态）；(2) 每天方差**相同**（否则是 $\sqrt{\sigma_1^2 + \dots + \sigma_h^2}$ ）。一句话： **$\ln$ 让收益可加，"不相关、同方差"让方差可加， $\sqrt{h}$ 是后者的结果。**

In practice • When $\sqrt{h}$ breaks down / 实操要点 • 什么时候 $\sqrt{h}$ 会失效

1. **Autocorrelation:** momentum (positive correlation) makes true 10-day volatility larger than $\sqrt{10}$ times daily; mean reversion makes it smaller. Test with the variance ratio $VR(h) = \text{Var}(r_{hD}) / (h \text{Var}(r_{1D}))$ against 1 (Lo–MacKinlay).
2. **Volatility clustering:** high σ today means high σ over the next days. Today's $\sigma \times \sqrt{10}$ understates risk while volatility rises and overstates it while it falls. EWMA implies a flat term structure; GARCH reverts to the long-run mean.
3. **Fat tails:** $\sqrt{h}$ scales σ , not quantiles directly. Summed fat-tailed daily returns move towards Normal (central limit theorem), so the 10-day tail is relatively thinner and 1-day VaR $\times \sqrt{10}$ **overstates** extreme quantiles.
4. **Mean:** the mean grows with h , σ with $\sqrt{h}$. Negligible at short horizons, the mean matters more as h grows: one reason VaR is a short-horizon measure.
5. **Constant positions:** $\sqrt{10}$ assumes no trading or hedging for ten days. In reality books are cut or hedged, which is why FRTB uses liquidity horizons by risk factor (10–120 days).
6. **Regulatory convenience:** Basel allows $\sqrt{10}$ to convert 1-day VaR to 10-day mainly because estimating 10-day returns directly is data-hungry: non-overlapping samples are one tenth the size, overlapping ones introduce serial correlation (§7.2). $\sqrt{10}$ is an accepted approximation, not an exact result.

What to check in practice is whether "uncorrelated, equal variance" holds for your data and horizon.

1. **自相关:** 动量 (正相关) 使真实 10 日波动率大于 $\sqrt{10}$ 倍, 均值回复 (负相关) 则小于。可用方差比 $VR(h) = \text{Var}(r_{hD}) / (h \text{Var}(r_{1D}))$ 检验是否接近 1 (Lo–MacKinlay)。
2. **波动率聚集:** 今天 σ 高意味着未来几天 σ 也高。用今天的 σ 乘 $\sqrt{10}$, 在波动率上升期低估、回落期高估。EWMA 隐含"未来波动率不变"的平坦期限结构; GARCH 则向长期均值回归。
3. **厚尾:** $\sqrt{h}$ 放大的是 σ , 不直接是分位数。厚尾日收益加总后按中心极限定理趋近正态, 10 日分布的尾部相对更"薄", 把 1 日 VaR 直接乘 $\sqrt{10}$ 会高估极端分位数。
4. **均值:** 均值按 h 、 σ 按 $\sqrt{h}$ 增长。期限短时均值可忽略; 期限越长, 均值占比越大——这也是 VaR 只用于短期的原因之一。
5. **头寸不变:** $\sqrt{10}$ 假设 10 天内不交易、不对冲。实际会减仓或对冲, 因此 FRTB 改为按风险因子流动性划分持有期 (10–120 天)。
6. **监管简便性:** Basel 允许用 $\sqrt{10}$ 把 1 日 VaR 换算为 10 日, 主要因为直接估计 10 日收益有数据问题: 不重叠样本只有十分之一, 重叠样本又引入序列相关 (见 §7.2)。 $\sqrt{10}$ 是"可接受的近似", 不是精确结论。

实务中要检查的, 是"不相关、同方差"这两个假设在你的数据和持有期上是否站得住。

1.4 Portfolio volatility 组合波动率

$$\sigma_{\Pi}^2 = \mathbf{w}' \Sigma \mathbf{w}, \quad \text{two assets: } \sigma_{\Pi}^2 = \sigma_X^2 + 2\rho_{XY}\sigma_X\sigma_Y + \sigma_Y^2 \quad (\text{in money terms})$$

Worked examples 1–2 • Analytical VaR / 例题 1–2 • 单资产与双资产解析 VaR

Ex. 1: a £5,000,000 position with 1% daily volatility. Annualised volatility $0.01\sqrt{252} \approx 15.9\%$; 10-day s.d. $0.01\sqrt{10} = 3.16\%$. 99%/10-day VaR = $2.326 \times 0.0316 \times 5,000,000 \approx \text{£}367,800$ (£368,400 with the rounded factor 2.33).

Ex. 2: two assets, £100,000 each, both 1% daily volatility (£1,000), correlation 0.3. $\sigma_{\Pi}^2 = 2.6 \times 10^6$, $\sigma_{\Pi} = \text{£}1,612.45$. 99%/5-day VaR = $2.33 \times 1612.45 \times \sqrt{5} \approx \text{£}8,401$. With correlation 1 the VaR would be £10,420; the difference is the diversification benefit.

例 1: 单一资产头寸 £5,000,000, 日波动率 1%。年化波动率 $0.01\sqrt{252} \approx 15.9\%$; 10 日标准差 $0.01\sqrt{10} = 3.16\%$ 。99%/10 日 VaR = $2.326 \times 0.0316 \times 5,000,000 \approx \text{£}367,800$ (用取整因子 2.33 得 £368,400)。

例 2: 两资产各 £100,000, 日波动率均为 1% (即 £1,000), 相关系数 0.3。 $\sigma_{\Pi}^2 = 1000^2 + 2(0.3)(1000)(1000) + 1000^2 = 2.6 \times 10^6$, $\sigma_{\Pi} = \text{£}1,612.45$ 。99%/5 日 VaR = $2.33 \times 1612.45 \times \sqrt{5} \approx \text{£}8,401$ 。若相关系数为 1, VaR 为 £10,420; 差额即分散化收益。

In practice • Do not annualise VaR / 实操要点 • 年化 VaR 没有意义

VaR and ES are **short-horizon** measures (1-day, 10-day). Over six months or a year the portfolio is rebalanced and distributional assumptions drift, so 6M or 1Y VaR carries little meaning. Use stress tests for the medium term (Part II) and drawdown and scenario analysis for the long term (Part III); they have no fixed distributional horizon but reflect what investors actually experience.

VaR 与 ES 是**短期**风险度量 (1 日、10 日)。组合在半年、一年内会调仓, 分布假设也不再稳定, 因此 6 个月或 1 年 VaR 的解释力很弱。中期风险应改用压力测试 (第二部分), 长期风险用回撤与情景分析 (第三部分); 这些方法没有固定的分布期限, 但更贴近投资者的真实体验。

2 Value at Risk as a Percentile

VaR: 作为分位数的风险度量

2.1 Definition and probability statement 定义与概率陈述

VaR answers: "Over a given horizon, what is the loss we are c -confident will not be exceeded?" In terms of P&L X , the 99% VaR is the 1st percentile of the distribution of X :

VaR 回答的问题是: "在给定的持有期内, 有 c 的把握, 亏损不会比多少更糟?" 用盈亏 X 表示, 99% VaR 就是 X 分布的 1% 分位数:

$$\Pr(X \leq \text{VaR}_c(X)) = 1 - c$$

$$\text{VaR}_c(X) = \sup\{x \in \mathbb{R} : F_X(x) \leq 1 - c\} \quad (\text{P\&L, negative number})$$

$$\text{VaR}_c(L) = \inf\{x \in \mathbb{R} : F_L(x) \geq c\} \quad (\text{loss } L = -X, \text{ positive number})$$

Deep dive · Why "VaR is a percentile" / 深度解析 · 为什么说"VaR 是一个分位数"

The definition $\Pr(X \leq \text{VaR}) = 1 - c$ is literally the definition of a percentile: rank all possible P&L outcomes from worst to best, and VaR is the point that cuts off the worst $1 - c$. Three angles:

1. **Ranking**: sort 500 daily returns; the 99% VaR is about the 5th worst ($500 \times 1\%$). No distributional assumption.
2. **Inverse CDF**: the CDF maps a value to a cumulative probability; VaR goes the other way, $\text{VaR} = F^{-1}(1 - c)$.
3. **Normal formula**: -2.326 is the standard Normal 1st percentile; $\text{VaR} = \mu - 2.326\sigma$ just shifts and scales it to your portfolio.

Being a percentile is also VaR's limitation: it marks where the tail **starts**, not how **deep** it goes.

定义 $\Pr(X \leq \text{VaR}) = 1 - c$ 本身就是分位数的定义：把所有可能的盈亏从最差排到最好，VaR 是恰好切下最差 $1 - c$ 部分的那个点。可以从三个角度理解：

1. **排序**：500 个历史日收益从小到大排序，99% VaR 约为第 5 差的值 ($500 \times 1\%$)。无需任何分布假设。
2. **逆 CDF**：CDF 把"数值"映射为"累积概率"；VaR 反过来，给定概率求数值，即 $\text{VaR} = F^{-1}(1 - c)$ 。
3. **正态公式**： -2.326 是标准正态的 1% 分位数； $\text{VaR} = \mu - 2.326\sigma$ 只是把它平移、缩放到你的组合上。

分位数的身份也决定了 VaR 的局限：它只标出尾部的**起点**，不描述尾部的深度。

Deep dive · CDF as a ranking table / 深度解析 · 用"排名表"理解 CDF 与逆 CDF

The CDF $F(x) = \Pr(X \leq x)$ gives the share of outcomes at or below x . Sort ten daily returns from smallest to largest (table in the right-hand column):

- **Forward (CDF)**: value in, cumulative share out. $F(-2\%) = 20\%$.
- **Inverse (inverse CDF)**: share in, value out. "Where do the worst 10% end?" $\rightarrow F^{-1}(10\%) = -3\%$, the 90% VaR.

The standard Normal Φ is the continuous version of the same table:

$\Phi(-2.326) = 1\%$ is the forward lookup (Excel `NORM.S.DIST(-2.326, TRUE)`) and $\Phi^{-1}(1\%) = -2.326$ the inverse (`NORM.S.INV(0.01)`). **The CDF answers "how common is this loss?"; the inverse CDF answers "how large is a loss this common?"; VaR is the latter.**

CDF 的定义 $F(x) = \Pr(X \leq x)$ ，意思是"收益不超过 x 的情况占多大比例"。把 10 天的日收益从小到大排好：

收益 x	-3%	-2%	-1.5%	-1%	-0.5%	0%	...	2%
$F(x)$	10%	20%	30%	40%	50%	60%	...	100%

- **正查 (CDF)**：给收益值，查累积比例。 $F(-2\%) = 20\%$ 。
- **反查 (逆 CDF)**：给比例，查收益值。"最差的 10% 截止在哪里?" $\rightarrow F^{-1}(10\%) = -3\%$ ，即 90% VaR。

标准正态的 Φ 是同一张表的连续版本： $\Phi(-2.326) = 1\%$ 是正查 (Excel `NORM.S.DIST(-2.326, TRUE)`)， $\Phi^{-1}(1\%) = -2.326$ 是反查 (`NORM.S.INV(0.01)`)。CDF 回答"这个亏损有多常见"，逆 CDF 回答"多常见的亏损对应多大的数"；VaR 是后者。

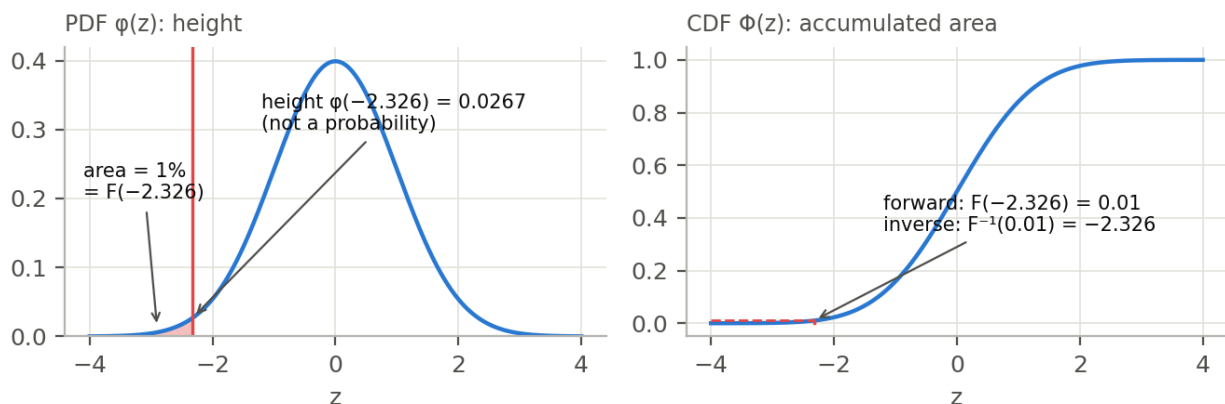


Figure 1. Left: the PDF is a height; probability is area under the curve. Right: the CDF is the accumulated area; VaR is read off it in reverse.

图 1 左：PDF 是曲线的高度，概率是曲线下的面积；右：CDF 是累积面积，VaR 在 CDF 上反查得到。

Deep dive • Why probability comes from the CDF, not the PDF / 深度解析 • 为什么概率看 CDF 而不看 PDF

For a continuous distribution, probability is **area** under the PDF; the CDF is the area accumulated up to x . The PDF itself is a **height**, and a height is not a probability.

1. **A single point has probability 0**: a return of exactly $-2.32635\ldots\%$ has probability zero; only intervals carry probability: $\Pr(a \leq X \leq b) = \int_a^b f(x) dx = F(b) - F(a)$.
2. **The PDF is a density**: probability per unit of x , like population per square kilometre; multiply by a width to get a probability. A density can exceed 1: with $\sigma = 1\%$, the Normal peak on a return axis is about 39.9. Anything that can exceed 1 cannot be a probability.
3. **Applied to VaR**: $\Pr(X \leq \text{VaR}) = 1\%$ asks for the area of the tail left of VaR, i.e. $F(\text{VaR}) = 1\%$, so the CDF (and its inverse) is used.
4. **The ϕ in the ES formula**: $\phi(-2.326) = 0.02665$ is **not** a 2.665% probability. It comes from the integral: tail mean $= \int x f(x) dx \div$ tail area $(1 - c)$; since the antiderivative of $z\phi(z)$ is $-\phi(z)$, ϕ appears as a height in the result. The denominator $1 - c$ is the probability, and it comes from the CDF.
5. **Exception: discrete distributions**: for discrete variables such as breach counts, each value has positive probability and the PMF is a probability. For $T = 250$, exactly 2 breaches has probability 25.7% (`BINOM.DIST(2,250,0.01,FALSE)`); traffic lights use the cumulative version (`TRUE`).

In one line: the PDF shows where probability is dense; the CDF tells you how much has accumulated. Asking for a probability means asking for an area.

对连续分布，概率是 PDF 曲线下的**面积**；CDF 就是累积到 x 的面积。PDF 本身只是曲线的高度，高度不是概率。

1. **单点概率为 0**：收益恰好等于 $-2.32635\ldots\%$ 的概率为 0，只有区间才有概率： $\Pr(a \leq X \leq b) = \int_a^b f(x) dx = F(b) - F(a)$ 。
2. **PDF 是密度**：单位是“每单位 x 的概率”，类似人口密度（人/平方公里），乘上宽度才得到概率。密度可以大于 1： $\sigma = 1\%$ 时，以收益为横轴的正态峰值高度约 39.9。能大于 1，就不可能是概率。
3. **放到 VaR 上**： $\Pr(X \leq \text{VaR}) = 1\%$ 问的是 VaR 左侧尾部的面积，即 $F(\text{VaR}) = 1\%$ ，因此用 CDF（反查用逆 CDF）。
4. **ES 公式里的 ϕ** ： $\phi(-2.326) = 0.02665$ 不是 2.665% 的概率。它来自积分：尾部均值 $= \int x f(x) dx \div$ 尾部面积 $(1 - c)$ ；因 $z\phi(z)$ 的原函数是 $-\phi(z)$ ，积分结果中 ϕ 以“高度”形式出现。分母 $1 - c$ 才是概率，来自 CDF。
5. **例外：离散分布**：突破次数等离散变量，每个取值有正概率，这时 PMF 就是概率。 $T = 250$ 时恰好 2 次突破的概率为 25.7% (`BINOM.DIST(2,250,0.01,FALSE)`)；交通灯用的是累积版本 (`TRUE`)。

一句话：PDF 描述概率“在哪里更密集”，CDF 告诉你“累积了多少”。问概率，就是问面积。

inf / Deep dive • Why supremum and infimum / 深度解析 • 为什么要用 sup

If the distribution is continuous and strictly increasing, the x solving $F(x) = 1 - c$ is unique and $F^{-1}(1 - c)$ suffices. Real P&L is often **discrete or mixed**: historical simulation has finitely many points, bond default is a 0/1 event, option books jump around strikes. The CDF is then a staircase: there may be **no** x with $F(x) = 1 - c$ exactly, or a **whole interval** of them.

The infimum says: “among all loss levels whose cumulative probability reaches at least c , take the smallest”. It guarantees a unique answer for any distribution. The supremum on P&L is the mirror image of the same idea.

Lower-case x is a specific value (dummy variable); upper-case X is the random variable. “ $x < \text{VaR}(X) < 0$ ” reads: a specific loss worse than VaR.

若分布连续且严格递增，满足 $F(x) = 1 - c$ 的 x 唯一，直接写 $F^{-1}(1 - c)$ 即可。但真实 P&L 往往是**离散或混合**的：历史模拟只有有限个样本点，债券违约是 0/1 事件，期权组合在行权价附近会出现跳跃。此时 CDF 呈阶梯状，可能**没有任何** x 恰好满足 $F(x) = 1 - c$ ，也可能有一整段区间都满足。

下确界 inf 的作用是：“在所有使累积概率达到或超过 c 的损失水平中，取最小的那个”。它保证定义在任何分布下都唯一存在。对 P&L 取上确界 sup 是同一件事的镜像写法。

小写 x 是具体取值（哑变量），大写 X 是整个随机变量。“ $x < \text{VaR}(X) < 0$ ”的意思是：某个具体亏损比 VaR 更糟。

2.2 The fundamental formula: standardise, then invert 基本公式：标准化与逆 CDF

Assume $X \sim N(\mu, \sigma^2)$. Standardising both sides of the event leaves the probability unchanged; applying Φ^{-1} then removes the probability, because $\Phi(z) = \Pr(Z \leq z)$ is the definition of the CDF:

$$\Pr\left(Z \leq \frac{\text{VaR} - \mu}{\sigma}\right) = 1 - c \implies \Phi\left(\frac{\text{VaR} - \mu}{\sigma}\right) = 1 - c \implies \boxed{\text{VaR}_c = \mu + \Phi^{-1}(1 - c) \sigma}$$

$\Phi^{-1}(1 - c)$ is the **Factor**, the standardised percentile. At 99%, $\Phi^{-1}(0.01) = -2.32635$ (Excel: `NORM.S.INV(0.01)`). Every parametric refinement of VaR swaps either the factor (Student-t, Cornish–Fisher, EVT) or the volatility (rolling window, EWMA, GARCH). **VaR = Factor $\times$ Volatility** is the backbone of this guide.

假设 $X \sim N(\mu, \sigma^2)$ 。把事件两边同时标准化，概率不变；再对两边施加 Φ^{-1} ，因为 $\Phi(z) = \Pr(Z \leq z)$ 本身就是 CDF 的定义，逆函数恰好把“概率”消去：

$\Phi^{-1}(1 - c)$ 称为因子（Factor），即标准化后的分位数。99% 时 $\Phi^{-1}(0.01) = -2.32635$ （Excel: `NORM.S.INV(0.01)`）。VaR 的一切参数化改进，本质上都是在换“因子”：Student-t 因子、Cornish–Fisher 因子、EVT 因子；以及在换“波动率”：滚动窗口、EWMA、GARCH。 **VaR = 因子 $\times$ 波动率** 是贯穿全书的骨架。

置信水平 c	尾部 1-c	正态因子 $\Phi^{-1}(1-c)$	正态 ES 因子 $\phi(z)/(1-c)$	ES/VaR	典型用途 Typical use
90%	10%	-1.2816	1.7550	1.369	内部限额 · internal limits
95%	5%	-1.6449	2.0627	1.254	基金 / UCITS 风险报告 · fund reporting
97.5%	2.5%	-1.9600	2.3378	1.193	FRTB ES、PRIIPs VaR
99%	1%	-2.3263	2.6652	1.146	Basel 内部模型、回溯 · Basel IMA, backtesting
99.9%	0.1%	-3.0902	3.3671	1.090	经济资本 · economic capital

In practice • Keep or drop μ ? / 实操要点 • 均值项 μ 要不要保留

A positive μ shrinks VaR (more optimistic). Because μ is so unstable, most banks, and the course's backtest demo, **drop μ** , which is conservative. If kept, use a long, robust estimate scaled by $\mu_{hD} = h\mu_{1D}$. The $-\frac{1}{2}\sigma^2 T$ term in the PRIIPs formula is not a mean but a log-normal drift correction (Itô), present even when $\mu = 0$.

正的 μ 会让 VaR 变小（更乐观）。由于 μ 估计极不稳定，多数银行与课程的回测演示都**直接去掉 μ** ，这是更保守的做法。若保留，应使用长期、稳健的估计，并用 $\mu_{hD} = h\mu_{1D}$ 缩放。PRIIPs 公式中的 $-\frac{1}{2}\sigma^2 T$ 并不是“均值”，而是对数正态下的漂移修正（伊藤引理），即使 $\mu = 0$ 也会出现。

Pitfall • What VaR does not say / 常见陷阱 • VaR 没告诉你的事

A 99% VaR says "1% of outcomes are worse" but **not how much worse**. Two portfolios can share a VaR while one has a thin tail and the other a very fat one. VaR is also not subadditive (bond example, §3.4). These two facts drove the regulatory move from VaR to ES.

99% VaR 只说明“有 1% 的情况比这更糟”，却不说明更糟多少。两个组合可以有相同的 VaR，但一个尾部平缓、另一个尾部极厚。VaR 也不满足次可加性（第 3.4 节的债券例子）。这两点正是监管从 VaR 转向 ES 的原因。

Regulatory mapping • VaR definition and parameters / 监管对应 • VaR 的定义与参数

框架	English	中文
UCITS	VaR mandatory for complex strategies; absolute VaR $\leq 20\%$ NAV or relative $\leq 2\times$ reference; 99%, 20 days, ≥ 1 year of data, daily (§8.1).	复杂策略必须用 VaR 法；绝对 VaR $\leq 20\%$ NAV 或相对 VaR ≤ 2 倍参考组合；99%、20 日、 ≥ 1 年数据、每日计算 (§8.1)。
AIFMD	No prescribed VaR; requires "appropriate" risk measurement documented in policy; if used, VaR and method are reported in Annex IV.	不规定 VaR 参数；要求“适当的”风险计量并写入风险管理政策；若采用，VaR 及其计算方法填入 Annex IV。
MiFID II	No VaR limit; PRIIPs uses a 97.5% VaR to set the MRM class, which feeds target-market and suitability decisions.	无 VaR 限额；PRIIPs 用 97.5% VaR 决定 MRM 等级，进而影响目标市场与适当性判断。

3 Expected Shortfall and Coherent Risk Measures

预期损失 ES 与一致性风险度量

3.1 Two equivalent definitions 两个等价定义

$$\text{ES}_c(X) = \mathbb{E}[X \mid X \leq \text{VaR}_c(X)] \quad (\text{conditional tail expectation})$$

$$\text{ES}_c(X) = \frac{1}{1-c} \int_0^{1-c} \text{VaR}_u(X) du \quad (\text{average of tail percentiles})$$

First definition: **given** that we are in the tail, what is the average loss? Second: average every VaR inside the tail (99%, 99.1%, 99.2%, ... 99.99%, ...). For continuous distributions they coincide; for discrete ones the second is more robust (always coherent). Both are **model-free**: the working formula depends on the distribution of X .

第一个定义：**已知**已经落入尾部，平均亏多少。第二个定义：把尾部内所有的 VaR（99%、99.1%、99.2%……99.99%……）取平均。对连续分布两者相同；对离散分布，第二个定义更稳健（它总是一致性风险度量）。两者都是模型无关的：具体算式取决于 X 的分布。

In short: **VaR marks the door to the tail; ES tells you what is inside.**

形象地说：VaR 标出通往尾部的门，ES 告诉你门后有什么。

3.2 Deriving Normal ES, step by step 正态分布下的 ES 推导（逐步）

Route A: from the average of tail percentiles. Rewrite the integral over the upper tail, substitute $\text{VaR}_u = \mu + \sigma\Phi^{-1}(1-u)$ and pull out the mean:

路线 A：从“尾部分位数平均”出发。先把积分改写为上尾形式，代入 $\text{VaR}_u = \mu + \sigma\Phi^{-1}(1-u)$ ，均值项直接提出：

$$\text{ES}_c = \frac{1}{1-c} \int_c^1 (\mu + \sigma\Phi^{-1}(1-u)) du = \mu + \frac{\sigma}{1-c} \int_c^1 \Phi^{-1}(1-u) du$$

Substitute $u = \Phi(z)$, $du = \varphi(z)dz$, limits $[c, 1] \mapsto [\Phi^{-1}(c), \infty)$, and use symmetry $\Phi^{-1}(1 - \Phi(z)) = -z$:

换元 $u = \Phi(z)$, $du = \varphi(z)dz$ ，积分限 $[c, 1] \mapsto [\Phi^{-1}(c), \infty)$ ；并利用对称性 $\Phi^{-1}(1 - \Phi(z)) = -z$ ：

$$= \mu - \frac{\sigma}{1-c} \int_{\Phi^{-1}(c)}^{\infty} z \varphi(z) dz$$

Key trick: the antiderivative of $z\varphi(z)$ is $-\varphi(z)$, since $\frac{d}{dz}e^{-z^2/2} = -ze^{-z^2/2}$. At the upper limit $\varphi(\infty) = 0$, hence:

关键技巧： $z\varphi(z)$ 的原函数是 $-\varphi(z)$ ，因为 $\frac{d}{dz}e^{-z^2/2} = -ze^{-z^2/2}$ 。上限处 $\varphi(\infty) = 0$ ，于是：

$$\text{ES}_c = \mu - \sigma \frac{\varphi(\Phi^{-1}(c))}{1-c}$$

$$c = 99\% : \frac{\varphi(-2.32635)}{0.01} = \frac{0.026652}{0.01} = 2.6652 \Rightarrow \text{ES}_{99\%} = \mu - 2.6652 \sigma$$

Route B: first tail moment $\div$ tail probability. Tail mean = $\int_{-\infty}^{\mu-n\sigma} xf(x) dx / \int_{-\infty}^{\mu-n\sigma} f(x) dx$. The denominator is the tail mass $1-c$, the

路线 B：从“尾部一阶矩 $\div$ 尾部概率”出发。尾部均值 = $\int_{-\infty}^{\mu-n\sigma} xf(x) dx / \int_{-\infty}^{\mu-n\sigma} f(x) dx$ 。分母是尾部概率质量 $1-c$ ，分子是尾部一阶矩。作换元 $x = \mu - z\sigma$ 后同样得到上式。两条路线殊途同归，面试时能讲清任一条即可。

numerator the first tail moment. The substitution $x = \mu - z\sigma$ yields the same result. Either route is fine for an interview.

The "inverse CDF inside the PDF" ($\varphi(\Phi^{-1}(c))$) looks odd until you plug in numbers: find the percentile -2.326 , then the density height there, 0.02665 .

"逆 CDF 套在密度里" ($\varphi(\Phi^{-1}(c))$) 初看奇怪, 代入数字就清楚了: 先求分位点 -2.326 , 再求该点的密度高度 0.02665 。

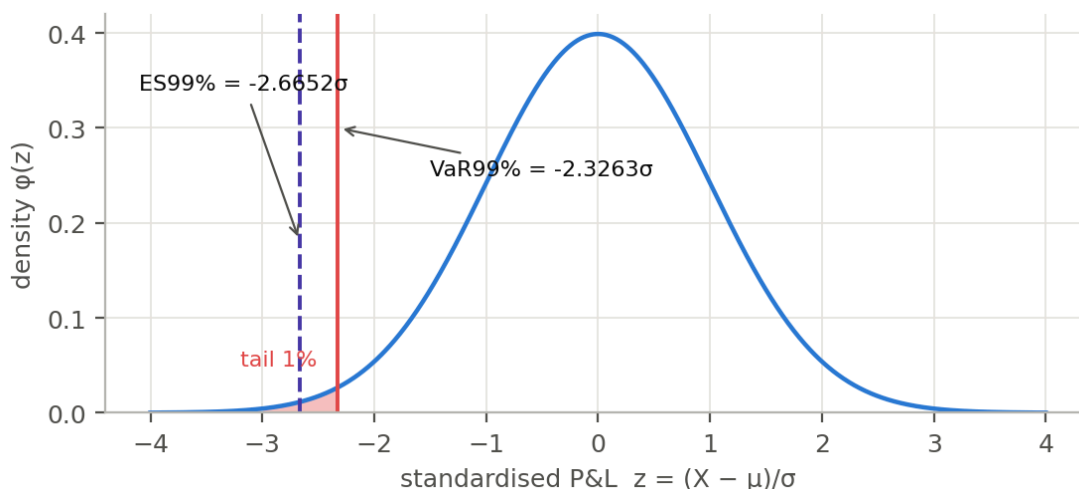


Figure 2. 99% VaR (solid) and 99% ES (dashed) under the standard Normal. ES is the centre of mass of the shaded tail.

图 2 标准正态下 99% VaR (实线) 与 99% ES (虚线)。ES 是阴影尾部的"重心"。

Worked example 3 • The tail beyond 2σ / 例题 3 • 超过 2σ 的尾部

The share of the standard Normal left tail beyond -2σ is $\Phi(-2) = 2.275\%$, i.e. confidence 97.725%. The mean of that tail is $-\varphi(2)/\Phi(-2) = -0.05399/0.02275 = -2.373$. In a Normal world the average loss beyond 2σ is only about 19% worse than 2σ itself: Normal tails are thin.

标准正态左尾超过 -2σ 的比例 $\Phi(-2) = 2.275\%$, 对应置信水平 97.725%。该尾部的平均值 $= -\varphi(2)/\Phi(-2) = -0.05399/0.02275 = -2.373$ 。可见在正态世界中, "超过 2σ 的平均损失"只比 2σ 本身坏约 19%——正态尾部非常薄。

3.3 The ES/VaR diagnostic ratio and tail heaviness ES/VaR 诊断比率与尾部厚度

$$\rho_c = \frac{\text{ES}_c}{\text{VaR}_c} \quad \text{Normal}(\mu=0) : \rho_c = \frac{\varphi(z_c)}{(1-c)|z_c|} \xrightarrow{c \rightarrow 1} 1; \quad \text{GPD tail} : \rho \xrightarrow{\text{VaR} \gg u} \frac{1}{1-\xi}$$

ρ is unit-free: how far the mass behind the door sits from the door. Under the Normal, ρ is moderate (1.146 at 99%) and **drifts slowly towards 1** as $c \rightarrow 1$ (table in §2.2): the deeper you go, the thinner a Normal tail looks. For power-law (fat) tails ρ converges to the constant $1/(1-\xi) > 1$: the tail never thins.

The numerator $\int x f(x) dx$ can explode for fat tails while the denominator (tail mass) stays finite; that is why ES can be undefined ($\xi \geq 1$) although VaR always exists. Hull's range $\xi \in [0.1, 0.4]$ keeps ρ between 1.1 and 1.7.

Note: the course slide describes Gaussian ρ as "fixed at every c "; strictly it is moderate and slowly decreasing towards 1, which is the contrast with the constant limit of fat tails.

ρ 无量纲, 衡量"门后"的质量相对"门口"有多远。正态分布下 ρ 适中 (99% 时 1.146), 并且随 $c \rightarrow 1$ 缓慢降向 1 (见 2.2 节表格): 越深入尾部, 正态尾部越"薄"。而幂律 (厚尾) 分布中, ρ 收敛到常数 $1/(1-\xi)$ 且大于 1: 尾部不会变薄。

ξ	尾部类型	$\rho \approx$
0.10	轻 (Hull 下限)	1.11
0.25	中等	1.33
0.40	重 (Hull 上限)	1.67
0.50	很重: $\text{ES} = 2 \times \text{VaR}$	2.00
$\rightarrow 1$	类 Cauchy: ES 不存在	$\rightarrow \infty$

3.4 Coherence and subadditivity 一致性风险度量与次可加性

Artzner et al. (1999) define a coherent risk measure $\rho(\cdot)$ (on positive losses) by four axioms:

- Monotonicity:** a portfolio with always-larger losses is not less risky;
- Subadditivity:** $\rho(X + Y) \leq \rho(X) + \rho(Y)$; merging does not add risk, i.e. diversification;
- Positive homogeneity:** double the position, double the risk;
- Translation invariance:** adding cash m reduces risk by m .

ES satisfies all four; VaR generally fails axiom 2 (except for Normal or elliptical distributions). Subadditivity lets ES capital be decomposed and aggregated cleanly across desks.

Artzner 等 (1999) 提出一致性风险度量 $\rho(\cdot)$ (以正损失计) 的四条公理:

- 单调性:** 损失总是更大的组合, 风险不更小;
- 次可加性:** $\rho(X + Y) \leq \rho(X) + \rho(Y)$, 合并不增加风险, 体现分散化;
- 正齐次性:** 头寸翻倍, 风险翻倍;
- 平移不变性:** 加入现金 m , 风险减少 m 。

ES 满足全部四条; VaR 一般不满足第 2 条 (正态或椭圆分布时例外)。次可加性使 ES 资本可以在交易台之间干净地分解与汇总。

Worked example 4 • Discrete defaults / 例题 4 • 离散违约: VaR 违反、ES 满足次可加性

Three independent bonds A, B, C, face value £1,000 each, default probability 0.5% each, loss given default = face value. The portfolio loss distribution is tabulated in the right-hand column.

VaR: a single bond defaults with $0.5\% < 1\%$, so $\text{VaR}_{99}(A) = \text{£}0$. For the portfolio the cumulative probability at £0 is only $98.51\% < 99\%$, so $\text{VaR}_{99}(ABC) = \text{£}1,000 > 0+0+0$. **VaR violates subadditivity:** diversification "increases" VaR.

ES: for one bond, half of the 1% tail is a £1,000 loss and half is £0, so $\text{ES}(A) = 0.005 \times 1000 / 0.01 = \text{£}500$. For the portfolio, 2 and 3 defaults occupy 0.0075% and 0.0000125% of probability; the remaining conditional mass 0.992525 is one default: $\text{ES}(ABC) = \text{£}1,007.49 \leq \text{£}1,500$. **ES is subadditive.**

Note how the calculation treats the tail as exactly 1% of probability, splitting the £1,000 atom where it straddles the 99th percentile. This is the "average of tail percentiles" definition at work, and it is why that definition is preferred for discrete P&L.

三只独立债券 A、B、C，面值各 £1,000，违约概率各 0.5%，违约损失为全部面值。组合损失分布：

违约数	损失	概率	累积
0	£0	98.5075%	98.5075%
1	£1,000	1.4850%	99.9925%
2	£2,000	0.0075%	≈ 100%
3	£3,000	0.0000125%	100%

VaR: 单只债券违约概率 $0.5\% < 1\%$ ，故 $\text{VaR}_{99}(A) = \text{£}0$ ；组合累积概率在 £0 处只有 $98.51\% < 99\%$ ，故 $\text{VaR}_{99}(ABC) = \text{£}1,000 > 0+0+0$ 。**VaR 违反次可加性**——分散投资反而"增加"了 VaR。

ES: 单只债券的 1% 尾部中，0.5% 是损失 £1,000，0.5% 是 £0， $\text{ES}(A) = 0.005 \times 1000 / 0.01 = \text{£}500$ 。组合的 1% 尾部中，2 次和 3 次违约分别占 0.0075% 和 0.0000125%，其余 0.992525 的条件概率为 1 次违约： $\text{ES}(ABC) = 0.992525(1000) + 0.0074625(2000) + 0.0000125(3000) = \text{£}1,007.49 \leq \text{£}1,500$ 。ES 满足次可加性。

3.5 Why FRTB chose 97.5% ES 为什么 FRTB 选择 97.5% ES

Under the Normal the 97.5% ES factor 2.338 almost equals the 99% VaR factor 2.326: for well-behaved distributions capital is roughly unchanged, but for fat-tailed books 97.5% ES is materially higher than 99% VaR. It penalises exactly the tail shape. That is the calibration logic behind the Basel move from 99% VaR to 97.5% ES.

The price is harder backtesting: ES is a conditional mean rather than a binary event and is not *elicitable* (Gneiting 2011), so there is no test as clean as counting breaches. FRTB therefore still backtests **VaR** at 97.5% and 99% while capital is measured with ES.

正态下 97.5% ES 因子 2.338 与 99% VaR 因子 2.326 几乎相等：对"正常"分布，资本水平大致不变；但对厚尾组合，97.5% ES 会显著高于 99% VaR——它惩罚的恰恰是尾部形状。这是 Basel 委员会从 99% VaR 过渡到 97.5% ES 的校准逻辑。

代价是回溯变难：ES 是条件均值而非二元事件，且在统计上不具备"可引出性" (elicitability, Gneiting 2011)，没有像数突破次数那样简洁的检验。因此 FRTB 仍然用 97.5% 与 99% 的 **VaR** 做回溯，用 ES 计量资本。

Regulatory mapping • Expected Shortfall / 监管对应 • ES

框架	English	中文
UCITS	ES not required; CESR encourages supplementing VaR for tail-heavy strategies (arbitrage, structured), with ES a common choice.	不要求 ES; CESR 鼓励对尾部风险明显的策略（套利、结构化）补充 VaR 以外的度量，ES 是常见选择。
AIFMD	Not required; managers may use ES as an internal limit metric, documented in the risk policy.	不要求；管理人可将 ES 作为内部限额指标，须在风险管理政策中说明。
MiFID II	Through IFR/CRR for large investment firms' trading books: the FRTB internal model uses 97.5% ES (EU from 2027).	通过 IFR/CRR 适用于大型投资公司的交易账簿：FRTB 内部模型法用 97.5% ES（欧盟 2027 年起）。

4 Volatility: Rolling Window, EWMA and GARCH

波动率建模：滚动窗口、EWMA 与 GARCH

4.1 Rolling-window standard deviation 滚动窗口标准差

The simplest approach: each day compute the sample s.d. of the latest n daily returns (Excel STDEV.S), shifting the window by one observation. The course demo uses $n = 21$ (about a month) and a backtest period of 200+ trading days.

- n too short (10–15): noisy estimates, jumpy VaR;
- n too long (60–100): insensitive to recent volatility, clustered breaches at the start of a crisis;
- the Normal factor presumes at least ~25 observations.

Weakness: equal weights inside the window, zero outside. A shock lifts volatility when it enters and drops it **abruptly** when it exits n days later (the "ghost effect").

最简单的做法：每天用最近 n 个日收益计算样本标准差（Excel STDEV.S），窗口每次移动 1 个观测。课程演示取 $n = 21$ （约 1 个月），回溯期 200 个以上交易日。

- n 太短（10–15）：估计噪声大，VaR 剧烈跳动；
- n 太长（60–100）：对近期波动变化不敏感，危机初期突破成串；
- 正态因子的前提是样本量至少约 25 个观测。

缺点：窗口内等权、窗口外权重为零。一次大冲击在进入窗口时让波动率跳升，在 n 天后突然离开时又让波动率骤降（"幽灵效应"）。

4.2 EWMA (RiskMetrics) EWMA (RiskMetrics)

$$\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) u_{t-1}^2 \iff \sigma_t^2 = (1 - \lambda) \sum_{i \geq 1} \lambda^{i-1} u_{t-i}^2, \quad \tau_{1/2} = \frac{\ln 0.5}{\ln \lambda}$$

EWMA stands for Exponentially Weighted Moving Average. It is a "probability filter": it weights a **prior** (yesterday's variance σ_{t-1}^2) against **news** (yesterday's squared return u_{t-1}^2) using λ .

- Low λ : more weight on news, very responsive, but more procyclical and noisy;
- High λ : more weight on the prior, slow to react, close to a long window.

RiskMetrics (J.P. Morgan, 1994) uses $\lambda = 0.94$ for daily data, a half-life of about 11.2 days; $\lambda = 0.97$ gives about 22.8 days, and observations older than 60–80 days barely matter. EWMA has no fixed window and hence no ghost effect.

EWMA 全称为 Exponentially Weighted Moving Average (指数加权移动平均)。它可视为一个"概率滤波器": 在先验 (昨天的方差估计 σ_{t-1}^2) 与新闻 (昨天收益的平方 u_{t-1}^2) 之间按 λ 加权。

- λ 低: 新闻权重重大, 对市场反应灵敏, 但更"顺周期"、更嘈杂;
- λ 高: 先验权重重大, 反应缓慢, 接近长窗口。

RiskMetrics (J.P. Morgan, 1994) 对日度数据取 $\lambda = 0.94$, 半衰期约 11.2 天; $\lambda = 0.97$ 时半衰期约 22.8 天, 60–80 天以前的观测几乎不再起作用。EWMA 没有固定窗口, 也就没有"幽灵效应"。

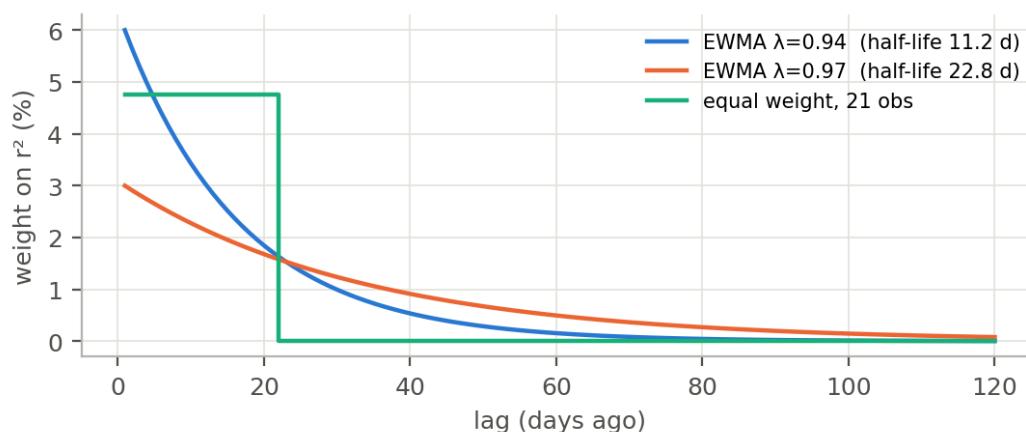


Figure 3. Weight on past squared returns. EWMA decays smoothly; the 21-day window drops to zero on day 22.
图 3 不同方案对过去平方收益的权重。EWMA 平滑衰减; 21 日窗口在第 22 天断崖式归零。

In practice • Three EWMA implementation details / 实操要点 • EWMA 的三个实现细节

1. **Initialisation:** seed σ_0^2 with the average squared return over a long history (e.g. the first 250 days). With a long enough burn-in the seed is immaterial ($\approx 150+$ days for $\lambda=0.97$).
 2. **Indexing:** $\sigma_{t|t-1}$ must use returns up to $t-1$ only to be a genuine forecast; including u_t gives an ex-post estimate and look-ahead bias in backtests. The Bank of England paper shows the two conventions lead to different FHS outcomes.
 3. **Choosing λ :** calibrate so breach counts match theory, or so devolatilised residuals look i.i.d., but beware over-fitting; the course workbook's RiskMetrics sheet uses $\lambda=0.88$ as a sensitivity illustration.
1. **初始化:** 用较长历史 (如前 250 日) 的平方收益平均作为 σ_0^2 。热身期 (burn-in) 足够长时, 初值影响可忽略 ($\lambda=0.97$ 时约需 150 日以上)。
 2. **索引约定:** $\sigma_{t|t-1}$ 只能用到 $t-1$ 的收益, 才是真正的预测; 若把当天收益 u_t 也放进去, 就是"事后"估计, 用于回测会产生前视偏差。英格兰银行论文指出两种约定会导致不同的 FHS 结果。
 3. **λ 的选择:** 可以用"使突破次数最接近理论值"或"使去波动化残差最接近 i.i.d."来校准, 但要警惕过度拟合; 课程 Excel 中的 RiskMetrics 页用了 $\lambda=0.88$ 作为敏感性演示。

In practice • Setting exponential weights with λ / 实操要点 • 用 λ 设定指数权重

1. Two equivalent computations: the recursion $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) u_{t-1}^2$ needs only yesterday's variance and return, one Excel formula filled down; the explicit form $\sigma_t^2 = \sum_i w_i u_{t-i}^2$ with $w_i = (1 - \lambda) \lambda^{i-1}$. With finite data (n observations) the weights sum to $1 - \lambda^n$, so **normalise:** $w_i = (1 - \lambda) \lambda^{i-1} / (1 - \lambda^n)$. At $\lambda = 0.94$ truncation at 250 days is negligible; at $\lambda = 0.99$ it is not.

2. Choosing λ :

Method	How
Industry convention	RiskMetrics: 0.94 daily, 0.97 monthly
From half-life	$\lambda = 0.5^{1/au_{1/2}}$; a 20-day half-life gives $\lambda \approx 0.966$
From mean look-back	mean weight lag $1/(1 - \lambda)$: about 17 days at 0.94, 33 at 0.97
Statistical fit	minimise the squared error of σ^2 forecasts against realised u^2 , or maximum likelihood
Backtest calibration	breach rate close to theory, without clustering

Covariance matrices need **one common λ** across assets, otherwise the matrix may not be positive semi-definite and portfolio variance can even turn negative; yet the Bank of England study finds different "optimal λ " by asset (WTI 1.00, Gasoil 0.958, NatGas 0.973), a trade-off to manage.

3. Initialisation: seed σ_0^2 with the average squared return over the first 1–2 years; allow a burn-in of about 5 half-lives (≈ 60 days at $\lambda = 0.94$, ≈ 115 at 0.97).

4. In historical simulation: give scenario i days ago a normalised weight, sort P&L from worst to best, and the scenario where cumulative weight first reaches 1% is the 99% VaR (deep dive after §5.2).

5. Other parameterisations: exponential weights can equally be set by half-life or effective window length, all interchangeable; GARCH adds a long-run term ω on top of the λ structure (§4.3).

1. 两种等价算法: 递推式 $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) u_{t-1}^2$ 每天只需昨天的方差与收益, Excel 一行公式下拉即可; 显式权重式 $\sigma_t^2 = \sum_i w_i u_{t-i}^2$, $w_i = (1 - \lambda) \lambda^{i-1}$ 。数据有限 (n 个观测) 时权重和只有 $1 - \lambda^n$, 须归一化: $w_i = (1 - \lambda) \lambda^{i-1} / (1 - \lambda^n)$ 。 $\lambda = 0.94$ 时 250 天截断影响可忽略, $\lambda = 0.99$ 时不可忽略。

2. 如何选 λ :

方法	做法
行业惯例	RiskMetrics: 日度 0.94, 月度 0.97
按半衰期	$\lambda = 0.5^{1/au_{1/2}}$; 半衰期 20 天 $\rightarrow \lambda \approx 0.966$
按平均回看期	权重平均滞后 $1/(1 - \lambda)$: 0.94 约 17 天, 0.97 约 33 天
统计优化	最小化 σ^2 预测与实际 u^2 的均方误差, 或极大似然
回溯校准	使突破率接近理论值且不成串

计算协方差矩阵时所有资产须用**同一个 λ** , 否则矩阵可能非半正定, 组合方差甚至为负; 但英格兰银行研究显示各资产"最优 λ "不同 (WTI 1.00、Gasoil 0.958、NatGas 0.973), 两者需要取舍。

3. 初始化: σ_0^2 取前 1–2 年平方收益的平均; 热身期约 5 个半衰期 ($\lambda = 0.94$ 约 60 天, 0.97 约 115 天)。

4. 用于历史模拟: 给第 i 天前的情景赋归一化权重, 按损益从最差到最好排序, 累加权重首次达到 1% 的情景即为 99% VaR (§ 5.2 深度解析)。

5. 其他设定方式: 指数加权也可用半衰期或有效窗口长度设定, 三者可互换; GARCH 则在 λ 的基础上加入长期均值项 ω (§ 4.3)。

4.3 GARCH(1,1) versus EWMA GARCH(1,1) 与 EWMA 的比较

$$\sigma_{t+1}^2 = \omega + \alpha r_t^2 + \beta \sigma_t^2, \quad \bar{\sigma}^2 = \frac{\omega}{1 - \alpha - \beta}, \quad \text{EWMA} = \text{GARCH with } \omega = 0, \alpha = 1 - \lambda, \beta = \lambda$$

GARCH stands for Generalized Autoregressive Conditional Heteroskedasticity: volatility varies over time (conditional heteroskedasticity) and depends on past squared returns and past variance (autoregressive); Bollerslev (1986) generalised Engle's (1982) ARCH model by adding the lagged variance term.

GARCH 全称为 Generalized Autoregressive Conditional Heteroskedasticity (广义自回归条件异方差): 波动率随时间变化 (条件异方差), 由过去的平方收益和过去的方差决定 (自回归); Bollerslev (1986) 在 Engle (1982) ARCH 模型的基础上加入过去方差项, 故称"广义"。

	EWMA	GARCH(1,1)
参数 Parameters	1 个: λ	3 个: ω, α, β (极大似然估计 MLE)
均值回复 Mean reversion	None; forecast is flat across horizons 无; 预测波动率在所有期限上保持不变	Yes, towards $\bar{\sigma}^2$ at speed $\alpha + \beta$ 有; 以 $\alpha + \beta$ 的速度回到长期方差 $\bar{\sigma}^2$
多期预测 Term structure	$\sigma_{hD} = \sigma \sqrt{h}$	$\sum_k [\bar{\sigma}^2 + (\alpha + \beta)^{k-1} (\sigma_{t+1}^2 - \bar{\sigma}^2)]$
实务评价 Practitioner view	Simple, transparent, uniform across assets; more procyclical 简单、透明、跨资产统一; 更顺周期	More parameters, unstable estimates; forecast pulled towards the long-run mean, may under-react in a crisis 更多校准参数, 参数不稳定; 预测被拉向长期均值, 危机中可能反应不足

Deep dive • Why practitioners prefer EWMA / 深度解析 • 为何从业者偏好 EWMA

GARCH's mean reversion is academically more "correct", but risk management cares about volatility over the **next week or two**. When a crisis erupts, GARCH forecasts are pulled back towards the long-run mean and look optimistic exactly when conservatism is needed; three parameters also need periodic re-estimation across hundreds of risk factors. EWMA has one parameter, no estimation error and is uniform across factors; these engineering virtues usually outweigh the theoretical shortcoming.

GARCH 的均值回复在学术上更"正确", 但风险管理关心的是**下一两周**的波动率。在危机爆发时, GARCH 的预测会被长期均值往回拉, 恰恰在最需要保守的时候显得乐观; 而且三个参数需要定期重估, 跨数百个风险因子维护成本高。EWMA 只有一个参数、没有估计误差、对所有因子一致, 这些工程性优点通常压倒了理论上的缺陷。

Regulatory mapping • Volatility estimation / 监管对应 • 波动率估计

框架	English	中文
UCITS	History ≥ 1 year (250 days), shorter only if volatility rises significantly; data refreshed at least quarterly. Weighting schemes such as EWMA should be justified in the RMP, including the effective observation period.	观测期 ≥ 1 年 (250 营业日), 仅在波动率显著上升时可缩短; 数据至少每季更新。若用 EWMA 等加权方案, 应在风险管理流程中论证其有效观测期与合规性。
AIFMD	Free choice of method, subject to independent validation and periodic review (Level 2 Art. 45); document the rationale for λ and window length.	方法自选, 但须独立验证并定期复核 (Level 2 第 45 条); 选择 λ 、窗口长度的依据要留档。
MiFID II	PRIPs Category 2 MRM estimates σ , skewness and kurtosis from 5 years of returns (at least 2 years if daily).	PRIPs 第 2 类产品 MRM 用 5 年历史收益 (日度至少 2 年) 估计 σ 与偏度、峰度。

5 Historical and Filtered Historical Simulation

历史模拟与过滤历史模拟

5.1 Historical simulation, step by step 历史模拟 (HS) 的步骤

1. Identify the risk factors driving portfolio value (equity prices, FX, curve nodes, implied vols, ...) and collect their historical changes over the lookback (e.g. 250 or 500 days);
2. Assume "tomorrow replays historical day i ": **revalue** today's positions under day i 's factor moves, giving N scenario P&Ls;
3. Sort the N P&Ls and take the $1 - c$ percentile as VaR; average the scenarios worse than VaR for ES.

Single-index shortcut: sort historical returns and take PERCENTILE(..., 0.01).

1. 确定驱动组合价值的风险因子 (股价、汇率、收益率曲线节点、隐含波动率等), 收集回看窗口内 (如 250 或 500 日) 的历史变动;
2. 假设"明天重演历史上的第 i 天", 用今天的头寸与第 i 天的风险因子变动对组合**重新估值**, 得到 N 个情景 P&L;
3. 把 N 个 P&L 排序, 取 $1 - c$ 分位数作为 VaR; 把比 VaR 更差的情景平均, 即为 ES。

单一指数的简化版本: 直接对历史收益排序, PERCENTILE(..., 0.01)。

Pitfall • A tail of 2.5 observations / 常见陷阱 • 尾部只有 2.5 个观测

A 1% tail of a 250-day window holds 2.5 observations; 300 days gives 3. VaR is set by one or two worst days and ES is an average of two or three numbers, so sampling error is large. Conventions for "which observation" also differ: Excel PERCENTILE.INC interpolates between neighbours, PERCENTILE.EXC uses position $(N + 1)p$, and many banks take the $\lceil Np \rceil$ -th worst. In small samples they can differ by tens of basis points; **document the choice**.

250 日窗口的 1% 尾部只有 2.5 个观测, 300 日也只有 3 个。VaR 实际上由一两个最差日决定, ES 是两三个数的平均——统计误差极大。此外, 不同软件对"第几个"有不同约定: Excel PERCENTILE.INC 在相邻观测之间线性插值, PERCENTILE.EXC 用 $(N + 1)p$ 的位置, 许多银行直接取"第 $\lceil Np \rceil$ 差"。三者在小样本下差异可达数十个基点, **必须在模型文档中写明**。

5.2 Plateaux and insensitivity HS 的"平台"与不敏感性

Equal-weighted HS VaR moves in **plateaux**: it changes only when the worst observation leaves the window or a worse one enters. Two consequences:

- after a shock, VaR stays elevated for a full window length (250 days) even if markets have calmed;
- when markets turn from calm to stressed, VaR reacts slowly and breaches cluster.

Moreover, no unfiltered HS model can ever produce a VaR **larger than the worst loss in its window**; it cannot react to stress beyond what it has "seen".

Regulation and practice: Basel III / FRTB require at least a 250-day lookback, and FRTB adds a stressed calibration window. Front-office internal VaR often uses 60–150 days for responsiveness (more procyclical); more commonly banks keep a long window and apply EWMA decay or volatility scaling to historical returns: the "best of both worlds".

等权 HS 的 VaR 呈**阶梯状平台**: 只有当最差观测滑出窗口、或更差的新观测进入时, VaR 才会变化。后果有二:

- 冲击发生后, VaR 会在整整一个窗口长度 (250 日) 内保持高位, 即使市场早已平静;
- 市场由平静转为动荡的初期, VaR 反应迟缓, 突破成串出现。

此外, 任何未过滤的 HS 模型, 其 VaR 永远**不可能超过窗口内最差一天的损失**, 无法对"前所未见"的压力作出反应。

监管与实践: Basel III / FRTB 要求至少 250 日回看, FRTB 还要求包含压力时期的校准窗口。前台内部 VaR 常用 60–150 日窗口以提高灵敏度 (更顺周期); 更常见的是保留长窗口, 但对历史收益施加 EWMA 衰减或波动率缩放——"两全其美"。

What the plateau effect is: historical simulation or an equal-weighted rolling window gives every observation inside the window the same weight and zero outside, so VaR moves only when one of the worst observations enters or leaves, producing step-like plateaux. Three consequences:

- **Slow reaction:** when calm turns to stress the window still holds calm data, VaR is too low and breaches cluster (as in Feb–Mar 2025 in §9.3);
- **Ghost effect:** an extreme shock keeps VaR high for a full window length, then VaR drops abruptly on day $n + 1$ when it leaves;
- **Ceiling:** unfiltered HS VaR can never exceed the worst loss in the window.

It appears in regulatory 250-day HS VaR (Basel, UCITS), equal-weighted rolling standard deviations (e.g. a 21-day window), and CCP and bilateral margin models built on HS.

How exponential weighting fixes it: the observation i days ago gets weight $(1 - \lambda)\lambda^{i-1}$, so old data fade smoothly instead of dropping off a cliff. The half-life is about 11 days at $\lambda = 0.94$ and about 23 days at 0.97. Main applications:

1. **EWMA volatility and covariance:** the volatility input to parametric VaR (§4.2);
2. **Filtered historical simulation:** devolatilise and revolatilise with EWMA volatility (§5.3);
3. **Age-weighted historical simulation** (Boudoukh–Richardson–Whitelaw, 1998): weight historical scenarios by λ^{i-1} and take a weighted quantile;
4. **Special case of GARCH:** EWMA is GARCH(1,1) without the long-run mean term (§4.3).

The price: exponential weighting is smoother and more responsive but more **procyclical**: VaR is low in calm periods and jumps in a crisis, and if it drives margin it concentrates margin calls in the crisis. In the Bank of England study, peak-to-trough at $\lambda = 0.95$ is about 1.8 times that of unfiltered HS (2.97 vs 1.69, \$5.4). A common compromise keeps a 250+ day window as a floor with exponential weighting on top.

平台效应是什么：历史模拟或等权滚动窗口给窗口内每个观测相同权重、窗口外权重为 0，所以 VaR 只在最差的几个观测进入或离开窗口时才变动，图上呈阶梯状平台。它带来三个后果：

- **反应迟缓：**市场由平静转向动荡时，窗口里还是平静期数据，VaR 偏低，突破成串（§9.3 的 2025 年 2–3 月即是如此）；
- **幽灵效应：**一次极端冲击让 VaR 在整整一个窗口长度内保持高位，到第 $n + 1$ 天离开窗口时 VaR 又突然下跌；
- **上限：**未过滤 HS 的 VaR 永远不会超过窗口内最差一天的损失。

它出现在监管要求的 250 日 HS VaR（Basel、UCITS）、等权滚动标准差（如 21 日窗口），以及用 HS 计算保证金的 CCP 与双边保证金模型中。

指数加权如何解决：第 i 天前的观测权重为 $(1 - \lambda)\lambda^{i-1}$ ，旧数据权重平滑衰减而不是断崖归零。 $\lambda = 0.94$ 时半衰期约 11 天，0.97 时约 23 天。主要应用：

1. **EWMA 波动率与协方差：**参数法 VaR 的波动率输入（§4.2）；
2. **过滤历史模拟：**用 EWMA 波动率去波动化、再波动化（§5.3）；
3. **按时间加权的歷史模拟**（Boudoukh–Richardson–Whitelaw, 1998）：给历史情景按 λ^{i-1} 赋权后取加权分位数；
4. **GARCH 的特例：**EWMA 等于没有长期均值项的 GARCH(1,1)（§4.3）。

代价：指数加权更平滑、更灵敏，但也更顺周期——平静期 VaR 很低，危机时迅速抬高；若用于保证金，会在危机中集中追加保证金。英格兰银行的研究中， $\lambda = 0.95$ 时的峰谷比约为未过滤 HS 的 1.8 倍（2.97 对 1.69，§5.4）。常见折中：保留 250 日以上长窗口作为下限，再叠加指数加权。

5.3 Filtered HS: devolatilise, then revolatilise 过滤历史模拟（FHS）：去波动化与再波动化

$$x_j^F(i) = x_j(i) \cdot \frac{\sigma_{Q,j}^\mu(t)}{\sigma_{P,j}^\lambda(i)} \quad \text{devol by the vol on day } i, \text{ revol by the vol on VaR day } t$$

The FHS idea (Boudoukh et al. 1998; Hull & White 1998): historical returns carry valuable information about tail shape, but they occurred under different volatility regimes. Standardise each historical return by its own-day volatility (devol, giving residuals), multiply by today's volatility (revol), and obtain the scenario set "as if history replayed at today's volatility"; then run ordinary HS.

The Bank of England paper compares two typical designs:

- **LTV** (long-term devol): devol with a 500-day equal-weighted volatility, revol with EWMA(0.97). All historical returns get the **same** factor, so skewness, kurtosis and correlations are preserved; the distribution is simply rescaled.
- **STV** (short-term devol, Hull–White type): each historical day is devolatilised by its own EWMA(0.97) vol. Factors differ by observation, so skewness, kurtosis, autocorrelation and cross-correlations **change**.

In the paper, WTI sample excess kurtosis rises from 3.56 to 7.35 under STV; most of UK NatGas autocorrelation disappears after scaling.

FHS (Boudoukh 等 1998; Hull & White 1998) 的思路：历史收益包含关于尾部形状的宝贵信息，但它们发生在不同的波动率环境中。先用当日波动率估计把每个历史收益“标准化”（devol，得到残差），再乘以今天的波动率（revol），得到“如果历史事件在今天的波动率下重演”的情景集，然后照常做 HS。

英格兰银行论文比较了两种典型设定：

- **LTV**（长期波动率去波动化）：用 500 日等权波动率统一去波动化、用 EWMA(0.97) 再波动化。所有历史收益乘以同一个因子，偏度、峰度、相关性完全保留，只是整体放大或缩小。
- **STV**（短期波动率去波动化，Hull–White 型）：每个历史日用自己的 EWMA(0.97) 波动率去波动化。每个观测的因子不同，因此改变了偏度、峰度、自相关和因子间相关性。

论文中的例子：WTI 的样本超额峰度从 3.56 被 STV 放大到 7.35；UK 天然气的自相关在缩放后大部分消失。

Deep dive · Why scale a 99th percentile by volatility? / 深度解析 · 为什么用波动率去缩放 99% 分位数

Volatility is driven mostly by the **centre** of the distribution, yet we care about the **tail**. We still scale by it because volatility needs far less data to estimate than a 99th percentile: for the same accuracy, vol scaling makes the model much more reactive. The implicit assumption is that the **ratio of volatility to the 99th percentile is stable**. If the tail lengthens or shrinks independently of the centre (e.g. jump risk rises while day-to-day volatility does not), it fails. The paper also finds devolatilised residuals are usually still non-Normal and not fully independent (Jarque–Bera statistics far above the 9.2 critical value): FHS often works partly because a "ghost" of non-normality survives devolatilisation.

波动率主要由分布的**中心**决定，而我们关心的是**尾部**。之所以仍用它缩放，是因为估计波动率所需的数据远少于估计 99% 分位数：用波动率缩放可以在相同精度下让模型灵敏得多。其隐含假设是"**波动率与 99% 分位数之比**"保持稳定。若尾部独立于中心而伸长或收缩（例如跳跃风险上升而日常波动不变），这一假设失效。论文还指出，去波动化后的残差往往仍不是正态、也不完全独立（Jarque–Bera 统计量远超临界值 9.2）——FHS 表现好，部分是因为残差中仍保留了非正态的"幽灵"信息。

5.4 Procyclicality 顺周期性

FHS gives low VaR in calm periods and raises it fast in turmoil. When VaR sets initial margin (CCPs or bilateral OTC), this timeliness produces concentrated margin calls during a crisis and amplifies liquidity stress. The paper measures it by:

- **Peak-to-trough**: max / min margin for a constant portfolio over the observation period;
- **n-day procyclicality**: the largest margin increase over any n days, as % of notional.

For a WTI position (historical data): unfiltered HS peak-to-trough 1.69; LTV($\lambda=0.95$) 2.97, LTV(0.97) 2.53, LTV(0.99) 2.02, STV 2.82. The smaller λ , the more reactive and the more procyclical. Largest 5-day increase: unfiltered 0.39%, LTV(0.95) 3.18%.

Mitigants: EMIR requires CCPs to adopt one of three tools, one being a **floor** at a 10-year equal-weighted VaR; one can also use a partial scaling factor $\frac{K+f}{K+1}$ ($K \geq 0$), between full scaling f and none.

FHS 在平静期给出低 VaR、在动荡期迅速抬高 VaR。若 VaR 用于计算初始保证金（CCP 或双边 OTC），这种"及时"会在危机中制造集中的追加保证金，放大流动性压力。论文用两个指标衡量：

- **峰谷比**：固定组合在观测期内最高保证金 / 最低保证金；
- **n 日顺周期性**：任意 n 日内保证金的最大增幅（占名义本金百分比）。

WTI 头寸的结果（历史数据）：未过滤 HS 的峰谷比 1.69；LTV($\lambda=0.95$) 2.97、LTV(0.97) 2.53、LTV(0.99) 2.02、STV 2.82。 λ 越小越灵敏，也越顺周期。5 日最大增幅：未过滤 0.39%，LTV(0.95) 3.18%。

缓释工具：EMIR 要求 CCP 采用三种措施之一，其中之一是以 10 年等权 VaR 作为**下限**；也可以用部分缩放因子 $\frac{K+f}{K+1}$ ($K \geq 0$)，在完全缩放 f 与不缩放 1 之间取折中。

Pitfall · The ratchet model and over-fitting / 常见陷阱 · "棘轮模型"与过度拟合

The paper builds a cynical "ratchet" VaR: 500-day HS that moves to a worse observation if there were ≥ 3 breaches in two years and to a better one if ≤ 1 . It has no model of the return process yet often passes Kupiec and even reacts faster than HS. Lesson: **passing a backtest $\neq$ a correct model**. Simple tests have low power; validation should also examine residuals, stress scenarios and parameter stability, and monitor calibration drift over time.

论文构造了一个"犬儒"的棘轮 VaR：用 500 日 HS，若过去两年突破 ≥ 3 次就改取更差的一个观测， ≤ 1 次就改取更好的一个。它毫无收益生成过程的模型，却常常能通过 Kupiec 检验，甚至比 HS 更快响应波动。教训：**通过回测 $\neq$ 模型正确**。简单检验的检验力很低，模型验证应同时考察残差性质、压力情景与参数稳定性，并持续监控校准是否漂移。

Regulatory mapping · HS and FHS / 监管对应 · 历史模拟与 FHS

框架	English	中文
UCITS	Parametric, HS, FHS and Monte Carlo are all acceptable if they capture all material risks (including option non-linearity); initial and periodic validation required.	参数法、HS、FHS、蒙特卡洛均可，只要能覆盖全部重大风险（含期权等非线性）；模型须首次及定期验证。
AIFMD	No method restriction; FHS procyclicality should be considered in liquidity and leverage management, especially for funds posting derivative margin.	同样不限方法；FHS 的顺周期性应在流动性与杠杆管理中考虑，尤其是使用衍生品保证金的基金。
MiFID II	PRIIPs Category 3 (non-linear) products use bootstrap simulation (10,000+ resamples of historical returns) for the 97.5% VaR, essentially an HS variant.	PRIIPs 第 3 类（非线性）产品用自助法模拟（10,000 次以上重抽样历史收益）计算 97.5% VaR，本质是 HS 的变体。

6 VaR Methods: Non-Normal Corrections, Simulation, Decomposition and Yield Curves

VaR 计算方法：非正态修正、模拟框架、风险分解与收益率曲线

6.1 The Cornish–Fisher expansion Cornish–Fisher 展开

$$z_{CF} = z + \frac{S}{6}(z^2 - 1) + \frac{K}{24}(z^3 - 3z) - \frac{S^2}{36}(2z^3 - 5z),$$

$$z = \Phi^{-1}(p), \quad \text{VaR} = -\frac{1}{2}\sigma^2 T + z_{CF} \sigma \sqrt{T}$$

Cornish–Fisher uses the first four moments to transform the standard Normal quantile z into one matching the data's shape: negative skewness S and positive excess kurtosis K both push the left-tail quantile further out. It replaces only the factor in the VaR formula; the volatility part is unchanged.

Substituting z gives the coefficients at each level (table in the right-hand column).

Cornish–Fisher 用前四阶矩把标准正态分位数 z 变换为"与数据形状匹配"的分位数：负偏度 S 与正超额峰度 K 都会把左尾分位数推得更远。它只替换 VaR 公式中的"因子"，波动率部分保持不变。

把 z 代入即得各置信水平的系数：

On the course S&P 500 sample (2013–2017), $\gamma_1 = -0.437$ and $\gamma_2 = 2.870$; plugged in directly they give a 99% CF factor of -3.247 , about 40% deeper than the Normal -2.326 .

p	z	S 系数	K 系数	S ² 系数
2.5%	-1.960	+0.474	-0.0687	+0.146
1%	-2.326	+0.735	-0.2338	+0.376

课程 S&P 500 样本（2013–2017）： $\gamma_1 = -0.437$ ， $\gamma_2 = 2.870$ ，直接代入得 99% CF 因子 -3.247 ，比正态的 -2.326 深约 40%。

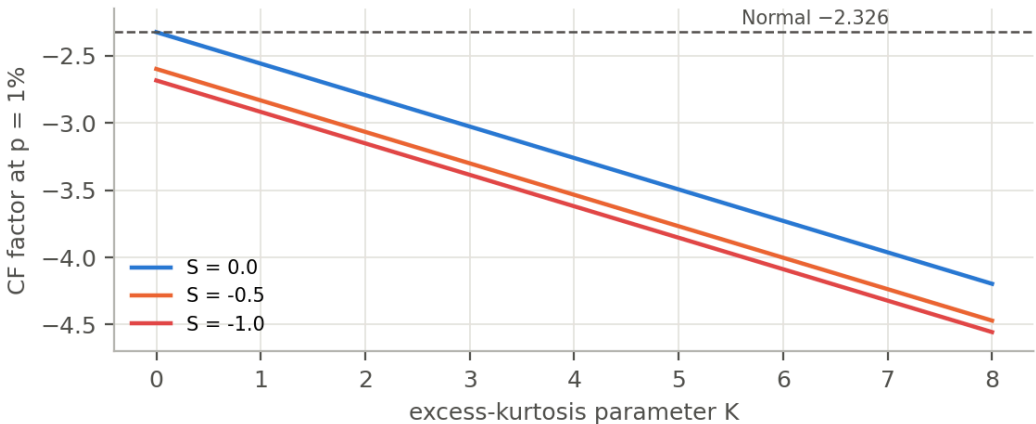


Figure 4. The 99% Cornish–Fisher factor as a function of K and S. Dashed: Normal factor.
图 4 99% Cornish–Fisher 因子随 K、S 的变化。虚线为正态因子。

Deep dive • S and K are parameters, not sample moments / 深度解析 • S、K 是参数，不是样本矩

After the CF transform, the **actual** skewness γ_1 and excess kurtosis γ_2 are complex non-linear functions of S, K , not S, K themselves. The rigorous route: compute sample γ_1, γ_2 , then **solve numerically** for S, K (e.g. `scipy.optimize.fsolve`); the solution is not unique and depends on starting values. Reference values (RiskConcile) are tabulated in the right-hand column.

For small $|\gamma_1|, \gamma_2$ the two nearly coincide, so direct substitution is an acceptable quick approximation.

Validity domain: the transform must be monotone so quantile order is preserved. This requires $|S|$ below about 2.49 and K within an S-dependent band; with $S = 0, 0 \leq K \leq 8$, and K never above 11.55. Implied limits on actual moments: $|\gamma_1| \leq 4.36, 0 \leq \gamma_2 \leq 43.28$. In practice $|\gamma_1| > 2$ is rare, corresponding to $|S|$ below 0.5–0.7.

经 CF 变换后的分布，其实际偏度 γ_1 与超额峰度 γ_2 是 S, K 的复杂非线性函数，并不等于 S, K 。严格做法是：计算样本 γ_1, γ_2 ，再数值求解对应的 S, K （如 `scipy.optimize.fsolve`）；解不唯一，依赖初值。对照 (RiskConcile)：

γ_1	γ_2	S	K
0.1	0.2	0.0958	0.1872
-0.2	0.5	-0.1821	0.4317
1.0	2.0	0.8833	3.5875

当 $|\gamma_1|, \gamma_2$ 较小时两者近似相等，直接代入是可接受的快速近似。

有效域：变换必须单调，分位数顺序才不会颠倒。这要求 $|S|$ 小于约 2.49，且 K 位于随 S 变化的区间内； $S = 0$ 时 $0 \leq K \leq 8$ ， K 最大不超过 11.55。对应的实际矩范围： $|\gamma_1| \leq 4.36, 0 \leq \gamma_2 \leq 43.28$ 。实务中 $|\gamma_1| > 2$ 很少见，对应 $|S|$ 低于 0.5–0.7。

In practice • Estimation window / 实操要点 • 估计窗口

Estimating skewness and kurtosis from 21–42 observations is meaningless: one or two extreme days swing them wildly (the course notebook’s 62-day rolling exploration confirms it). In practice estimate S, K over 200–500+ days and hold them as **periodically recalibrated parameters**; use the short window only for σ . RiskConcile’s 90-day rolling analysis of the FXI ETF shows short-window sample moments periodically leaving the validity domain, distorting VaR.

用 21–42 个观测估计偏度和峰度毫无意义——一两个极端日就能让它们剧烈摆动（课程 Python 笔记本中 62 日滚动窗口的探索也证实了这一点）。实务上用 200–500 日甚至更长的窗口估计 S, K ，并把它们作为**定期重校准的参数**固定使用；短窗口只用于估计 σ 。RiskConcile 对 FXI ETF 的 90 日滚动分析显示，短窗口的样本矩会周期性地落到有效域之外，导致 VaR 失真。

6.2 Cornish–Fisher VaR in PRIIPs PRIIPs 的 Cornish–Fisher VaR

$$\text{VaR}_{\text{return}} = \sigma \sqrt{N} \left(-1.96 + 0.474 \frac{\mu_1}{\sqrt{N}} - 0.0687 \frac{\mu_2}{N} + 0.146 \frac{\mu_1^2}{N} \right) - \frac{1}{2} \sigma^2 N$$

The European Supervisory Authorities (ESMA, EBA, EIOPA), in the PRIIPs RTS (final draft March 2016), use a 97.5% Cornish–Fisher VaR for Category 2 products, converted to a “VaR-equivalent volatility” (VEV) and mapped to a 1–7 market-risk class. Here μ_1, μ_2 are the skewness and excess kurtosis of **daily** returns and N the number of trading days in the recommended holding period; the $1/\sqrt{N}$ and $1/N$ scaling reflects how skewness and kurtosis decay under aggregation of i.i.d. returns.

欧洲监管三局（ESMA、EBA、EIOPA）在 PRIIPs 监管技术标准（2016 年 3 月终稿）中，对第 2 类产品采用 97.5% Cornish–Fisher VaR，再换算为 “VaR 等效波动率”（VEV）并映射为 1–7 级市场风险等级。其中 μ_1, μ_2 是日度收益的偏度和超额峰度， N 是建议持有期内的交易日数； $1/\sqrt{N}$ 与 $1/N$ 的缩放反映了 i.i.d. 假设下偏度和峰度随期限加总而衰减。

Pitfall • The sign of the S^2 term / 常见陷阱 • S^2 项的符号

The course notes and the RiskConcile note write the term as " $-0.146S^2$ ". Substituting $z = -1.96$ into the expansion: $-\frac{1}{36}(2z^3 - 5z) = -\frac{1}{36}(-15.06 + 9.80) = +0.146$. The PRIIPs text also has **+0.146**. Trust the derivation; copying the minus sign into a spreadsheet understates VaR for strongly skewed products.

课程讲义与 RiskConcile 白皮书将该式写作 " $-0.146S^2$ ". 按展开式代入 $z = -1.96$: $-\frac{1}{36}(2z^3 - 5z) = -\frac{1}{36}(-15.06 + 9.80) = +0.146$ 。PRIIPs 原文亦为 **+0.146**。请以推导为准；若在 Excel 中照抄负号，偏度较大的产品 VaR 会被低估。

6.3 Extreme value theory: peaks over threshold 极值理论 (EVT)：超额值峰值法

$$\text{VaR}_q = u + \frac{\beta}{\xi} \left\{ \left[\frac{n}{n_u} (1 - q) \right]^{-\xi} - 1 \right\}, \quad \text{ES}_q = \frac{\text{VaR}_q + \beta - \xi u}{1 - \xi}$$

Gnedenko–Pickands–Balkema–de Haan: for a high enough threshold u , the excess loss $L - u$ converges to a generalised Pareto distribution (GPD) with scale β and shape ξ . n is the total number of observations and n_u the number above u .

In practice: set u near the empirical 95th percentile (course workbook: 500 scenarios, $u = 160$, $n_u = 22$); fit β, ξ by maximum likelihood: compute $\ln g_i$ for each exceedance and maximise the sum over β, ξ (Excel Solver). Hull suggests $\xi \in [0.1, 0.4]$. EVT's value is **extrapolation**: it can estimate quantiles beyond the worst observed loss, which HS cannot.

Gnedenko–Pickands–Balkema–de Haan 定理：对足够高的阈值 u ，超额损失 $L - u$ 的分布收敛到广义帕累托分布 (GPD)，参数为尺度 β 与形状 ξ 。 n 为总观测数， n_u 为超过 u 的观测数。

实操：阈值常取经验分布的 95% 分位数附近（课程 Excel：500 个情景， $u = 160$ ， $n_u = 22$ ）；用极大似然估计 β, ξ ：对每个超额观测计算 $\ln g_i = \ln \left[\frac{1}{\beta} (1 + \xi \frac{x_i - u}{\beta})^{-1/\xi - 1} \right]$ ，对 β, ξ 最大化其和（Excel 规划求解）。Hull 建议 $\xi \in [0.1, 0.4]$ 。EVT 的价值在于**外推**：可以估计比样本中最差观测更极端的分位数，这是 HS 做不到的。

6.4 Method overview 方法总览

方法 Method	因子 Factor	波动率 Volatility	优点 Strengths	弱点 Weaknesses
Normal analytical 正态解析	$\Phi^{-1}(1-c)$	滚动 / EWMA	Simple, decomposable, closed-form ES 简单、可分解、闭式 ES	Understates fat tails 低估厚尾
Student-t	$t^{-1} \cdot \sqrt{(v-2)/v}$	滚动 / EWMA	One parameter for fat tails; suits single stocks 一个参数捕捉厚尾；适合单只股票	Symmetric; v unstable 对称； v 估计不稳
Cornish–Fisher	$z_{\text{CF}}(S,K)$	滚动 / EWMA	Skew and kurtosis; used by regulators 纳入偏度峰度；监管采用	Validity domain; needs long window for S,K 有效域受限；S,K 需长窗口
历史模拟 HS	empirical 经验分位数	implicit 隐含于样本	Model-free; full revaluation 无分布假设；非线性头寸全重估	Plateaux, slow, few tail points 平台、迟钝、尾部样本少
过滤 HS / FHS	residual empirical 残差经验分位数	EWMA revol EWMA 再波动化	Long history + current vol 兼顾长历史与当前波动	Alters moments and correlations; procyclical 改变高阶矩与相关；顺周期
EVT (GPD)	GPD 分位数	—	Extrapolates the far tail 可外推极端尾部	Threshold-sensitive; data hungry 阈值选择敏感；数据需求大

6.5 Three VaR frameworks: parametric, historical and Monte Carlo 三种 VaR 计算框架：参数法、历史模拟与蒙特卡洛

All three compute the same quantity, the $1 - c$ quantile of portfolio P&L. They differ only in **where the distribution comes from**.

- **Parametric** (variance–covariance, delta-normal): assume a distribution, estimate $\sigma_P = \sqrt{\mathbf{w}' \Sigma \mathbf{w}}$, $\text{VaR} = \mu + \text{factor} \times \sigma$. Variants use Normal, Student-t or Cornish–Fisher factors, and delta-gamma approximations for options (Chs 2–4, §6.1).
- **Historical simulation**: assume "tomorrow replays a past day", apply each of the last N days' factor moves to today's positions with full revaluation, sort and take the quantile. Variants: equal-weighted, age-weighted, filtered HS (Ch. 5).
- **Monte Carlo**: specify a stochastic model for risk factors (multivariate Normal or t, GARCH, jumps, copulas), simulate 10,000–100,000 scenarios, fully revalue each and take the quantile.

三种方法计算的是同一个量——组合损益分布的 $1 - c$ 分位数。区别只在于**这个分布从哪里来**。

- **参数法** (Parametric / Variance–Covariance / Delta-Normal)：假设损益服从某个分布，估计 $\sigma_P = \sqrt{\mathbf{w}' \Sigma \mathbf{w}}$ ， $\text{VaR} = \mu + \text{因子} \times \sigma$ 。变体有正态、Student-t、Cornish–Fisher 因子，以及期权组合的 Delta-Gamma 近似（第 2–4 章、§ 6.1）。
- **历史模拟** (Historical Simulation)：假设“明天重演历史上的某一天”，把过去 N 天的风险因子变动施加到今天的持仓上全重估，排序取分位数。变体有等权 HS、按时间加权 HS、过滤 HS（第 5 章）。
- **蒙特卡洛** (Monte Carlo Simulation)：为风险因子设定随机模型（多元正态或 t、GARCH、跳跃、copula），模拟 1 万至 10 万个情景，逐一全重估后取分位数。

	参数法 Parametric	历史模拟 Historical	蒙特卡洛 Monte Carlo
Source of distribution 分布来源	Assumed (Normal, t, CF) 假设（正态、t、CF）	Actual historical moves 历史实际变动	Specified stochastic model 设定的随机模型
Fat tails, skew 厚尾与偏度	Needs a corrected factor 需修正因子	Automatic within the window 自动包含窗口内的厚尾	Depends on the model 取决于模型
Non-linearity 非线性	Poor; delta-gamma approximation 差；Delta-Gamma 近似	Good (full revaluation) 好（全重估）	Good, including path dependence 好，也能处理路径依赖
Correlation 相关性	Covariance matrix, may be low in calm periods 协方差矩阵，平静期可能偏低	Historical co-movement 历史真实联动	Model-set, can be stressed 模型设定，可做压力相关
Beyond history 超出历史的情景	Yes (extrapolates) 可以（外推）	No; capped at the worst day 不能；不超过窗口最差日	Yes 可以
Data and compute 数据与计算量	Least, fastest 最少、最快	Needs complete history 需完整历史	Heaviest (N revaluations) 最大（N 次全重估）
Transparency 透明度	High 高	High, easy to explain 高，易于向董事会解释	Low, most model risk 低，模型风险最大
Typical weakness 典型问题	Understates tails 低估尾部	Plateaux, few tail points 平台效应、尾部样本少	Misspecification, sampling error 模型设错、抽样误差

Worked example 5 • One dataset, three methods / 例题 5 • 同一组数据，三种方法

The last 500 S&P 500 daily returns (Jan 2016–Jan 2018, daily volatility 0.64%), 1-day 99%: see the table in the right-hand column.

The Normal parametric result is the most optimistic; historical simulation captures the sample's actual tail. Monte Carlo depends on the model: an unconstrained maximum-likelihood t fit gives about 1.9 degrees of freedom, implying infinite variance and a meaningless ES, a vivid example of model risk.

标普 500 最近 500 个日收益（2016-01 至 2018-01，日波动率 0.64%），1 日 99%：

方法	VaR	ES
参数法（正态）	1.49%	1.71%
历史模拟	1.87%	2.55%
蒙特卡洛（t 分布，自由度 4，按样本 σ 校准）	1.68%	2.31%

正态参数法最乐观；历史模拟抓到了样本中的真实尾部。蒙特卡洛的结果取决于模型：若不设约束直接用极大似然拟合 t 分布，自由度只有约 1.9，意味着方差无穷大、ES 失去意义——这是模型风险的直观例子。

In practice • Risk-factor mapping and Monte Carlo sampling error / 实操要点 • 风险因子映射与蒙特卡洛抽样误差

- **Equities:** map single stocks to an index via β , but keep single-name factors for concentrated holdings or idiosyncratic risk is understated.
- **Bonds and rates:** map cash flows to key tenors (e.g. 3M, 1Y, 2Y, 5Y, 10Y, 30Y) with PV01 per tenor as the exposure; RiskMetrics mapping splits each cash flow between adjacent vertices preserving present value and variance. Add spread factors (CS01) for credit.
- **FX:** map net exposure by currency; non-base-currency assets carry both an asset and an FX factor (the cross term in §11.2).
- **Options:** delta-gamma-vega in parametric VaR, or delta-gamma-Cornish-Fisher using the first four moments of the quadratic P&L in the CF expansion. With heavy option books, switch to full-revaluation HS or Monte Carlo.
- **Monte Carlo sampling error:** the quantile's standard error is about $\sqrt{p(1-p)/N}/f(q)$, roughly ± 0.13 percentage points at $N = 10,000$ in Example 5. Use enough paths or variance reduction (antithetic variates, importance sampling), and fix the random seed for reproducibility.

- **股票：**个股可按 β 映射到指数，但集中持仓应保留个股因子，否则低估特质风险。
- **债券与利率：**现金流映射到关键期限点（如 3M、1Y、2Y、5Y、10Y、30Y），以各期限 PV01 为敞口；RiskMetrics 的映射在保持现值和方差不变的前提下，把现金流拆到相邻两个期限点。信用债另加利差因子（CS01）。
- **外汇：**按币种净敞口映射；非基础货币的资产同时受资产价格与汇率两个因子影响（§ 11.2 的交叉项）。
- **期权：**参数法用 Delta-Gamma-Vega；也可用 Delta-Gamma-Cornish-Fisher，即用二次型损益的前四阶矩代入 CF 展开。含大量期权时应改用全重估的 HS 或蒙特卡洛。
- **蒙特卡洛抽样误差：**分位数标准误差约为 $\sqrt{p(1-p)/N}/f(q)$ 。例题 5 中 $N = 10,000$ 时约 ± 0.13 个百分点；需要足够多的模拟次数，或用对偶变量、重要性抽样等方差缩减技术，并固定随机种子以保证结果可复现。

Choosing a method and regulatory use 如何选择与监管应用

框架 Framework	方法要求 Method rule	要点 Key points
UCITS （CESR/10-788）	All three allowed 三种都允许	Must capture all material risks including option gamma/vega and basis (e.g. swap–bond spread); independent validation after development; monthly backtesting. Limits apply whatever the method. 须覆盖全部重大风险，包括期权的 Gamma、Vega 与基差风险（如掉期与债券利差）；首次开发后须由独立于建模的一方验证；每月回溯。VaR 限额与方法无关。
AIFMD	Manager's choice 自选	Documented in the risk policy and independently validated; Annex IV reports the VaR and its method (Ch. 17). 写入风险管理政策并独立验证；Annex IV 填报 VaR 数值与计算方法（第 17 章）。
MiFID II / PRIIPs	Set by product category 按产品类别指定	Category 2 (linear) uses parametric Cornish–Fisher VaR; Category 3 (non-linear) uses bootstrap simulation (10,000+ resamples of historical returns), between HS and Monte Carlo. 第 2 类（线性）用参数法（Cornish–Fisher VaR）；第 3 类（非线性）用自助法模拟（10,000 次以上重抽样历史收益），介于 HS 与蒙特卡洛之间。
Banks (and CRR investment firms) 银行 Basel / FRTB（含 CRR 下投资公司）	Not prescribed 不规定方法	Mostly HS, some Monte Carlo; FRTB PLA effectively requires full revaluation of non-linear positions, which pure parametric models struggle to pass. 多数用 HS，部分用蒙特卡洛；FRTB 的 PLA 测试实际上要求非线性头寸全重估，纯参数法很难通过。
EMIR (CCP margin) EMIR（CCP 保证金）	Not prescribed 不规定方法	Mainly HS and FHS, with procyclicality controls (§5.4). 以 HS、FHS 为主，须控制顺周期性（§ 5.4）。

In practice: linear books (equities, bonds, spot FX) can use parametric VaR with a t or CF factor; options and structured products need full-revaluation HS or FHS; path-dependent products (barriers, autocallables) need Monte Carlo. Most banks and alternative UCITS run HS/FHS as the main engine, use parametric VaR for fast decomposition (§6.6), and Monte Carlo for complex products and long horizons.

实务选择：线性组合（股票、债券、外汇现货）用参数法即可，但因子应做 t 或 CF 修正；含期权与结构化产品用 HS 或 FHS 全重估；路径依赖产品（障碍期权、自动赎回票据）用蒙特卡洛。多数银行与另类 UCITS 以 HS/FHS 为主力，参数法用于快速分解（§ 6.6），蒙特卡洛用于复杂产品与长期限。

6.6 VaR decomposition: marginal, component and incremental VaR VaR 分解：边际、成分与增量 VaR

$$MVaR_i = \frac{\partial VaR}{\partial w_i} = z \frac{(\Sigma \mathbf{w})_i}{\sigma_P}, \quad CVaR_i = w_i MVaR_i, \quad \sum_i CVaR_i = VaR, \quad IVaR = VaR(\mathbf{w} + \Delta \mathbf{w}) - VaR(\mathbf{w}) \approx \Delta \mathbf{w}' MVaR$$

VaR is positively homogeneous, so by Euler's theorem the **component VaRs** (position × marginal VaR) add up exactly to portfolio VaR. This answers three daily questions:

- **Marginal VaR:** how much does VaR change per extra €1 of an asset? Guides which trade reduces risk most;
- **Component VaR:** where does portfolio risk come from? Used for risk budgets and limit allocation;
- **Incremental VaR:** how much would a new trade change VaR? Used in pre-trade approval; recompute directly for large trades rather than using the marginal approximation.

For HS or Monte Carlo, take each position's average P&L in scenarios where portfolio P&L is near VaR (or decompose ES the same way, which is more stable).

VaR 是正齐次函数，由欧拉定理，各头寸的**成分 VaR**（头寸 × 边际 VaR）恰好加总为组合 VaR。这回答了三个日常问题：

- **边际 VaR：**多持有 €1 某资产，VaR 变化多少？用于判断加仓还是减仓更降风险；
- **成分 VaR：**组合风险来自哪里？用于风险预算与限额分配；
- **增量 VaR：**一笔新交易会 让 VaR 变化多少？用于交易前审批。大额交易应直接重算，而非用边际近似。

对 HS 或蒙特卡洛，成分 VaR 可取"组合损益接近 VaR 的情景中各头寸损益的平均"（或对 ES 做同样分解，结果更稳定）。

Worked example 6 • Decomposing VaR for three assets / 例题 6 • 三资产组合的 VaR 分解

€100m portfolio: equities €50m (16% annual volatility), government bonds €30m (5%), credit €15m (6%); correlations equity–govt –0.2, equity–credit 0.5, govt–credit 0.4. 1-day 99% parametric VaR = **€1.23m**; decomposition in the table in the right-hand column.

Stand-alone VaRs sum to €1.52m against €1.23m for the portfolio; the €0.29m gap (about 19%) is the diversification benefit. Equities are half the notional but 92% of the risk; government bonds, negatively correlated with equities, contribute almost nothing. Incremental VaR: another €10m of equities adds about €0.23m of VaR; another €10m of government bonds adds only about €0.004m.

€1 亿组合：股票 €5,000 万（年化波动率 16%）、政府债 €3,000 万（5%）、信用债 €1,500 万（6%）；相关系数 股票–政府债 –0.2、股票–信用 0.5、政府债–信用 0.4。1 日 99% 参数法 VaR = **€123 万**。

头寸	单独 VaR	成分 VaR	占比
股票	€117 万	€114 万	92.4%
政府债	€22 万	€0.7 万	0.6%
信用债	€13 万	€8.6 万	7.0%
合计	€152 万	€123 万	100%

单独 VaR 之和 €152 万与组合 VaR €123 万之差（€29 万，约 19%）就是分散化收益。股票占名义本金一半，却贡献了 92% 的风险；政府债因与股票负相关，成分 VaR 几乎为零。增量 VaR：再加 €1,000 万股票，VaR 增加约 €23 万；再加 €1,000 万政府债，仅增加约 €0.4 万。

6.7 Illiquid assets and data gaps 非流动资产与数据缺口

AIFs often hold private equity, real estate and private credit without daily prices. VaR computed on appraisal series badly understates risk because appraisals are smoothed. Common treatments:

- **Proxies:** map to listed comparables (private equity → sector small-cap index with a leverage-adjusted β ; private credit → high-yield or leveraged-loan index);
- **Unsmoothing:** apply the Geltner adjustment $r_t = (r_t^* - \phi r_{t-1}^*) / (1 - \phi)$ to appraisal returns r_t^* , with ϕ the first-order autocorrelation, to restore suppressed volatility;
- **Look-through:** funds of funds and multi-layer structures should look through to underlying holdings; if impossible, use fund-level return proxies with a conservative buffer;
- **Missing data:** for new issues or mismatched holidays, backfill, proxy or interpolate, and record the scope and effect of each treatment;
- **Complementary measures:** for such assets, pair VaR with stress tests, liquidity buckets and drawdown analysis (Parts II and III).

AIF 常持有私募股权、房地产、私募信贷等缺少日度价格的资产。直接用其估值序列计算 VaR 会严重低估风险，因为评估价格是平滑的。常见处理：

- **代理序列：**用上市可比资产映射（私募股权 → 同行业小盘股指数乘以杠杆调整后的 β ；私募信贷 → 高收益债或杠杆贷款指数）；
- **去平滑：**对评估收益 r_t^* 做 Geltner 调整 $r_t = (r_t^* - \phi r_{t-1}^*) / (1 - \phi)$ ， ϕ 为一阶自相关，恢复被平滑掉的波动；
- **穿透：**组合基金或多层结构应穿透到底层持仓；无法穿透时用基金层面的收益代理并加保守缓冲；
- **缺失数据：**新上市证券或节假日不一致时，用回填、代理或插值，并记录每种处理的范围与影响；
- **补充指标：**对这类资产，VaR 应与压力测试、流动性分层和回撤分析一起使用（第二、三部分）。

6.8 Multi-factor models and multi-factor VaR 多因子模型与多因子 VaR

$$r_i = \alpha_i + \sum_k \beta_{ik} f_k + \varepsilon_i, \quad \Sigma = B \Omega B' + D,$$

$$\text{VaR} = z \sqrt{\mathbf{b}' \Omega \mathbf{b} + \mathbf{w}' D \mathbf{w}}, \quad \mathbf{b} = B' \mathbf{w}$$

A multi-factor model explains many asset returns with a few common factors (market, rates, spreads, industries, styles). Its core job is **dimension reduction**: the risk of thousands of assets becomes the risk of dozens of factors plus each asset's specific risk. B holds factor exposures, Ω the factor covariance ($K \times K$, $K \ll N$), D the diagonal specific-risk matrix (residuals assumed uncorrelated).

多因子模型用少数共同因子（市场、利率、利差、行业、风格等）解释大量资产的收益，核心作用是**降维**：几千个资产的风险归结为几十个因子的风险加上各资产的特质风险。 B 为因子暴露， Ω 为因子协方差（ $K \times K$ ， $K \ll N$ ）， D 为特质风险对角矩阵（假设残差互不相关）。

Type	Factor source	Examples
Macroeconomic	Observable economic and market variables	GDP, inflation, rates, oil, USD
Fundamental / style	Asset characteristics	Industry, country, value, size, momentum, quality (Barra, Axioma-type models)
Statistical	Extracted from return data	Principal components, e.g. level, slope and curvature of the yield curve

类型	因子来源	例子
宏观因子	可观测经济与市场变量	GDP、通胀、利率、油价、美元
基本面 / 风格因子	资产特征	行业、国家、价值、规模、动量、质量（Barra、Axioma 类商业模型）
统计因子	从收益数据中 提取	主成分分析（PCA），如利率曲线的水平、斜率、曲率

Multi-factor models in regulatory frameworks 监管体系中的多因子模型

体系 Framework	如何使用因子 How factors are used
FRTB standardised approach FRTB 标准法（SBM）	A regulator-specified multi-factor model: prescribed risk factors (e.g. 10 GIRR tenors, credit spreads, equity, commodity, FX), delta/vega/curvature sensitivities times prescribed risk weights, aggregated with prescribed correlations. EU from 2027; also used for K-NPR by CRR investment firms. 本质上是监管规定的多因子模型：预设风险因子（如一般利率风险 10 个期限点、信用利差、股票、商品、外汇），计算 Delta、Vega、曲率敏感度，乘以规定风险权重，再按规定相关系数加总： $K_b = \sqrt{\sum W S_k^2 + \sum \sum \rho_{kl} W S_k W S_l}$ 。欧盟 2027 年起适用；CRR 下投资公司的 K-NPR 同样使用。
FRTB internal models FRTB 内部模型法	Each risk factor must pass the eligibility test (RFET); failures are NMRFs capitalised by stress scenarios; stressed ES is calibrated on a reduced factor set explaining at least 75% of full-set ES. 每个风险因子须通过可建模性测试（RFET），不通过的为不可建模风险因子（NMRF），单独按压力情景计提；压力期 ES 用"精简风险因子集"校准，须解释全因子集 ES 的至少 75%。
UCITS（CESR/10-788）	The VaR model must cover every risk factor with more than negligible influence; in practice, map positions to factors (§6.5). VaR 模型须覆盖所有影响不可忽略的风险因子；实务上即把持仓映射到因子（§ 6.5）。
AIFMD Annex IV	Net equity delta, bucketed DV01, CS01 and vega are factor exposures in all but name (Ch. 17). 报告净股票 Delta、按期限分段的 DV01、CS01、Vega，本质上是一组因子敞口（第 17 章）。
EMIR uncleared margin EMIR 非清算保证金	UCITS and AIFs typically compute initial margin with ISDA SIMM, a standardised multi-factor sensitivity model similar in structure to SBM. UCITS 与 AIF 作为金融对手方普遍用 ISDA SIMM 计算初始保证金：标准化的多因子敏感度模型，结构与 SBM 类似。
Solvency II（insurance） Solvency II（保险）	The standard formula charges interest, equity, property, spread, currency and concentration modules and aggregates with correlation matrices; internal models often use factor-based economic scenario generators. 标准公式按利率、股票、房地产、利差、汇率、集中度等模块计提再以相关矩阵汇总；内部模型多用基于因子的经济情景生成器。
EBA / ECB stress tests EBA / ECB 压力测试	Macro factor paths mapped through satellite models to credit losses and market shocks (§10.4, §14.2). 宏观因子路径经"卫星模型"映射到信用损失与市场冲击（§ 10.4、§ 14.2）。
US SEC Rule 18f-4（reference） 美国 SEC Rule 18f-4（参照）	Fund derivatives VaR limits in the UCITS mould: relative VaR ≤ 200% of a designated reference portfolio, or absolute VaR ≤ 20% of net assets; 99%, 20 days, at least 3 years of data. 基金衍生品 VaR 限额与 UCITS 思路相同：相对 VaR ≤ 指定参考组合的 200%，或绝对 VaR ≤ 净资产 20%；99%、20 日、至少 3 年数据。

Implementing multi-factor VaR 多因子 VaR 的实施

1. **Choose factors**: market, industry and style for equities; tenors and spreads for bonds; plus FX.
2. **Estimate exposures B** : time-series regression or characteristics; DV01, CS01 and delta directly for bonds and derivatives.
3. **Estimate factor covariance Ω** : typically EWMA with shrinkage or positive-definiteness repair; stress key correlations where needed.
4. **Estimate specific risk D** : residual variances; do not ignore it for concentrated holdings.
5. **Compute**: parametric with the formula above; for HS or Monte Carlo, replay or simulate factor returns, bootstrap or simulate residuals separately, and fully revalue.

Why factors: 1,000 stocks need about 500,000 covariances, 40 factors only about 800, and more stable; new or data-poor securities get risk through their exposures (§6.7); VaR splits into systematic and specific parts and by factor (§6.6); stress tests and VaR share one factor set.

1. **选因子**：股票用市场、行业、风格；债券用期限点与利差；另加汇率。
2. **估暴露 B** ：时间序列回归或资产特征；债券与衍生品直接用 DV01、CS01、Delta。
3. **估因子协方差 Ω** ：常用 EWMA，并做收缩或正定性修正；必要时对关键相关性做压力设定。
4. **估特质风险 D** ：残差方差；集中持仓的特质风险不可忽略。
5. **计算**：参数法用上式；HS 或蒙特卡洛则重演或模拟因子收益，残差另行自助抽样或模拟，再全重估。

为什么用因子：1,000 只股票约需估计 50 万个协方差，40 个因子只需约 800 个且更稳定；新证券或缺数据的证券可通过暴露估风险（§ 6.7）；VaR 可拆为系统性与特质部分并按因子分解（§ 6.6）；压力测试与 VaR 共用一套因子，口径一致。

Worked example 12 • Multi-factor VaR / 例题 12 • 多因子 VaR

€100m equity portfolio of 40 stocks. Exposures: market $\beta = 1.1$, value 0.3, size -0.2 ; annual factor volatilities: market 16%, value 6%, size 5%; each stock's specific volatility 25%, about 4.0% at portfolio level after equal-weight diversification. Results in the table in the right-hand column.

The portfolio is essentially "levered market beta": the most effective risk reduction is hedging the market factor with index futures, not swapping stocks.

€1 亿股票组合，持有 40 只股票。因子暴露：市场 $\beta = 1.1$ 、价值 0.3、规模 -0.2 ；因子年化波动率：市场 16%、价值 6%、规模 5%；每只股票特质波动率 25%，等权分散后组合特质波动率约 4.0%。

结果	数值
组合年化波动率	17.8%
其中系统性 / 特质	17.3% / 4.0%
1 日 99% VaR	约 €260 万
方差贡献：市场因子	约 96%
方差贡献：特质风险	约 5%
方差贡献：风格因子	≈ 0 （规模 -0.9% ）

这个组合本质上是"带杠杆的市场 β "：降风险最有效的是用股指期货对冲市场因子，而不是换股。

Pitfall • Limits of multi-factor models / 常见陷阱 • 多因子模型的局限

- **Omitted factors:** when new common drivers emerge in a crisis (liquidity, crowded trades) the model understates risk;
- **Unstable exposures and correlations:** correlations rise in crises and the "uncorrelated specific risk" assumption breaks;
- **Fat tails:** the parametric version still needs a t or Cornish–Fisher correction;
- **Validation:** check the bias statistic (standard deviation of realised return ÷ forecast volatility, ideally about 1) and backtest regularly.
- **遗漏因子:** 危机中出现新的共同驱动（流动性、拥挤交易）时模型低估风险；
- **暴露与相关性不稳定:** 危机中相关性上升，"特质风险互不相关"的假设失效；
- **厚尾:** 参数法版本仍需 t 或 Cornish–Fisher 修正；
- **验证:** 定期检查偏差统计量（实际收益 ÷ 预测波动率的标准差，理想值约为 1），并做回测。

6.9 Yield curves: construction and use in risk management 收益率曲线：构建与在风险管理中的应用

$$DF(t) = e^{-z(t)t}, \quad f(t_1, t_2) = \frac{z(t_2)t_2 - z(t_1)t_1}{t_2 - t_1}, \quad DV01_k = -\frac{\partial V}{\partial r_k} \times 0.0001$$

Curves underpin fixed-income and derivative valuation and interest-rate risk: **zero rates** $z(t)$, **discount factors** $DF(t)$ and **forward rates** $f(t_1, t_2)$ are three views of one curve. In risk management each tenor point is a risk factor.

Multi-curve framework: since 2008 one currency needs several curves. The **discount curve** follows the collateral rate: cash-collateralised derivatives under a CSA are discounted at the overnight rate (€STR, SOFR, SONIA), i.e. OIS discounting; **projection curves** are built per floating index (e.g. 3M and 6M EURIBOR). USD and GBP LIBOR have ceased and contracts moved to risk-free rates. The CSA collateral currency sets the discount curve; where several currencies are eligible, the poster holds a cheapest-to-deliver option, and cross-currency collateral brings in FX basis.

Curve construction 曲线构建

1. **Choose inputs:** deposits or OIS at the short end, futures or FRAs in the middle, swaps at the long end; liquid and snapped at the same time.
2. **Bootstrap:** start at the shortest tenor and use known discount factors to solve for the next, so every input reprices exactly.
3. **Interpolate:** log-linear discount factors (simple, piecewise-flat forwards), monotone convex (smooth, no negative-forward spikes), splines (smoother but can oscillate locally). The choice directly affects forwards and key-rate sensitivities.
4. **Check:** zero repricing error on inputs; no spikes in forwards; floors in negative-rate regions; long-end extrapolation (insurers use Smith–Wilson converging to an ultimate forward rate).

曲线是固定收益、衍生品估值和利率风险的基础：**零息利率** $z(t)$ 、**贴现因子** $DF(t)$ 与**远期利率** $f(t_1, t_2)$ 是同一条曲线的三种写法。风险管理里，曲线的每个期限点都是一个风险因子。

多曲线框架: 2008 年后，同一货币要用多条曲线。**贴现曲线**由抵押品利率决定——有 CSA、以现金抵押的衍生品按隔夜利率（€STR、SOFR、SONIA）贴现，即 OIS 贴现；**预测曲线**按浮动利率指数分别构建（如 3 个月与 6 个月 EURIBOR）。美元和英镑 LIBOR 已停止，相关合约已转向无风险利率。CSA 约定的抵押品币种决定贴现曲线；可用多种币种交抵押品时，交出方有"最便宜交付"选择权，跨币种还涉及外汇基差。

1. **选输入工具:** 短端用存款或 OIS，中段用期货或 FRA，长端用利率互换；须流动、同一时点报价。
2. **逐段剥离 (bootstrap):** 从最短期开始，每一步用已知的贴现因子求出下一个期限的贴现因子，使每个输入工具被精确重新定价。
3. **插值:** 对数线性贴现因子（简单、远期分段平坦）、单调凸插值（远期平滑且无负值跳动）、样条（更平滑但有局部振荡风险）。插值方法会直接影响远期利率和关键期限敏感度。
4. **检查:** 输入工具重新定价误差为零；远期曲线没有异常尖峰；负利率区间的下限设置；长端外推（保险业用 Smith–Wilson 方法向终极远期利率收敛）。

其他曲线 Other curves	用途 Use
Credit (CDS spreads → hazard rates, survival) 信用曲线（CDS 利差 → 违约强度、生存概率）	CS01, CVA (§8.7), credit bond valuation CS01, CVA（§ 8.7）、信用债估值
Repo and dividend curves 回购与股息曲线	Equity forwards, futures basis, equity options 股票远期、期货基差、股票期权定价
Inflation curve 通胀曲线	Inflation-linked bonds and swaps; inflation scenarios (§14) 通胀挂钩债与互换；通胀情景（§ 14）
Commodity forward curve 商品远期曲线	Contango/backwardation, roll yield; map futures VaR by maturity point 升水 / 贴水、展期收益；期货的 VaR 按到期点映射
Volatility term structure and surface 波动率期限结构与曲面	Option valuation and bucketed vega (§10.3.2, §18.4) 期权估值与 Vega 分段（§ 10.3.2、§ 18.4）

Using curves in risk management 曲线在风险管理中的应用

- **Key-rate DV01**: bump one tenor by 1bp at a time, rebuild the curve and reprice to get per-tenor sensitivities. State whether you bump **par quotes** or **zero rates**; results differ.
- **Map to regulatory buckets**: the 10 FRTB GIRR tenors (0.25, 0.5, 1, 2, 3, 5, 10, 15, 20, 30 years); the 4 UCITS duration-netting buckets (§8.6); bucketed DV01 in Annex IV (Ch. 17).
- **Principal components (PCA)**: curve moves are typically explained by three components: **level** (parallel, about 80–90% of variance), **slope** (steepening/flattening) and **curvature** (butterfly). Use them to reduce dimensions, design scenarios and set limits.
- **Curve scenarios**: parallel, steepener, flattener, twist, butterfly. Basel IRRBB prescribes six standard shocks: parallel up/down, steepener, flattener, short up/down. Short shocks decay as $e^{-t/4}$, long shocks build as $1 - e^{-t/4}$; steepener $= -0.65|\Delta R_{\text{short}}| + 0.9|\Delta R_{\text{long}}|$, flattener $= +0.8|\Delta R_{\text{short}}| - 0.6|\Delta R_{\text{long}}|$. After the Basel Committee's 2024 recalibration, EUR sizes from 1 January 2026 are parallel 225bp, short 350bp, long 200bp (implemented in the EU through EBA standards; check the latest text). The same shapes work well as interest-rate stress scenarios for funds.

- **关键期限 DV01**: 逐个期限点上移 1bp、重新构建曲线并重估，得到各期限的敏感度。须说明是冲击**输入报价** (par) 还是**零息利率**，两者结果不同。
- **映射到监管分段**: FRTB 一般利率风险 10 个期限点 (0.25、0.5、1、2、3、5、10、15、20、30 年); UCITS 久期轧差 4 个区间 (§ 8.6); Annex IV 按期限分段报告 DV01 (第 17 章)。
- **主成分分析 (PCA)**: 利率曲线变动通常可由三个主成分解释——**水平** (平移, 约占 80–90% 的方差)、**斜率** (陡峭化 / 平坦化)、**曲率** (蝶式)。可用来降维、设计情景和设限额。
- **曲线情景**: 平移、陡峭化、平坦化、扭曲、蝶式。巴塞尔银行账簿利率风险 (IRRBB) 规定了六种标准冲击: 平行上移 / 下移、陡峭化、平坦化、短端上移 / 下移。短端冲击按 $e^{-t/4}$ 衰减, 长端按 $1 - e^{-t/4}$ 增加; 陡峭化 $= -0.65|\Delta R_{\text{short}}| + 0.9|\Delta R_{\text{long}}|$, 平坦化 $= +0.8|\Delta R_{\text{short}}| - 0.6|\Delta R_{\text{long}}|$ 。巴塞尔委员会 2024 年重新校准后, 欧元自 2026 年 1 月 1 日起为平行 225bp、短端 350bp、长端 200bp (欧盟通过 EBA 技术标准落地, 以最新文本为准)。这些形状同样适合作为基金的利率压力情景。

Worked example 14 • Curve risk hidden by parallel DV01 / 例题 14 • 平行 DV01 掩盖的曲线风险

Portfolio: receive-fixed 2-year swap €200m (loses €39k per +1bp), pay-fixed 10-year swap €50m (gains €44k per +1bp), 30-year government bond €20m (duration 20, loses €40k per +1bp). Net parallel DV01: loses €35k per +1bp. Results under IRRBB shapes with 2026 EUR parameters are in the table in the right-hand column.

The three positions largely offset in parallel DV01, so rate risk looks small; yet a short-end rise loses almost as much as a parallel rise, a flattener loses €3.83m while a steepener makes money. Parallel DV01 alone badly understates curve risk here, so limits and stress tests should be set by key rate or principal component.

组合: 收取固定的 2 年期互换 €2 亿 (利率每升 1bp 亏 €3.9 万)、支付固定的 10 年期互换 €5,000 万 (每升 1bp 赚 €4.4 万)、30 年期国债 €2,000 万 (久期 20, 每升 1bp 亏 €4 万)。合计平行 DV01: 每升 1bp 亏 €3.5 万。用 IRRBB 形状、欧元 2026 年参数:

情景	2 年	10 年	30 年	损益
平行上移	+225	+225	+225	-€788 万
平行下移	-225	-225	-225	+€788 万
陡峭化	-67	+147	+180	+€188 万
平坦化	+123	-87	-120	-€383 万
短端上移	+212	+29	0	-€702 万
短端下移	-212	-29	0	+€702 万

三笔头寸的平行 DV01 大部分相互抵消, 看起来利率风险不大; 但短端上移几乎和平行上移亏得一样多, 平坦化亏 €383 万而陡峭化反而赚钱。只看平行 DV01 会严重低估这类组合的曲线风险, 因此限额和压力测试都应按关键期限或主成分来设定。

Regulatory mapping • Non-Normal corrections / 监管对应 • 非正态修正

框架	English	中文
UCITS	Retail share classes' PRIIPs KID uses Cornish–Fisher VaR (§6.2); the pre-2023 KIID used volatility bands (SRRI).	零售份额类别的 PRIIPs KID 使用 Cornish–Fisher VaR (§6.2); 2023 年前的 KIID 用波动率区间 (SRRI)。
AIFMD	Retail-facing AIFs are also in PRIIPs scope; professional AIFs choose freely, and EVT is common in tail-risk reporting.	面向零售的 AIF 同样适用 PRIIPs; 专业投资者 AIF 可自选方法, EVT 常用于尾部风险报告。
MiFID II	Distributors use the SRI for target markets: $VEV = [\sqrt{(3.842 - 2 \cdot VaR)} - 1.96] / \sqrt{T}$; MRM classes 1–7 break at 0.5%, 5%, 12%, 20%, 30%, 80%.	分销商以 SRI 界定目标市场: $VEV = [\sqrt{(3.842 - 2 \cdot VaR)} - 1.96] / \sqrt{T}$, MRM 1–7 级阈值 0.5%、5%、12%、20%、30%、80%。

7 Backtesting: Procedure, Tests and Traffic Lights

VaR 回测: 步骤、统计检验与交通灯

7.1 The logic of a backtest 回测的逻辑

Picture this: each morning you compute a 99%/10-day VaR using **only data available then** and "seal" it; ten trading days later you compare it with the realised return over that period and check whether the worst-loss prediction was breached. If the model is right, breaches occur about 1% of the time and **independently**. A backtest tests exactly these two properties: **frequency** (unconditional coverage) and **independence**.

想象每天早上, 你只用**截至当时的**数据计算 99%/10 日 VaR 并"封存"; 10 个交易日后, 把它与这段时间实际实现的收益对比, 看"最坏损失"的预测是否被突破。若模型正确, 突破应以约 1% 的频率**独立地**发生。回测检验的正是这两点: **频率** (无条件覆盖) 与**独立性**。

7.2 Step by step (course Excel / Python demo) 逐步操作 (课程 Excel / Python 演示)

$$\text{VaR}_t = \Phi^{-1}(0.01) \times \sigma_t \sqrt{10} \ (+10\mu), \quad r_{10D,t} = \ln(S_{t+10}/S_t), \quad \text{breach}_t = 1\{r_{10D,t} < \text{VaR}_t\}$$

1. **Returns:** the day-1 price is only used to compute the first log-return.

2. **Rolling σ :** 21 returns are needed for the first (daily) s.d.; the window moves by one observation (returns 1–21, 2–22, ...).

3. **VaR:** from the day of the 21st return, compute VaR_t daily.

4. **Forward return:** $r_{10D,t}$ uses prices 10 days after t ; the last 10 days have none.

5. **Eligible comparisons:** $T = N_{\text{prices}} - 21 - 10$. Course data: 1,250 prices → 1,219 comparisons.

6. **Statistics:** (a) count and share of breaches; (b) conditional breach probability $\Pr(\text{breach}_t \mid \text{breach}_{t-1})$; (c) comparison with the binomial.

7. **Plot:** VaR_t and $r_{10D,t}$ together with breaches marked; look for clusters.
1. **收益:** 第 1 天的价格只用于算第一个对数收益。

2. **滚动 σ :** 需要 21 个收益才能算第一个 (日频) 标准差; 窗口每次移 1 个观测 (第 1–21、2–22……个收益)。

3. **VaR:** 从第 21 个收益所在日起, 每天算出 VaR_t 。

4. **前瞻收益:** $r_{10D,t}$ 使用 t 之后 10 天的价格, 最后 10 天没有前瞻收益。

5. **有效比较数:** $T = N_{\text{价格}} - 21 - 10$ 。课程数据 1,250 个价格 → 1,219 次比较。

6. **统计:** (a) 突破次数与比例; (b) 条件突破概率 $\Pr(\text{breach}_t \mid \text{breach}_{t-1})$; (c) 与二项分布比较。

7. **作图:** VaR_t 与 $r_{10D,t}$ 同图、标出突破点, 观察是否成串。

Pitfall • Overlapping 10-day returns / 常见陷阱 • 重叠的 10 日收益
Consecutive 10-day forward returns share 9 days of data, so breaches cluster mechanically: one large drop breaches several consecutive comparisons. The binomial "independent breaches" assumption fails and independence tests reject by construction. Regulatory backtesting therefore compares **1-day** VaR with **1-day** P&L (MAR32). To backtest a 10-day VaR, use non-overlapping periods (one tenth of the sample) or overlap-adjusted tests.

相邻两天的 10 日前瞻收益共享 9 天数据, 因此突破天然成串: 一次大跌会让连续好几天的比较同时突破。这使得"突破相互独立"的二项分布假设不成立, 独立性检验会被机械性地拒绝。监管回溯因此使用**1 日** VaR 对比 **1 日** P&L (MAR32)。若必须回溯 10 日 VaR, 可只用不重叠区间 (样本缩小为 1/10), 或用考虑重叠的调整检验。

7.3 The binomial distribution and Basel traffic lights 二项分布与 Basel 交通灯

$$X \sim \text{Binomial}(T, p = 1 - c), \quad \mathbb{E}[X] = Tp, \quad \Pr(X \leq k) = \sum_{j=0}^k \binom{T}{j} p^j (1 - p)^{T-j} \quad (\text{Excel: BINOM.DIST}(k, T, p, \text{TRUE}))$$

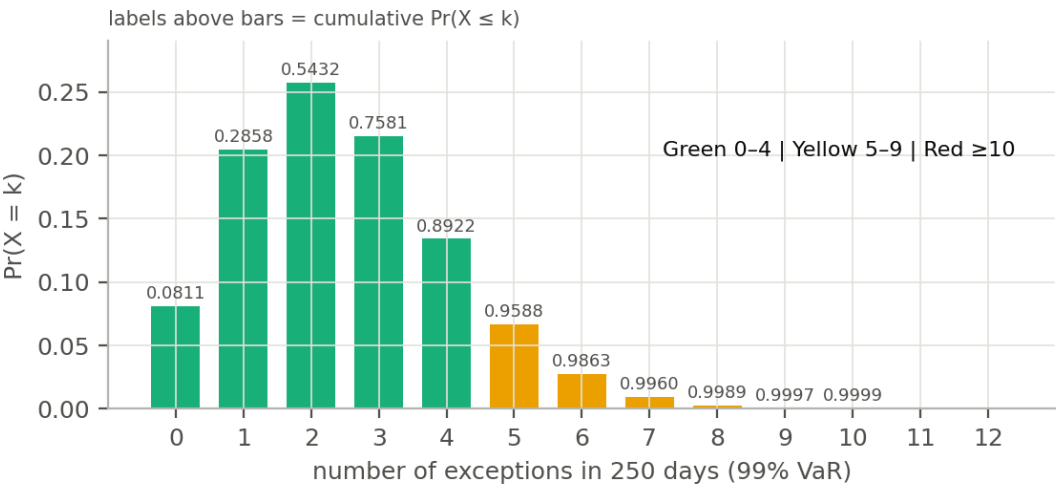


Figure 5. Distribution of breach counts for T=250, p=1%; numbers above bars are cumulative probabilities.
图 5 T=250, p=1% 时突破次数的分布; 柱上数字为累积概率。

Basel (1996) rule: with k breaches, the model is **green** if $\Pr(X \leq k) < 95\%$, **yellow** if $95\% \leq \Pr(X \leq k) < 99.99\%$, **red** if $\geq 99.99\%$. For $T = 250$: green 0–4, yellow 5–9, red ≥ 10 .

The zone drives a capital multiplier applied to the VaR-based charge (Basel 2.5: 3 + plus factor; FRTB MAR32 bank-wide: 1.5 rising to 2.0). In the red zone a supervisor may also withdraw model approval.

Basel (1996) 规则: 若观察到 k 次突破, 且 $\Pr(X \leq k) < 95\%$, 为绿区; $95\% \leq \Pr(X \leq k) < 99.99\%$ 为黄区; $\geq 99.99\%$ 为红区。对 $T = 250$: 绿 0–4, 黄 5–9, 红 ≥ 10 。

突破数	区域	Basel 2.5 乘数	FRTB 乘数
0–4	绿	3.00	1.50
5	黄/琥珀	3.40	1.70
6	黄/琥珀	3.50	1.76
7	黄/琥珀	3.65	1.83
8	黄/琥珀	3.75	1.88
9	黄/琥珀	3.85	1.92
≥ 10	红	4.00	2.00

Pitfall • Off-by-one zone boundaries / 常见陷阱 • 区域边界差一

Using `scipy.stats.binom.ppf(0.95, 250, 0.01) = 5` and writing "green ≤ 5 " is off by one: `ppf` returns the smallest k whose cumulative probability **reaches** 95%, and that k already belongs to yellow. Correct: green if $k < \text{ppf}(0.95)$, red if $k \geq \text{ppf}(0.9999)$. Also, one course slide gives the yellow threshold as 0.9599; it should be **0.9999**. The slide's breach-probability table is in fact the **probability mass** for $T=252$, $p=5\%$ (`BINOMDIST(..., FALSE)`), not cumulative probabilities for a 99% VaR.

若用 `scipy.stats.binom.ppf(0.95, 250, 0.01)` 得到 5，并写成"绿区 ≤ 5 "，就错了一格：`ppf` 返回的是"使累积概率**达到** 95% 的最小 k "，而该 k 本身已属于黄区。正确写法是"绿区 $k < \text{ppf}(0.95)$ "、"红区 $k \geq \text{ppf}(0.9999)$ "。另：课件有一处把黄区阈值写成 0.9599，应为 **0.9999**；课件中的"突破概率表"实际是 $T=252$ 、 $p=5\%$ 的**概率质量** (`BINOMDIST(..., FALSE)`)，并非 99% VaR 的累积概率。

7.4 Kupiec proportion-of-failures test Kupiec 比例失败检验 (POF)

$$\text{LR}_{\text{POF}} = -2 \ln \frac{(1-p)^{T-x} p^x}{(1-\hat{p})^{T-x} \hat{p}^x}, \quad \hat{p} = \frac{x}{T}, \quad \text{LR}_{\text{POF}} \sim \chi_1^2 \text{ (5\% critical value 3.841)}$$

Null: the true breach probability equals p . The likelihood ratio compares how likely the observed breaches are under the theoretical p versus the observed frequency $\hat{p}$. The test is **two-sided**: too many breaches (aggressive model) and too few (over-conservative, wasted capital) are both rejected.

Example: $T = 249$, $x = 11$, $\hat{p} = 4.42\%$. $\text{LR} = 15.96$, $p\text{-value } 6.5 \times 10^{-5}$: a strong rejection.

原假设：真实突破概率等于 p 。似然比检验比较"按理论概率 p "与"按观测频率 $\hat{p}$ "两种情况下观察到这组突破的可能性。它是**双侧**的：突破太多（模型激进）与太少（模型过度保守、资本浪费）都会被拒绝。

例： $T = 249$, $x = 11$, $\hat{p} = 4.42\%$ 。 $\text{LR} = 15.96$, $p\text{ 值} = 6.5 \times 10^{-5}$ ，强烈拒绝。

7.5 Christoffersen independence and conditional coverage Christoffersen 独立性与条件覆盖

$$\pi_0 = \frac{n_{01}}{n_{00} + n_{01}}, \quad \pi_1 = \frac{n_{11}}{n_{10} + n_{11}}, \quad \text{LR}_{\text{ind}} = -2 \ln \frac{(1-\pi)^{n_{00}+n_{10}} \pi^{n_{01}+n_{11}}}{(1-\pi_0)^{n_{00}} \pi_0^{n_{01}} (1-\pi_1)^{n_{10}} \pi_1^{n_{11}}} \sim \chi_1^2, \quad \text{LR}_{\text{cc}} = \text{LR}_{\text{POF}} + \text{LR}_{\text{ind}} \sim \chi_2^2$$

n_{ij} counts days with state i yesterday and j today ($1 = \text{breach}$). The course's "probability of a breach given a breach in the previous period" is π_1 . For a good model $\pi_1 \approx \pi_0 \approx p$; $\pi_1 \gg p$ signals clustered breaches and a model that reacts too slowly to volatility changes. The conditional coverage test LR_{cc} (5% critical value 5.991) tests frequency and independence jointly.

n_{ij} 为"前一天状态 i 、当天状态 j "的天数 ($1 = \text{突破}$)。课程中的"给定上期突破时本期突破的条件概率"就是 π_1 。若模型良好, $\pi_1 \approx \pi_0 \approx p$; 若 $\pi_1 \gg p$, 说明突破成串, 模型对波动率变化反应太慢。条件覆盖检验 LR_{cc} (5% 临界值 5.991) 同时检验频率与独立性。

In practice • Power and sample size / 实操要点 • 检验力与样本量

With 250 observations a year and 2.5 expected breaches, every test has low power: a model whose true breach rate is 2% has a sizeable chance of staying green. With few breaches the Christoffersen statistic is also unstable (n_{11} is usually 0 or 1). So: (1) use as long a history as possible (years, many portfolios); (2) backtest several levels (97.5% and 99%); (3) combine with P&L attribution, residual diagnostics and qualitative review rather than a single p -value.

一年 250 个观测、期望突破 2.5 次，任何检验的检验力都很低：一个真实突破率为 2% 的模型，有相当大的概率仍落在绿区。突破极少时，Christoffersen 统计量也不稳定 (n_{11} 往往为 0 或 1)。因此：(1) 用尽可能长的历史（多年、多组合）；(2) 同时回溯多个置信水平（97.5% 与 99%）；(3) 结合 P&L 归因、残差诊断与定性审查，而不是只看一个 p 值。

7.6 Three ways to improve VaR 改进 VaR 的三个方向

1. **Mean:** has μ been dropped? A positive μ biases VaR towards optimism.
2. **Volatility:** responsive enough? Too procyclical? GARCH adds parameters and pulls forecasts towards the long-run mean; practitioners use EWMA but must manage procyclicality.
3. **Factor:** the Normal percentile is "good enough" but inadequate for empirical P&L; Cornish–Fisher or t factors improve it.

1. **均值：**是否已去掉 μ ？保留一个正的 μ 会让 VaR 偏乐观。
2. **波动率：**够不够灵敏？是否过于顺周期？GARCH 引入更多参数且预测被拉向长期均值；实务常用 EWMA，但要管理顺周期性。
3. **因子：**正态分位数"够用"，但对经验 P&L 并不充分；Cornish–Fisher 或 t 因子可以改进。

Regulatory mapping • Backtesting / 监管对应 • 回溯

框架	English	中文
UCITS	Monthly backtest of 1-day VaR at least; more than 4 overshootings at 99% in 250 days must be reported to senior management and the regulator, with a model review (consistent with the Basel 0–4 green zone).	至少每月回溯 1 日 VaR；250 日内 99% 突破超过 4 次须报告高管层与主管机关，并审查模型（与 Basel 绿区 0–4 次一致）。
AIFMD	Level 2 Art. 45 requires periodic backtests "where appropriate", with no numeric threshold; practice often borrows the UCITS 4-overshooting standard.	Level 2 第 45 条要求"适当时"定期回溯，无量化门槛；实务中多参照 UCITS 的 4 次标准。
MiFID II	Investment firms using CRR internal models follow bank backtesting rules (1-day 99% VaR vs actual and hypothetical P&L, traffic-light multiplier).	使用 CRR 内部模型的投资公司适用银行回溯规则（1 日 99% VaR 对比实际与假设 P&L，交通灯乘数）。

8 Regulation of Short-Term Risk Measurement

短期风险的监管要求

The three EU regimes differ sharply in how hard they bite on short-term measurement: **UCITS** has binding VaR limits and a backtesting threshold; **AIFMD** prescribes process, leverage calculation and reporting but no VaR cap; **MiFID II** itself sets no VaR rules; its requirements arrive through the PRIIPs KID, client loss reporting and the investment-firm prudential regime (IFR/IFD). This chapter covers each, then the bank framework (Basel/FRTB) as a reference point.

三套欧盟规则对短期风险计量的要求强度差别很大：**UCITS** 有硬性的 VaR 上限与回溯门槛；**AIFMD** 规定流程、杠杆计算与报告，但不设 VaR 上限；**MiFID II** 本身不规定 VaR，相关要求通过 PRIIPs KID、客户损失通知与投资公司资本规则（IFR/IFD）体现。本章逐一展开，最后附银行框架（Basel/FRTB）作为参照。

8.1 UCITS: global exposure UCITS：全球风险暴露

项目 Item	要求 Requirement（CESR/10-788；Directive 2010/43/EU）
方法选择 Method	Default commitment approach : global exposure ≤ 100% of NAV (conversion, netting and hedging rules in §8.6). VaR is mandatory for complex strategies, non-negligible exotic derivatives, or where commitment does not capture the risk. 默认 承诺法 ：全球风险暴露 ≤ 100% NAV（换算、轧差与对冲细则见 § 8.6）。策略复杂、奇异衍生品敞口不可忽略、或承诺法不能充分反映风险时， 必须用 VaR 法 并结合压力测试。
VaR 上限 Limits	Relative VaR ≤ 2 × reference-portfolio VaR; absolute VaR ≤ 20% of NAV. 相对 VaR ≤ 2 × 参考组合 VaR; 绝对 VaR ≤ 20% NAV。
标准参数 Parameters	99% one-tailed; 20 business days; ≥ 1-year history (shorter only if volatility has risen significantly); data updated at least quarterly; calculated at least daily. 99% 单尾；持定期 20 个营业日；观测期 ≥ 1 年（除非波动率显著上升可缩短）；数据至少每季更新；至少每日计算。
替代参数 Alternatives	Confidence ≥ 95% and horizon ≤ 1 month allowed, with the 20% limit rescaled by Normal quantiles and √t: 95%/5-day ≈ 7.07% . 置信水平 ≥ 95%、持定期 ≤ 1 个月，20% 上限按正态分位数与 √t 同比缩放：95%/5 日 ≈ 20% × (1.645/2.326) × √(5/20) = 7.07% 。
回溯 Backtesting	At least monthly, 1-day VaR vs actual daily change; more than 4 overshootings at 99% in 250 days must be reported to senior management and the competent authority, with a model review. 至少每月，用 1 日 VaR 对比组合实际日变动；250 日内 99% 突破 超过 4 次 须报告高管层与主管机关（爱尔兰为 Central Bank of Ireland），并审查模型。
模型治理 Governance	The model must capture all material risks (including non-linearity), with initial and periodic validation; the method is documented in the risk management process filed with the regulator. 模型须覆盖全部重大风险（含非线性），首次与定期验证；方法写入风险管理流程（RMP）并报备。
披露 Disclosure	VaR funds disclose expected leverage (sum of notionals) in the prospectus; retail share classes provide a PRIIPs KID (replacing the KIID since 2023). VaR 法基金须在招股书披露预期杠杆（名义本金之和）；零售份额类别提供 PRIIPs KID（2023 年起取代 KIID）。

In practice • Running UCITS VaR day to day / 实操要点 • UCITS VaR 的日常运作

1. **Reference portfolio for relative VaR**: unleveraged and, in principle, free of derivatives (except to express shorts in long/short strategies or for currency-hedged indices); its risk profile must match the fund's objectives, policies and limits; if the risk/return profile changes often or no reference portfolio can be defined, relative VaR must **not** be used; switch to absolute VaR. Selection and maintenance belong in the risk management process and the reference portfolio is disclosed in the prospectus.

2. **Holding-period conversion**: with a 1-day VaR, rescale the 20% limit by $VaR_{x\text{ days}} \approx VaR_{t\text{ days}} \sqrt{x/t}$ (assuming i.i.d. factor returns), or equivalently multiply the 1-day VaR by √20 and compare with 20%. Relative VaR needs no adjustment since both sides scale.

3. **Limit-usage monitoring**: compute "VaR ÷ limit" daily. Example: the \$6.6 portfolio has a 1-day 99% VaR of €1.23m, ×√20 ≈ €5.50m = 5.5% of NAV, i.e. 27.5% of the absolute VaR cap. Set internal warning (e.g. 80%) and hard (e.g. 90%) lines to keep a buffer.

4. **Annual report**: disclose at least the **lowest, highest and average** VaR-limit usage over the financial year and the model and inputs (model, confidence level, holding period, data length); disclose leverage employed (sum of notionals). The prospectus discloses expected leverage and the possibility of higher levels.

5. **Risk management process (RMP)**: method, parameters, reference portfolio, backtesting and stress-testing procedures and escalation are written into the RMP filed with the regulator; material changes require an RMP update.
1. **相对 VaR 的参考组合**：须无杠杆，原则上不含衍生品（多空策略用衍生品表达空头、或货币对冲指数除外）；风险特征须与基金的投资目标、政策和限额一致；若风险收益特征经常变化或无法定义参考组合，**不得使用**相对 VaR，应改用绝对 VaR。参考组合的确定与维护须纳入风险管理流程，并在招股书中披露。

2. **持定期换算**：用 1 日 VaR 时，按 $VaR_{x\text{ days}} \approx VaR_{t\text{ days}} \sqrt{x/t}$ 对 20% 上限做同比缩放（前提是风险因子收益独立同分布）。等价地，也可把 1 日 VaR 乘以 √20 后与 20% 比较。相对 VaR 两边同时缩放，无需调整。

3. **限额使用率监控**：每日计算"VaR ÷ 限额"。例：\$ 6.6 组合 1 日 99% VaR €123 万，× √20 ≈ €550 万 = 5.5% NAV，使用绝对 VaR 上限的 27.5%。可设内部预警线（如 80%）与硬限额（如 90%），留出缓冲。

4. **年报披露**：至少披露财年内 VaR 限额使用率的**最低、最高与平均值**，以及模型与输入（计算模型、置信水平、持定期、数据长度）；披露期间实际杠杆（名义本金之和）。招股书披露预期杠杆及可能更高的情形。

5. **风险管理流程文件（RMP）**：方法、参数、参考组合、回溯与压力测试程序、上报流程写入 RMP 并报备主管机关；方法或参数的重大变更须更新 RMP。

Deep dive • Absolute versus relative VaR / 深度解析 • 绝对 VaR 与相对 VaR

Absolute VaR: the fund's own VaR as a percentage of NAV, compared with no benchmark; $\leq 20\%$ of NAV at 99%/20 days. Suited to absolute-return, multi-strategy and alternative UCITS without a fitting benchmark. Example: a 20-day VaR of 12% of NAV uses 60% of the cap.

Relative VaR: the **ratio** of fund VaR to reference-portfolio VaR, ≤ 2 . Suited to benchmark-driven funds using derivatives for overlays or leverage. Example: fund VaR 12%, reference portfolio (unleveraged MSCI World) VaR 8%, ratio 150%.

	Absolute VaR	Relative VaR
Yardstick	Fixed 20% of NAV	Reference VaR $\times 2$
Cap moves with markets?	No	Yes
In a crisis	Cap reached sooner; forced deleveraging possible	Reference VaR rises too, allowed VaR expands
In calm markets	Room for heavy leverage within 20%	Bounded by $2\times$
Horizon conversion	1-day VaR $\times \sqrt{20}$	Both sides scale; none needed
Prerequisite	None	An eligible reference portfolio

Two common mistakes: (1) relative VaR is **not** tracking-error VaR: the former is a ratio of two stand-alone VaRs, the latter the VaR of the active return "fund minus benchmark", measuring deviation from the benchmark; (2) the method is fixed in the RMP and prospectus and cannot be switched opportunistically to make compliance easier.

绝对 VaR: 基金自身 VaR 占 NAV 的百分比，不与任何基准比较；99%/20 日下 $\leq 20\%$ NAV。适用于绝对收益、多策略、另类 UCITS 等没有合适基准的基金。例：基金 20 日 VaR = 12% NAV，使用上限的 60%。

相对 VaR: 基金 VaR 与参考组合 VaR 的**比值** ≤ 2 倍。适用于有明确基准、用衍生品做叠加或杠杆的基金。例：基金 VaR 12%、参考组合（无杠杆 MSCI World）VaR 8%，比值 150%。

	绝对 VaR	相对 VaR
尺子	固定 20% NAV	参考组合 VaR $\times 2$
上限是否随市场变	不变	随市场波动变化
危机时	更易触及上限，可能被迫降杠杆	参考组合 VaR 同步上升，允许的 VaR 放大
平静期	可在 20% 内大幅加杠杆	受 2 倍约束
持有期换算	1 日 VaR 须 $\times \sqrt{20}$	分子分母同时缩放，无需换算
前提	无	须有合格参考组合

两个常见误区: (1) 相对 VaR **不等于**跟踪误差 VaR——前者是两个独立 VaR 的比值，后者是"基金减基准"主动收益的 VaR，衡量偏离基准的风险；(2) 方法选择须写入风险管理流程与招股书，不能为了更容易合规而临时切换。

Pitfall • Absolute VaR and leverage / 常见陷阱 • 绝对 VaR 与杠杆

The absolute VaR cap limits "loss in normal markets", not leverage itself. ESMA's 2025 analysis found alternative UCITS on absolute VaR ran average gross leverage of about 800% of NAV, above hedge-fund AIFs at 448%. When volatility is low VaR shrinks and funds can lever up within the 20% cap; when volatility returns they may be forced into procyclical deleveraging. VaR limits must therefore be paired with stress testing (Part II) and leverage monitoring.

绝对 VaR 上限约束的是"正常市场下的损失"，不是杠杆本身。ESMA 2025 年的分析显示，采用绝对 VaR 法的另类 UCITS 平均总杠杆约 800% NAV，高于对冲基金 AIF 的 448%。低波动期 VaR 变小，基金可以在不违反 20% 上限的情况下加杠杆；一旦波动回升，就可能被迫顺周期去杠杆。因此 VaR 限额必须与压力测试（第二部分）和杠杆监控一起使用。

8.2 AIFMD: process, leverage and reporting AIFMD：流程、杠杆与报告

- **Risk function** (AIFMD Art. 15; Level 2 Arts 38–45): functionally separate from portfolio management; limits for market, credit, liquidity, counterparty and operational risk; "where appropriate", periodic backtests of risk measurement and stress tests and scenario analyses. Whether to use VaR is the manager's choice, recorded in the risk management policy.
- **Leverage** (Level 2 Arts 7–8): compute under both the **gross** and **commitment** methods; set and disclose a maximum per AIF; commitment leverage above $3\times$ NAV is "substantial" and triggers extra reporting. Regulators may cap leverage under Art. 25 (and ESMA's 2020 guidelines).
- **Annex IV reporting:** market risk profile (net equity delta, DV01, CS01, vega, VaR and its method, etc.), liquidity profile, stress-test results and monthly returns. See Chapter 17 for the step-by-step preparation process.
- **AIFMD II** (2024/927, Level 1 applicable from 16 April 2026; Ireland S.I. 181/2026): open-ended AIFs select at least two liquidity management tools; loan-originating AIFs capped at 175% (open-ended) and 300% (closed-ended) leverage. The new reporting template follows Level 2 measures after October 2027.
- If sold to retail investors, an AIF also needs a PRIIPs KID (see §6.2).
- **风险管理职能**（AIFMD 第 15 条；Level 2 第 38–45 条）：与投资管理能力分离；为市场、信用、流动性、交易对手与操作风险设定限额；"适当时"定期回溯风险计量方法，并做压力测试与情景分析。选用 VaR 与否由管理人决定并写入风险管理政策。
- **杠杆**（Level 2 第 7–8 条）：必须同时按**总额法**与**承诺法**计算；为每只 AIF 设定最高杠杆并向投资者披露；承诺法杠杆超过 NAV 的 **3 倍**即为"实质性杠杆"，须额外报送。监管机构可依第 25 条（及 ESMA 2020 年指引）施加杠杆上限。
- **Annex IV 报告**：包括市场风险概况（净股票 Delta、DV01、CS01、Vega、VaR 及其计算方法等）、流动性概况、压力测试结果与月度收益。逐步准备流程见第 17 章。
- **AIFMD II**（2024/927, Level 1 自 2026 年 4 月 16 日起适用；爱尔兰 S.I. 181/2026）：开放式 AIF 须选定至少两种流动性管理工具；发放贷款的 AIF 杠杆上限开放式 175%、封闭式 300%。新版报告模板随 Level 2 细则晚于 2027 年 10 月。
- 若向零售投资者销售，AIF 同样需要 PRIIPs KID（见 § 6.2）。

8.3 MiFID II: clients, products and firm capital MiFID II: 客户、产品与公司资本

- **Clients:** suitability and appropriateness must reflect risk tolerance and capacity to bear losses; discretionary or leveraged accounts must be told by end of business day when value falls **10%** (and each further 10%) since the last report (Delegated Regulation 2017/565 Art. 62).
- **Products:** product governance uses the SRI (1–7) to express target-market risk tolerance; the SRI combines the PRIIPs MRM (97.5% Cornish–Fisher VaR → VEV, §6.2) with the credit risk class.
- **Firm capital** (IFR/IFD): investment firms with trading books compute K-NPR using CRR market-risk methods. EU FRTB applies from **1 January 2027**, with transitional relief to end-2029 under the June 2026 delegated act (e.g. PLA as monitoring only, weekly ES).
- **客户层面:** 合适性与适当性评估须考虑风险承受度和承受损失能力; 全权委托或含杠杆头寸的账户, 价值较上次报告下跌**10%** (及此后每 10%) 须在当日营业结束前通知客户 (Delegated Regulation 2017/565 第 62 条)。
- **产品层面:** 产品治理以 SRI (1–7) 界定目标市场的风险承受度; SRI 来自 PRIIPs 的 MRM (97.5% Cornish–Fisher VaR → VEV, § 6.2) 与信用风险等级。
- **公司资本** (IFR/IFD): 持有交易账簿的投资公司, K-NPR 沿用 CRR 市场风险方法。欧盟 FRTB 自 **2027 年 1 月 1 日** 起适用, 2026 年 6 月授权法案给予至 2029 年底的过渡性豁免 (如 PLA 暂作监测、ES 可按周计算)。

8.4 Reference: bank and margin frameworks 参照: 银行与保证金框架

框架 Framework	度量 Measure	要点 Key points
Basel 1996 / 2.5	99% 10 日 VaR (+ 压力 VaR, 2009)	Capital = multiplier (3 + plus) × VaR; 1-day 99% backtest with traffic lights; ≥250 days of data. 资本 = 乘数 (3 + 附加因子) × VaR; 1 日 99% VaR 回溯, 交通灯; ≥250 日数据。
FRTB (Basel III 市场风险, MAR30–33)	97.5% ES, liquidity horizons 10–120 days, stressed calibration 97.5% ES, 流动性期限 10–120 日, 压力期校准	Desk-by-desk IMA approval: backtesting + PLA; bank-wide 1-day 99% VaR vs actual and hypothetical P&L over 250 days. 内部模型需逐台审批: 回溯 + P&L 归因 (PLA); 银行层面 1 日 99% VaR 对比实际与假设 P&L, 250 日。
PRIIPs RTS	97.5% Cornish–Fisher VaR → VEV → 市场风险等级 1–7	Standardised risk disclosure for retail packaged products. 面向零售投资产品的标准化风险披露。
EMIR (CCP 保证金)	通常 99%+ VaR/ES, 常用 FHS	Must limit procyclical changes: buffer, stressed weighting, or a 10-year lookback floor. 须限制顺周期变化: 缓冲、压力权重或 10 年回看下限。

Deep dive • Bank market-risk capital formulas / 深度解析 • 银行市场风险资本公式

Basel 2.5 (from 2009): $\text{capital} = \max\{\text{VaR}_{t-1}, m_c \overline{\text{VaR}}_{60}\} + \max\{s \text{VaR}_{t-1}, m_s s \overline{\text{VaR}}_{60}\}$, with VaR and stressed VaR at 99%/10 days, $\bar{\tau}_{60}$ the 60-business-day average, and multipliers $m_c, m_s \geq 3$ plus the backtesting add-on (§7.3). Stressed VaR is calibrated to a 12-month period of significant stress so capital does not collapse in calm markets.

FRTB (EU from 2027): 97.5% ES cascaded by liquidity horizon $LH_j \in \{10, 20, 40, 60, 120\}$ days using the formula in the right-hand column, where $\text{ES}_T(P, j)$ shocks only factors with liquidity horizon $\geq LH_j$. ES is calibrated on a reduced factor set over a stress period and scaled by the current full/reduced ratio, then blended with the sum of asset-class ES at $ho = 0.5$ (partial diversification), plus stressed capital for non-modellable risk factors (NMRF) and the default risk charge (DRC).

Basel 2.5 (2009 年起): 资本 = $\max\{\text{VaR}_{t-1}, m_c \overline{\text{VaR}}_{60}\} + \max\{s \text{VaR}_{t-1}, m_s s \overline{\text{VaR}}_{60}\}$, 其中 VaR 与压力 VaR 均为 99%/10 日, $\bar{\tau}_{60}$ 为过去 60 个营业日均值, 乘数 $m_c, m_s \geq 3$ 加回溯附加因子 (§ 7.3)。压力 VaR 用 12 个月的显著压力期数据校准, 使资本在平静期不至过低。

FRTB (欧盟 2027 年起): 按流动性期限 $LH_j \in \{10, 20, 40, 60, 120\}$ 日分层计算 97.5% ES:

$$\text{ES} = \sqrt{\text{ES}_T(P)^2 + \sum_{j \geq 2} \left(\text{ES}_T(P, j) \sqrt{\frac{LH_j - LH_{j-1}}{T}} \right)^2}, T = 10$$

$\text{ES}_T(P, j)$ 只冲击流动性期限 $\geq LH_j$ 的风险因子。ES 用"精简因子集 × 压力期"校准, 并按"全因子集 / 精简因子集"在当前期的比率放大; 再与分资产类别 ES 之和按 $ho = 0.5$ 加权 (部分承认分散化), 加上不可建模风险因子 (NMRF) 的压力资本与违约风险资本 (DRC)。

8.5 Three P&L definitions under FRTB FRTB 中的三种 P&L

- **Actual P&L (APL):** all realised P&L including intraday trading and daily valuation adjustments (excluding fees and commissions);
 - **Hypothetical P&L (HPL):** revalue yesterday's closing positions with today's market data, excluding intraday trades and fees; it measures the effect of market moves on yesterday's book;
 - **Risk-theoretical P&L (RTPL):** P&L produced by the risk model's valuation engine using only modelled risk factors; it measures whether the model explains P&L.
 - **实际 P&L (APL):** 包含日内交易、费用以外的全部实际损益与每日估值调整;
 - **假设 P&L (HPL):** 把前一日收盘头寸用今日市场数据重新估值, 排除日内交易与费用——衡量"市场变动对昨日头寸的影响";
 - **风险理论 P&L (RTPL):** 由风险模型的估值引擎、仅用模型内风险因子算出的 P&L——衡量"模型能否解释 P&L"。
- 回溯:** VaR 对比 APL 与 HPL。交易台层面, 12 个月内 99% 突破 >12 次或 97.5% 突破 >30 次, 即失去内部模型资格。
- P&L 归因 (PLA):** 比较 RTPL 与 HPL, 使用 Spearman 秩相关 (绿区 >0.80, 红区 <0.70) 和 Kolmogorov–Smirnov 统计量 (绿区 <0.09, 红区 >0.12)。红区交易台退回标准法, 琥珀区加收资本。

In practice • The hard part is the P&L itself / 实操要点 • 真正的难点在 P&L 本身

Textbooks backtest a single index; real books hold equities, options, commodity futures, TRS and rates products. Producing clean, timely HPL and RTPL per desk, consistent with the risk system (snapshot timing, valuation-model consistency, market-data sources, stripping fees and valuation adjustments), often takes longer than the VaR formula itself. Aligning valuation and risk teams on "same timestamp, same curves, same models" is a prerequisite for passing PLA.

教科书只回溯一个指数；真实组合可能包含股票、期权、商品期货、总收益互换、利率产品。为每个交易台准备干净、及时、与风险系统口径一致的 HPL 和 RTPL，涉及头寸快照时点、估值模型一致性、市场数据来源、费用与估值调整的剔除，这往往比 VaR 公式本身更耗时。估值与风险团队对齐"同一时点、同一曲线、同一模型"是通过 PLA 的前提。

8.6 UCITS commitment approach: conversion, netting and hedging UCITS 承诺法：换算、轧差与对冲

UCITS not using VaR measure global exposure with the **commitment approach**: convert each derivative into the equivalent market value of its underlying, apply permitted netting and hedging, sum, and keep the total $\leq 100\%$ of NAV. Exposure from efficient portfolio management (repo, securities lending) where cash collateral is reinvested above the risk-free return also counts towards the same cap (CESR/10-788).

未采用 VaR 法的 UCITS 用**承诺法**计量全球风险暴露：把每个衍生品换算成标的资产的等值市场头寸，经允许的轧差与对冲后加总，结果须 $\leq 100\%$ NAV。高效投资组合管理（回购、证券借贷）中，若把现金抵押品再投资于收益高于无风险利率的资产，所产生的敞口也须计入同一上限（CESR/10-788）。

工具 Instrument	换算方法 Conversion
Equity index / stock futures 股指 / 股票期货	Contracts $\times$ multiplier $\times$ index level or price 合约数 $\times$ 合约乘数 $\times$ 指数点位或股价
Bond futures 国债期货	Contracts $\times$ notional size $\times$ market price of cheapest-to-deliver bond 合约数 $\times$ 名义合约规模 $\times$ 最便宜可交割券市价
Plain vanilla options 普通期权	Contracts $\times$ multiplier $\times$ underlying price $\times$ delta (maximum delta for barriers) 合约数 $\times$ 乘数 $\times$ 标的价格 $\times$ Delta（障碍期权取最大 Delta）
Interest rate swaps 利率互换	Market value of underlying (or fixed-leg notional) 标的市值（或固定端名义本金）
CDS (protection seller) CDS（卖出保护）	Higher of reference-asset market value and notional 参考资产市值与名义本金中较高者
CDS (protection buyer) CDS（买入保护）	Market value of reference asset 参考资产市值
Total return swaps 总收益互换	Underlying market value; both legs for non-basic TRS 标的市值；非标准 TRS 为两端市值之和
FX forwards / currency swaps 外汇远期 / 货币互换	Notional of the currency leg(s) 货币端名义本金
Convertibles and embedded derivatives 可转债等嵌入衍生品	Referenced shares $\times$ share price $\times$ delta 对应股数 $\times$ 股价 $\times$ Delta
Variance swaps 方差互换	Variance notional $\times$ current variance 方差名义本金 $\times$ 当前方差

- **Excluded:** (1) performance swaps that fully exchange the return of held assets for another reference return without extra leverage or optionality; (2) derivatives which, combined with cash held, equal a direct holding of the underlying (e.g. index futures plus equivalent cash), creating no incremental exposure.
- **Netting** (same underlying): derivatives on the same underlying, or a derivative and its underlying asset, may be netted even with different maturities, provided the aim is to eliminate risk.
- **Hedging** (not necessarily the same underlying) must meet all five criteria: aim to offset risk, not generate return; verifiable risk reduction at fund level; offsets both general and specific risk; same asset class; effective in stressed markets. Recognised: duration hedging of bonds with an IRS; beta hedging a diversified equity portfolio with index futures. **Not recognised:** hedging a share with a correlated different share, alpha-preserving beta hedges, merger arbitrage, cross-asset hedges (e.g. credit protection against equities).
- **Duration netting** (interest-rate derivatives): allocate to four buckets (0–2y, 2–7y, 7–15y, 15y+), convert to equivalent positions at the target duration (derivative MtM $\div$ target duration $\times$ duration). Net fully within a bucket; count 40% of amounts netted between adjacent buckets, 75% one bucket apart, 100% for the most remote buckets and for unnetted positions. Not allowed for arbitrage strategies or hedging arrangements.

- **不计入：**(1) 把组合持仓的收益完全换成其他参考资产收益、且无额外杠杆或期权特征的业绩互换；(2) 衍生品与所持现金组合后等同于直接持有标的（如股指期货 + 等额现金），不产生增量敞口。
- **轧差**（同一标的）：同一标的的衍生品之间、衍生品与标的资产之间可以轧差，即使到期日不同；目的必须是消除风险。
- **对冲**（不必同一标的）：须同时满足五条——目的是抵消风险而非获取收益；组合层面风险可验证地下降；同时抵消一般风险与特定风险；属于同一资产类别；在压力市场中仍然有效。认可例子：债券组合 + 利率互换的久期对冲、分散化股票组合 + 股指期货的 β 对冲。**不认可：**用相关的另一只股票对冲、保留 α 的 β 对冲、并购套利、跨资产类别对冲（如用信用保护对冲股票）。
- **久期轧差**（利率衍生品）：按剩余期限分入 0–2 年、2–7 年、7–15 年、15 年以上四个区间，先换算为目标久期下的等值头寸（衍生品市值 $\div$ 目标久期 $\times$ 久期）。同一区间内完全轧差；相邻区间轧差部分计入 40%；隔一个区间计入 75%；最远区间计入 100%；未轧差部分计入 100%。不得用于套利策略或对冲安排。

Worked example 7 • Commitment calculation / 例题 7 • 承诺法计算

NAV €100m: physical equities €40m, cash €60m; derivatives and their conversion as in the table in the right-hand column. Total commitment **€53m = 53% of NAV**, within the 100% cap.

If the €20m net long futures are fully cash-backed as an index-replication strategy, they can be excluded as cash-equivalent, cutting commitment to 33%. Physical equities do not count (they are not derivatives), but a short index future used to beta-hedge them can be offset if it meets the five hedging criteria.

NAV €1 亿：实物股票 €4,000 万、现金 €6,000 万。衍生品：

头寸	换算	计入
买入 Euro Stoxx 50 期货（12 月）	€3,000 万	€2,000 万（同标的轧差）
卖出 Euro Stoxx 50 期货（3 月）	-€1,000 万	
买入欧元兑美元远期（非对冲）	€1,500 万	€1,500 万
买入指数看涨期权，名义 €2,000 万，Delta 0.4	€800 万	€800 万
卖出 CDS 保护，名义 €1,000 万（参考债市值 €950 万）	€1,000 万	€1,000 万
合计		€5,300 万 = 53% NAV

在 100% 上限之内。若净多头期货 €2,000 万由等额现金支持、构成指数复制策略，可按“现金等价”不计入，承诺敞口降为 33%。实物股票本身不计入承诺法（它不是衍生品），但若用空头股指期货对它做 β 对冲并满足五条标准，该空头期货的敞口可以抵消。

8.7 Counterparty risk and CVA 交易对手风险与 CVA

$$CE = \max \left(\sum_{k \in \text{netting set}} MtM_k - C, 0 \right), \quad CVA \approx (1 - R) \sum_i EE(t_i) [PD(t_{i-1}, t_i)] DF(t_i), \quad \lambda \approx \frac{s}{1 - R}$$

Market risk measures losses from price moves; **counterparty risk** measures losses when a counterparty defaults while a contract is in the money. Key concepts:

- **Current exposure (CE)**: sum of MtM within a netting set (ISDA master agreement) less collateral *C*, floored at zero;
- **Potential future exposure (PFE)**: a high quantile (e.g. 95%) of future exposure; **expected exposure (EE)** is its mean and EPE its time average;
- **CVA**: the valuation adjustment for possible counterparty default, with recovery *R* and hazard rate λ approximated from the CDS spread *s*;
- **Wrong-way risk**: default probability rising with exposure (e.g. buying sovereign CDS protection from a bank in that country) badly understates CVA;
- **Mitigants**: netting, collateral agreements (CSA), daily variation margin (VM), initial margin (IM), central clearing.

市场风险度量价格变动造成的损失；**交易对手风险**度量对手在合约盈利时违约造成的损失。关键概念：

- **当前敞口（CE）**：同一净额结算集合（ISDA 主协议）内各合约市值之和，扣除抵押品 *C*，负值取 0；
- **潜在未来敞口（PFE）**：未来某时点敞口的高分位数（如 95%）；**预期敞口（EE）** 是其均值，EPE 是 EE 的时间平均；
- **CVA**：因对手可能违约而对衍生品估值做的扣减，*R* 为回收率， λ 由 CDS 利差 *s* 近似推出；
- **错向风险**：对手违约概率与敞口同向上升（如向某国银行买入该国主权 CDS 保护），会使 CVA 被严重低估；
- **缓释**：净额结算、抵押品协议（CSA）、每日变动保证金（VM）、初始保证金（IM）、中央清算。

Worked example 8 • Estimating CVA / 例题 8 • CVA 估算

A 5-year interest rate swap with constant expected exposure EE = €1m (uncollateralised); counterparty 5-year CDS spread 120bp, recovery 40%, risk-free rate 3%. Hazard rate $\lambda = 0.012/0.6 = 2\%$, 5-year cumulative default probability 9.5%. Summing quarterly gives **CVA ≈ €53k**, about 5.3% of the exposure; the shortcut "EE × spread × annuity" gives €55k. With a daily-VM CSA, EE falls to the price move over the margin period of risk (tens of thousands of euros) and CVA shrinks to a small fraction.

一笔 5 年期利率互换，预期敞口 EE 恒为 €100 万（未抵押）；对手 5 年 CDS 利差 120bp，回收率 40%，无风险利率 3%。违约强度 $\lambda = 0.012/0.6 = 2\%$ ，5 年累计违约概率 9.5%。按季度离散求和，CVA ≈ €5.3 万，约为敞口的 5.3%。快速近似“EE × 利差 × 年金因子”得 €5.5 万。若签署每日 VM 的 CSA，EE 降至追加保证金间隔内的价格变动（约几万欧元），CVA 随之降至原来的一小部分。

框架 Framework	交易对手风险要求 Counterparty-risk requirements
UCITS	Exposure to any single OTC derivative counterparty $\leq 5\%$ of assets, or 10% for credit institutions (UCITS Directive Art. 52); measured as positive MtM net of eligible collateral, including margin not protected by client-money rules. Collateral must be liquid, valued daily, of good credit quality, weakly correlated with the counterparty, diversified (ESMA 2014/937: generally $\leq 20\%$ of NAV per issuer), held by a third-party custodian and enforceable at any time; non-cash collateral may not be sold, reinvested or pledged, cash collateral only placed in prescribed low-risk assets; a haircut policy is required. 对单一 OTC 衍生品对手的风险敞口 $\leq 5\%$ 资产，对手为信用机构时 $\leq 10\%$（UCITS 指令第 52 条）；敞口按正市值扣除合格抵押品计算，未受客户资金规则保护的保证金也计入。抵押品须满足流动性、每日估值、发行人信用质量、与对手低相关、分散化（ESMA 2014/937 指引：单一发行人一般 $\leq 20\%$ NAV）、第三方托管、可随时执行等要求；非现金抵押品不得出售、再投资或质押，现金抵押品只能投资于规定的低风险资产；须有折扣率（haircut）政策。
AIFMD	No numeric cap; due diligence and ongoing monitoring of counterparties with internal limits; Annex IV reports top-5 counterparty exposures in both directions, collateral and clearing shares (Ch. 17); collateral reuse counts in leverage. 无数量上限；须对交易对手做尽职调查与持续监控，设定交易对手限额；Annex IV 报告前 5 大交易对手的双向敞口、抵押品与清算比例（第 17 章）；抵押品再利用计入杠杆。
EMIR (UCITS and AIFs are financial counterparties) EMIR (UCITS 与 AIF 均为金融对手方)	Standardised OTC derivatives above the clearing threshold must be centrally cleared; uncleared derivatives require daily variation margin and, above the size threshold, initial margin (usually ISDA SIMM); trades are reported to a trade repository. 超过清算门槛的标准化 OTC 衍生品须中央清算；未清算衍生品须每日交换变动保证金，规模超过门槛者还须交换初始保证金（通常用 ISDA SIMM 计算）；交易须向交易报告库报告。
MiFID II / IFR	Investment firms dealing on own account hold K-TCD (trading counterparty default) capital under the IFR: $\text{exposure} \times \text{risk factor} \times \text{CVA multiplier}$; large firms under CRR follow bank rules. 自营交易的投资公司按 IFR 的 K-TCD（交易对手违约）计提资本：敞口 $\times$ 风险因子 $\times$ CVA 乘数；适用 CRR 的大型公司按银行规则。
Banks 银行 Basel III	Default exposure via SA-CCR ($\text{EAD} = 1.4 \times (\text{replacement cost} + \text{PFE})$) or internal models; separate CVA risk capital (BA-CVA or SA-CVA). 违约风险敞口用 SA-CCR（$\text{EAD} = 1.4 \times (\text{重置成本} + \text{PFE})$）或内部模型；另计 CVA 风险资本（BA-CVA 或 SA-CVA）。

9 Reproducible Case Studies

案例研究（可复现）

9.1 Replicating the course backtest: S&P 500, 99%/10-day, 21-obs rolling σ 复现课程回测：S&P 500，99%/10 日，21 日滚动 σ

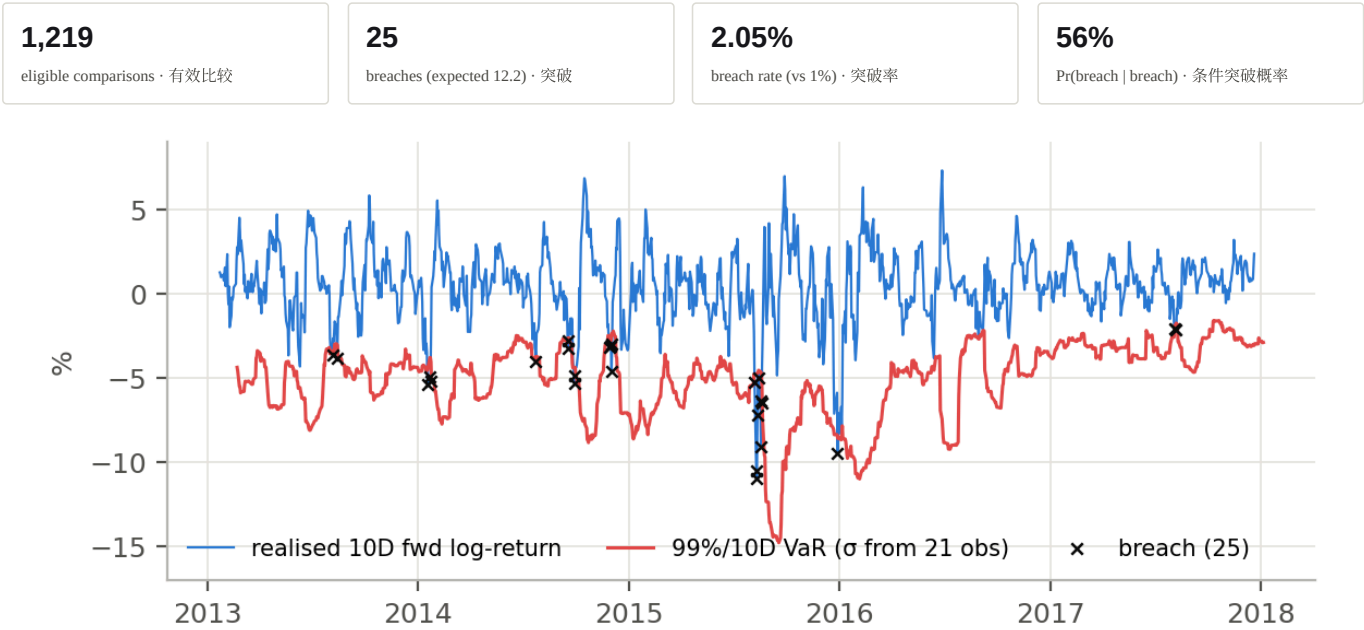


Figure 6. 99%/10-day VaR (red) and 10-day forward return (blue), Jan 2013–Jan 2018; × marks breaches.
图 6 2013-01 至 2018-01 的 99%/10 日 VaR（红）与 10 日前瞻收益（蓝），× 为突破。

Recomputing independently from the 1,250 closes in the course workbook, following §7.2, reproduces the course notebook exactly: 1,219 comparisons, 25 breaches, 2.05%. Interpretation:

- the breach rate is about twice the theoretical level, so the 21-obs Normal VaR is aggressive;
- breaches cluster (Jan 2014, Sep–Dec 2014, the **August 2015** renminbi-devaluation shock). The 56% conditional probability, far above 1%, mixes genuine volatility clustering with the mechanical effect of overlapping 10-day windows;
- before August 2015 σ was low and the model could not "see" the coming shock; afterwards VaR deepened sharply and then snapped back 21 days later: the classic window effect.

用课程 Excel 中的 1,250 个收盘价、按 7.2 节步骤独立重算，结果与课程 Python 笔记本完全一致：1,219 次比较、25 次突破、2.05%。解读：

- 突破率约为理论值的两倍，说明 21 日正态 VaR 偏激进；
- 突破高度集中（2014 年 1 月、2014 年 9–12 月、2015 年 8 月人民币贬值冲击）。条件突破概率 56% 远高于 1%，一部分来自真实的波动率聚集，另一部分是 10 日重叠窗口的机械效应；
- 2015 年 8 月之前 σ 处于低位，模型“看不见”即将到来的冲击；冲击之后 VaR 大幅加深，又在 21 日后骤然回落——典型的窗口效应。

9.2 Comparing 1-day VaR models (Jan 2014–Jan 2018, T = 999) 1 日 VaR 模型对比（2014-01 至 2018-01，T = 999）

模型 Model	突破 x	x/T	LR _{POF} (p)	LR _{ind} (p)	最近 250 日	区域 Zone	平均 VaR
Rolling σ (21), Normal 滚动 σ (21) + 正态	23	2.30%	12.51 (0.0004)	2.57 (0.11)	5	黄 Yellow	-1.62%
EWMA 0.94, Normal EWMA 0.94 + 正态	21	2.10%	9.31 (0.002)	3.17 (0.08)	4	绿 Green	-1.65%
EWMA 0.94 + Cornish–Fisher*	11	1.10%	0.10 (0.75)	8.15 (0.004)	3	绿 Green	-2.31%
Historical simulation 250 历史模拟 250	15	1.50%	2.20 (0.14)	5.63 (0.02)	3	绿 Green	-2.14%
Filtered HS (EWMA 0.94) 过滤 HS (EWMA 0.94)	14	1.40%	1.45 (0.23)	6.17 (0.01)	3	绿 Green	-2.17%

* The CF factor uses full-sample skewness -0.437 and excess kurtosis 2.870 (not inverted to S,K) and so contains look-ahead bias; illustration only. All models use data up to $t - 1$ for day t ; 1-day returns do not overlap. Expected breaches: 9.99.
* CF 因子使用全样本偏度 -0.437、超额峰度 2.870（未反解 S,K），含前视偏差，仅作参考。所有模型在第 t 日只使用截至 $t - 1$ 的数据；1 日收益不重叠。期望突破 9.99 次。

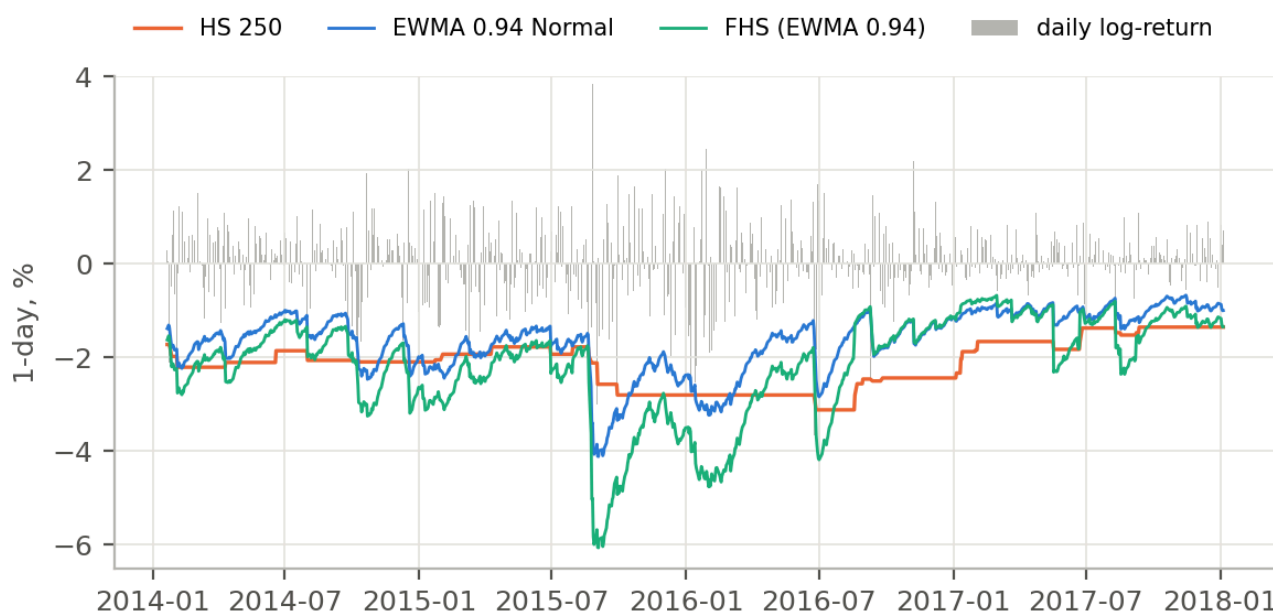


Figure 7. Three 1-day 99% VaRs: HS 250 plateaux, fast-reacting EWMA, and FHS combining the empirical tail with current volatility.

图7 三种1日99% VaR: HS 250的阶梯平台、EWMA的快速反应、FHS兼具经验尾部与当前波动。

Takeaways: (1) With a Normal factor, breaches run at about twice the theoretical rate whatever the volatility model: the problem is the factor, not the volatility. (2) Switching to a CF factor or to empirical quantiles (HS, FHS) passes the frequency test. (3) Yet the independence test rejects for CF, HS and FHS: with only 11–15 breaches, two consecutive pairs inflate π_1 ; in small samples LR_{ind} is unstable and should be read alongside the chart. (4) HS VaR stays on a high plateau for about a year after August 2015, while FHS eases back as markets calm: more accurate, but more procyclical.

结论: (1) 正态因子无论配哪种波动率，突破都约为理论的两倍——问题在“因子”而非“波动率”。(2) 换成 CF 因子或经验分位数（HS、FHS）后，频率检验都能通过。(3) 但独立性检验对 CF、HS、FHS 反而拒绝：这些模型突破只有 11–15 次，其中 2 对连续发生， π_1 被放大；在小样本下 LR_{ind} 不稳定，应结合图形判断。(4) HS 的 VaR 在 2015 年 8 月后维持约一年的高位平台，FHS 则在市场平静后逐步回落——更“准确”，但也更顺周期。

9.3 Reading an S&P 500 HS VaR backtest (Jan 2025–Jan 2026) 解读：S&P 500 历史模拟 VaR 回溯（2025-01 至 2026-01）

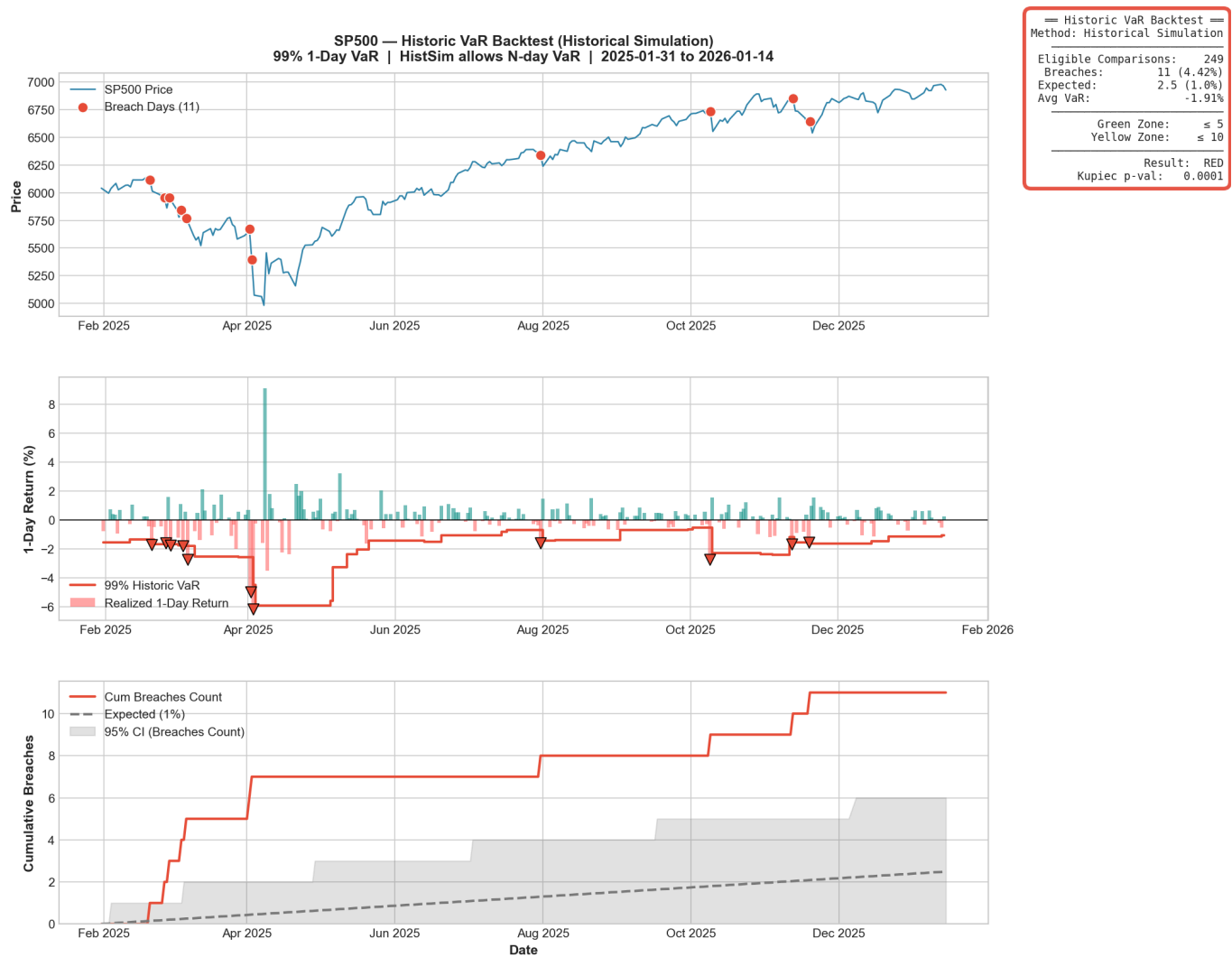


Figure 8. 99% 1-day HS VaR backtest: price and breach days (top), 1-day returns vs VaR (middle), cumulative breaches vs the binomial 95% band (bottom).

图 8 99% 1 日历史模拟 VaR 回溯：价格与突破日（上）、1 日收益与 VaR（中）、累计突破与二项分布 95% 置信带（下）。

Number check: 249 comparisons, 11 breaches (4.42%), 2.49 expected. Independently recomputed Kupiec LR = 15.96, $p = 6.5 \times 10^{-5}$ (the chart's 0.0001 is rounded); $\Pr(X \geq 11) = 5.2 \times 10^{-5}$. The red verdict is correct.

Zone boundaries: the chart states "Green ≤ 5 , Yellow ≤ 10 "; under the Basel rule it should read **green 0–4, yellow 5–9, red ≥ 10** (the off-by-one pitfall of §7.3). The verdict here is unaffected, but counts of 5 or 10 would be misclassified.

What the chart shows:

- **A short lookback:** after the April 2025 tariff shock VaR sits near -6% for about seven weeks and is back near -2% by late May. With a 250-day equal-weighted HS the extreme April days would stay in VaR until April 2026, which suggests a window of roughly one to two months (to be confirmed against the code). With only 30–60 observations the 1% quantile is essentially "the worst recent day" (or an interpolation between the two worst), so **every new worst day is necessarily a breach**: the structural reason for 11 breaches.
- **Clustered breaches:** five breaches in Feb–Mar fall within about five weeks, as markets turned from calm to falling while the window held only January's low-volatility data. A classic independence failure; add a Christoffersen test.
- **Plateaux and steps:** the flat segments and steps in the middle panel are the signature of HS.

Suggestions: extend the window to ≥ 250 days (the regulatory minimum) or move to FHS (EWMA 0.94–0.97 revol); document the percentile interpolation convention; report 97.5% ES and a stressed VaR alongside; for a portfolio, backtest against HPL.

数字核对：249 次比较、11 次突破（4.42%），期望 2.49 次。独立重算 Kupiec LR = 15.96, $p = 6.5 \times 10^{-5}$ （图中 0.0001 为四舍五入）； $\Pr(X \geq 11) = 5.2 \times 10^{-5}$ 。结论为红区，正确。

区域边界：图中写“绿区 ≤ 5 、黄区 ≤ 10 ”，按 Basel 规则应为绿 0–4、黄 5–9、红 ≥ 10 （见 7.3 节的差一陷阱）。本例结论不受影响，但在 5 次或 10 次时会误判。

读图要点：

- **回看窗口偏短：**2025 年 4 月关税冲击后，VaR 在约 -6% 维持了约 7 周，5 月下旬即回到 -2% 附近。若是 250 日等权 HS，4 月的极端日会在 VaR 中停留到 2026 年 4 月。由此推断窗口约为 1–2 个月（需与代码核对）。窗口仅 30–60 个观测时，1% 分位数基本就是“近期最差一天”（或最差两天之间的插值），**每当出现新的最差日就必然突破**——这是 11 次突破的结构性原因。
- **突破成串：**2–3 月的 5 次突破集中在约 5 周内，发生在从平静转为下跌的阶段，窗口里只有 1 月的低波动数据；典型的独立性失败，建议补充 Christoffersen 检验。
- **平台与阶梯：**中图红线的水平段与台阶，是 HS 的特征行为。

改进建议：窗口延长至 ≥ 250 日（监管最低要求），或改用 FHS（EWMA 0.94–0.97 再波动化）；报告中写明分位数插值约定；同时报告 97.5% ES 与压力期 VaR；将回溯改用 HPL（若为组合）。

Medium-Term Risk Testing · weeks to one year

第二部分 中期风险测试

Medium-term testing asks "can the portfolio survive a severe but plausible shock?". It does not rest on a probability distribution: it fixes a shock, computes the loss, liquidity gap and limit breaches, and pre-plans hedging, deleveraging or liquidity management tools.

中期测试回答"在严重但合理的冲击下，组合能否承受"。它不依赖概率分布，而是给定一个冲击，计算损失、流动性缺口与限额突破，并据此预先安排对冲、降杠杆或启用流动性管理工具。

10 Stress Testing: Theory

压力测试：理论

10.1 How stress testing complements VaR 压力测试与 VaR 的分工

VaR is a **probability statement**: from past distributions, "on 99% of normal days we will not lose more than this". A stress test is a **conditional statement**: "if X happens, we lose this much", with no probability attached. They answer different questions:

- VaR marks the door to the tail; a stress test walks through it and examines losses beyond VaR;
- VaR depends on its window and cannot see crises absent from it (§5.2, "never beyond the worst day in the window"); stress tests can impose unprecedented shocks;
- VaR assumes stable correlation; a stress test can impose correlation breakdown explicitly (bonds and equities falling together);
- VaR usually ignores liquidity and redemptions; a stress test can shock assets and liabilities together.

VaR 是一个**概率陈述**：用过去的分布推断"正常市场下 99% 的日子不会亏超过多少"。压力测试是一个**条件陈述**："如果发生 X，会亏多少"——不附带概率。两者回答不同的问题：

- VaR 刻画分布的"门口"，压力测试直接走进门后，考察 VaR 之外的尾部；
- VaR 依赖历史窗口，窗口里没有的危机它看不见（§ 5.2 的"不会超过窗口最差日"）；压力测试可以设定前所未见的冲击；
- VaR 假设相关性稳定，压力测试可以显式设定"相关性失效"（股债同跌）；
- VaR 通常不含流动性与赎回，压力测试可以同时冲击资产与负债两侧。

10.2 Types of stress test 压力测试的类型

类型 Type	做法 How	优点 Strength	局限 Limitation
Sensitivity 敏感性	One factor moved by a fixed amount 单一风险因子移动固定幅度（股票 -10%，利率 +100bp）	Transparent, additive, easy to limit 透明、可加总、便于限额	Ignores co-movement 忽略因子联动
Historical 历史情景	Replay factor moves from real crises 重演真实危机中各因子的变动（2008、2020、2022）	Credible, internally consistent 可信、内部一致	The next crisis will differ 下一次危机不会重复上一次
Hypothetical 假设情景	Narrative-driven consistent shocks 按叙事设定一组一致的冲击（滞胀、地缘冲突）	Forward-looking, targets vulnerabilities 前瞻、可针对组合弱点	Subjective, hard to calibrate 主观，幅度难校准
Reverse 反向压力测试	Fix an unacceptable loss, solve for the most plausible scenario 先定"不可接受的损失"，再反求导致它的最可能情景	Reveals hidden concentrations 揭示隐藏的集中风险	Many solutions; plausibility is model-based 多解；合理性判断依赖模型
Stressed VaR 压力 VaR	VaR calibrated on a stress period 用压力时期（如 2008）的数据校准 VaR	Same framework as VaR 与 VaR 同框架，可比	Still a single-period quantile 仍是单期分位数
Liquidity (LST) 流动性压力测试	Asset liquidation time/cost + redemptions and margin calls 资产端变现时间与成本 + 负债端赎回、保证金追加	Core fund regulatory requirement 基金监管的核心要求	Market-depth data are scarce 市场深度数据难得

10.3 From shocks to P&L: sensitivities and full revaluation 从冲击到损益：敏感度与全重估

$$\Delta P \approx \sum_i \delta_i \Delta x_i + \frac{1}{2} \sum_{i,j} \Gamma_{ij} \Delta x_i \Delta x_j,$$

bond:
$$\frac{\Delta P}{P} \approx -D \Delta y + \frac{1}{2} C (\Delta y)^2,$$

option:
$$\Delta V \approx \Delta \Delta S + \frac{1}{2} \Gamma (\Delta S)^2 + \mathcal{V} \Delta \sigma$$

There are two ways to map shocks to P&L. **Sensitivities** use a Taylor expansion: fast, suited to daily monitoring and many scenarios. **Full revaluation** reprices every position under shocked market data: exact but slow.

损益映射有两种方式。**敏感度法**用泰勒展开近似，速度快，适合日常监测与大量情景；**全重估**用定价模型在冲击后的市场数据下重新估值，精确但耗时。

Deep dive · Why Taylor fails for large shocks / 深度解析 · 为什么大冲击下泰勒近似会失效

A Taylor expansion is accurate only for small moves. Stress shocks are large: at +250bp the bond convexity term matters; at -30% equity a short out-of-the-money put becomes deep in the money, delta moves from -0.2 towards -1, and a gamma approximation badly understates the loss. Barrier options and autocallables can even jump. Rule of thumb: **sensitivities are fine for linear products; positions with options or material convexity need full revaluation in stress tests.**

泰勒展开只在"小变动"附近准确。压力情景的冲击往往很大：利率 +250bp 时债券凸性项已不可忽略；股票 -30% 时，虚值卖出看跌期权会变成深度实值，Delta 从 -0.2 变成接近 -1，Gamma 近似会严重低估亏损。障碍期权、自动赎回票据等路径依赖产品甚至会出现跳跃。经验法则：**线性产品可用敏感度，含期权或凸性显著的头寸在压力测试中必须全重估。**

10.3.1 The Taylor expansion explained 泰勒展开详解

$$f(x + \Delta x) \approx f(x) + f'(x) \Delta x + \frac{1}{2} f''(x) (\Delta x)^2 + \frac{1}{6} f'''(x) (\Delta x)^3 + \dots$$

A complicated function can be approximated near a point by a polynomial whose coefficients are its derivatives there. The **zeroth** term is the current value; the **first** is the tangent (direction and speed); the **second** corrects for curvature (how the tangent leaves the curve); higher terms refine further but fade quickly when Δx is small. Think of driving: first order is "where will I be in a minute at this speed", second order adds "am I accelerating or braking". In risk management Δx is the market shock: the smaller it is, the more first order suffices; the larger, the more higher terms matter, until the expansion itself breaks down.

一个复杂函数在某点附近可以用多项式近似，系数就是该点的各阶导数。**零阶**是当前值；**一阶**是切线（方向与速度）；**二阶**是曲率修正（切线偏离曲线的部分）；更高阶继续修正，但 Δx 小时贡献迅速变小。类比开车：一阶是"按当前车速开一分钟到哪"，二阶是"再考虑正在加速还是减速"。风险管理里的 Δx 就是市场冲击：冲击越小，一阶越够用；冲击越大，越需要高阶项，直到泰勒展开本身失效。

泰勒项 Term	债券 Bonds	期权 Options	风险度量 Risk measure
一阶 First order	Duration, DV01 久期 D、DV01	Delta, Vega, Theta	Delta-normal VaR, sensitivity stress Delta-Normal VaR、敏感性压力测试
二阶 Second order	Convexity 凸性 C	Gamma, cross-gamma Gamma、交叉 Gamma	Delta-gamma VaR, CF correction Delta-Gamma VaR、CF 修正

Worked example 9 · When the Taylor approximation is good enough / 例题 9 · 泰勒近似何时够用

Bond: 10-year zero-coupon at 3% (modified duration 9.71, convexity 103.7); results in the first table in the right-hand column.

Short put: spot 100, strike 90, volatility 20%, 3 months (delta -0.135, gamma 0.0217); seller's P&L in the second table.

The bond's convexity correction still works for large shocks. The option, at -30%, moves from out of the money to deep in the money, delta runs from -0.135 towards -1, the delta approximation understates the loss by almost 80%, and even delta-gamma understates it by about 28%.

Log returns: $\ln(1 + x) \approx x - x^2/2$: 9.53% vs 9.50% at +10%; -35.7% vs -34.5% at -30%. This is also where the $-\frac{1}{2}\sigma^2$ term in log-normal models comes from.

债券：10 年期零息债，收益率 3%（修正久期 9.71，凸性 103.7）。

利率	精确	一阶	一阶 + 二阶
+100bp	-9.21%	-9.71%	-9.19%
+300bp	-24.96%	-29.13%	-24.46%

卖出看跌期权：股价 100、行权价 90、波动率 20%、3 个月（Delta -0.135，Gamma 0.0217），卖方损益：

股价	精确	Delta	Delta-Gamma
-5	-1.00	-0.67	-0.95
-10	-2.88	-1.35	-2.43
-30	-19.30	-4.05	-13.81

债券的凸性修正在大冲击下仍然有效；期权在 -30% 时由虚值变为深度实值，Delta 从 -0.135 趋近 -1，Delta 近似低估亏损近 80%，加上 Gamma 仍低估约 28%。

对数收益： $\ln(1 + x) \approx x - x^2/2$ 。+10% 时 9.53% 对 9.50%；-30% 时 -35.7% 对 -34.5%。这也是对数正态模型中 $-\frac{1}{2}\sigma^2$ 项的来源。

Rule of thumb: small shocks and linear products → first order; convexity or gamma → at least second order; large shocks with options or path dependence → full revaluation. Periodically compare the Taylor approximation with full revaluation and escalate when the gap exceeds a threshold (e.g. 5%), which is in effect what the FRTB P&L attribution test checks.

实务规则：冲击小、产品线性 → 一阶；有凸性或 Gamma → 至少二阶；大冲击 + 期权或路径依赖 → 全重估。定期用全重估对比泰勒近似，误差超过阈值（如 5%）就升级为全重估——这正是 FRTB 损益归因测试（PLA）实际在检验的东西。

10.3.2 Full revaluation under large shocks: options and path-dependent products 大冲击下的全重估：期权与路径依赖产品

Full revaluation skips derivative approximations: feed the full shocked market data into the same pricing models used for daily valuation, reprice every position, and take P&L = shocked value - base value. Steps:

- 1. **Base snapshot:** freeze all valuation-date market data: spot, yield curves, credit spreads, implied volatility surfaces, dividends and borrow costs, correlations, FX.
- 2. **Build shocked market data:**
 - spot: move by relative or absolute shocks;

全重估是指：不用导数近似，而是把冲击后的全部市场数据输入与日常估值相同的定价模型，重新计算每个头寸的价值，损益 = 冲击后价值 - 基准价值。步骤如下：

- 1. **基准快照：**固定估值日的全部市场数据——现货、收益率曲线、信用利差、隐含波动率曲面、股息与债券成本、相关性、汇率。
- 2. **构造冲击后的市场数据：**
 - 现货：按相对或绝对冲击移动；

- volatility surfaces: implied vols usually rise and skew steepens in sell-offs, so shock vols too (e.g. equity -30% with ATM vol +15 points), and choose how the surface moves with spot: sticky strike or sticky moneyness/delta;
 - curves: shift, steepen or twist, then re-bootstrap, checking rate floors;
 - shock credit spreads, correlations and dividends alongside;
 - check the shocked surface is arbitrage-free (no calendar or butterfly arbitrage), or models may return nonsense.
3. **Reprice** with the same models and parameters as daily valuation (AIFMD Art. 19 valuation, UCITS valuation policy), so risk and valuation stay consistent.
4. **Time**: instantaneous shocks ignore time decay; multi-day scenarios should include theta, carry and roll-down, and the **path** of the shock (see path-dependent products below).
5. **Explain P&L**: split full-revaluation P&L into delta, gamma, vega and an unexplained residual; a large residual flags an unreliable Taylor approximation or a model problem.

- 波动率曲面：股市下跌时隐含波动率通常上升、偏斜变陡，情景应同时冲击波动率（如股票 -30% 配合平值波动率 +15 个点）；须选定曲面随现货移动的规则——黏性行权价（sticky strike）或黏性价值状态（sticky moneyness/delta）；
 - 曲线：平移、陡峭化、扭曲后重新拟合（bootstrap），检查负利率下限；
 - 信用利差、相关性、股息等一并冲击；
 - 检查冲击后的曲面无套利（无日历套利、无蝶式套利），否则定价模型可能给出荒谬结果。
3. **重新定价**：使用与日常估值相同的模型与参数（AIFMD 第 19 条估值、UCITS 估值政策），保证风险与估值口径一致。
4. **时间维度**：瞬时冲击不计时间流逝；若情景持续多日，应计入 Theta、票息与期限滚动，并考虑冲击发生的**路径**（见下文路径依赖产品）。
5. **损益解释**：把全重估损益分解为 Delta、Gamma、Vega 等可解释部分与“未解释”残差；残差大说明泰勒近似不可靠或模型有问题。

Worked example 10 • Vanilla option: joint spot and volatility shock / 例题 10 • 普通期权：现货与波动率联合冲击

The short put of Example 9, fully revalued with Black–Scholes; seller's P&L in the table in the right-hand column.

The vol shock alone loses about three times the premium (0.71): vega peaks near the money and out of the money. After a large spot fall the option is deep in the money, vega is small, and the extra vol shock adds little. Vega must be shocked together with spot in scenarios, not read off as a stand-alone sensitivity.

沿用例题 9 的卖出看跌期权（Black–Scholes 全重估），卖方损益：

情景	损益
仅波动率 +15 个点（20% → 35%）	-2.07
股价 -30%，波动率不变	-19.30
股价 -30%，波动率升至 35%	-19.76

单独的波动率冲击亏损约为期权费（0.71）的 3 倍——Vega 在平值和虚值附近最大。股价大跌后期权已深度实值，Vega 变小，波动率冲击的额外影响有限。结论：Vega 风险要在情景中和现货一起冲击，不能只看单独的敏感度。

Path-dependent products (barriers, autocallables, Asians, lookbacks) depend on the **path** of the underlying, not just the end point, so they rarely have closed forms and are priced by Monte Carlo or PDE methods. Key points in full revaluation:

- **Barrier status**: first check whether a barrier has already been hit; knocked-in or knocked-out products are priced as their new structure;
- **Path of the shock**: for multi-day scenarios (e.g. a 30% fall over 10 days), simulate that path and check barriers at the contractual monitoring frequency (daily, closing, at expiry) rather than only the end point; jumping straight to the end point can miss a knock-in along the way;
- **Observation dates and features**: autocall dates, coupon barriers, memory coupons and knock-in barriers must each be modelled;
- **Common random numbers**: use the **same random draws** for base and shocked valuations, or Monte Carlo noise swamps the true P&L difference;
- **Enough paths**: payoffs are discontinuous near barriers, so use more paths or a Brownian-bridge correction for discrete monitoring;
- **Greeks near barriers**: delta and gamma change violently or jump near a barrier, which is exactly where Taylor approximations fail worst.

路径依赖产品（障碍期权、自动赎回票据、亚式期权、回望期权）的价值取决于标的**走过的路径**，不只取决于终点，因此通常没有闭式解，要用蒙特卡罗或偏微分方程（PDE）数值定价。全重估时的要点：

- **障碍状态**：先确认障碍是否已被触发（已敲入 / 敲出的产品按新的结构定价）；
- **冲击的路径**：若情景持续多日（如 10 日内逐步下跌 30%），应模拟这条路径并按合同的观察频率（每日、收盘价、期末）检查障碍，而不是只看终点——瞬时跳到终点可能漏掉途中的敲入；
- **观察日与特殊条款**：自动赎回票据的赎回观察日、票息障碍、记忆票息、敲入障碍须逐一建模；
- **公共随机数**：基准估值与冲击后估值使用**同一组随机数**，否则两个蒙特卡罗估计的噪声会淹没真实的损益差；
- **足够的路径数**：障碍附近收益不连续，需要更多路径或布朗桥修正（纠正离散观察的偏差）；
- **障碍附近 Greeks 失真**：Delta、Gamma 在障碍附近剧烈变化甚至跳跃，泰勒近似在这里最不可靠。

Worked example 11 • Full revaluation of a barrier option / 例题 11 • 障碍期权的全重估

One-year down-and-in put: strike 100, barrier 70 (daily monitoring), volatility 20%, zero rates. Monte Carlo with 100,000 paths and common random numbers for base and shocks. Base value 2.50, delta -0.26, gamma 0.054 (finite differences). Value changes in the table in the right-hand column.

Three observations: (1) delta-gamma **overstates** the change here (by 17–25%), the opposite direction to Example 9, so Taylor errors cannot be assumed conservative in advance; (2) spot moves only 2 points from 71 to 69, yet crossing the barrier makes the value jump by about 2.9, a discontinuity no Taylor expansion can express; (3) higher volatility raises the knock-in probability and adds another 3.7. Investors in reverse convertibles or autocallables are effectively **short** such options and bear the opposite P&L.

1 年期向下敲入看跌期权：行权价 100、障碍 70（每日观察）、波动率 20%、利率 0。蒙特卡洛 10 万条路径，基准与冲击使用公共随机数。基准价值 2.50，Delta -0.26，Gamma 0.054（有限差分）。期权价值变化：

现货	全重估	Delta-Gamma
85	+8.03	+10.01
75	+19.80	+23.42
71（未敲入）	+25.80	+30.30
69（已敲入，变为普通看跌）	+28.66	—
75，波动率升至 30%	+23.46	—

三点观察：(1) Delta-Gamma 在这里高估了变化（误差 17–25%），方向与例题 9 相反——泰勒误差的方向取决于产品，无法事先保守假设；(2) 现货从 71 到 69 只差 2 点，但跨过障碍后价值跳升约 2.9，这种不连续性泰勒展开无法表达；(3) 波动率上升进一步提高敲入概率，价值再增 3.7。持有反向可转换或自动赎回票据的投资者实际上卖出了这类期权，损益方向相反。

In practice • Making full revaluation tractable / 实操要点 • 让全重估算得动

- **Spot-vol grids**: pre-compute full revaluations of each product on a spot × vol grid (e.g. spot -50% to +50%, vol -10 to +30 points) and interpolate per scenario. Standard in risk systems, and suited to the hundreds or thousands of HS or Monte Carlo scenarios.
- **Tiering**: sensitivities for linear positions, closed-form revaluation for vanillas, grids or Monte Carlo for path-dependent products.
- **Surrogate models**: for very complex products, regression or machine-learning pricing proxies, validated regularly against the full model.
- **Compute**: parallelise, cache intermediate results, run overnight batches; stress tests can use more paths than daily VaR.
- **Structured products held by funds**: do not rely only on issuer quotes; use an independent model or third-party valuation, documented in the valuation policy. PRIIPs Category 3 VaR and performance scenarios require simulation anyway.
- **Regulatory mapping**: UCITS VaR models must capture option gamma and vega (CESR/10-788); AIFMD requires independent model validation (Level 2 Art. 45) consistent with valuation policy (Art. 19); FRTB PLA effectively requires full revaluation of non-linear positions.

- **现货-波动率网格**：预先在“现货 × 波动率”网格（如现货 -50% 至 +50%、波动率 -10 至 +30 个点）上全重估每个产品，情景计算时插值取值。这是风险系统的标准做法，适合 HS 或蒙特卡洛的成百上千个情景。
- **分层处理**：线性头寸用敏感度，普通期权用闭式全重估，路径依赖产品用网格或蒙特卡洛。
- **替代模型**：对极复杂产品，可用回归或机器学习拟合的定价代理，但须定期用完整模型验证。
- **计算资源**：并行计算、缓存中间结果、夜间批处理；压力测试可比日常 VaR 用更多路径。
- **基金持有的结构化产品**：不应只依赖发行人报价；应有独立的定价模型或第三方估值，并写入估值政策。PRIIPs 第 3 类产品的 VaR 与业绩情景本身就要模拟。
- **监管对应**：UCITS 要求 VaR 模型捕捉期权的 Gamma、Vega 风险（CESR/10-788）；AIFMD 要求模型独立验证（Level 2 第 45 条）并与估值政策一致（第 19 条）；FRTB 的 PLA 测试事实上要求非线性头寸全重估。

10.4 Conditional scenarios: propagating core shocks 条件情景：从核心因子推出其余因子

$$\mathbb{E}[\mathbf{x}_s \mid \mathbf{x}_c] = \Sigma_{sc} \Sigma_{cc}^{-1} \mathbf{x}_c \quad (\text{jointly Normal factors, zero mean})$$

Real portfolios may carry thousands of risk factors, but a scenario usually specifies only a few **core factors** (equity index, 10-year rate, credit spread, USD). The remaining **satellite factors** (single stocks, other tenors, other currencies) are derived by conditional expectation from the covariance matrix: given the core move, the satellites move "on average" by this much. This keeps the scenario internally consistent and is standard for supervisory and bank internal scenarios.

Caution: correlations shift in crises (in 2022 equities and bonds fell together as their correlation turned positive). Propagating with a calm-period Σ understates joint declines, so a stressed covariance matrix, or explicitly set key correlations, is common.

实际组合可能有上千个风险因子，但情景通常只设定少数**核心因子**（如股指、10 年利率、信用利差、美元）。其余**卫星因子**（个股、其他期限、其他货币）用条件期望从协方差矩阵推出：核心因子动多少，卫星因子“平均而言”跟着动多少。这保证了情景内部一致，也是监管情景和银行内部情景的标准做法。

注意：危机中相关性会变化（例如 2022 年股债同跌，股债相关由负转正）。用平静期的 Σ 推导卫星因子会低估同向下跌，因此常用压力期协方差矩阵，或对关键相关性单独设定。

10.5 Plausibility and reverse stress testing 合理性与反向压力测试

How plausible is a scenario? The **Mahalanobis distance** measures how far it lies from "normal" in standard-deviation units, accounting for correlation. Reverse stress testing asks: among all scenarios producing loss L , which is most plausible (smallest Mahalanobis distance)? For a linear portfolio $\Delta P = \mathbf{w}'\mathbf{x}$, a Lagrange multiplier gives a closed form:

一个情景有多“合理”？常用**马氏距离**衡量它离“正常”有多远（以标准差为单位，已考虑相关性）。反向压力测试问：在所有能造成损失 L 的情景中，哪一个最合理（马氏距离最小）？对线性组合 $\Delta P = \mathbf{w}'\mathbf{x}$ ，用拉格朗日乘子可得闭式解：

$$d(\mathbf{x}) = \sqrt{\mathbf{x}'\Sigma^{-1}\mathbf{x}}, \quad \min_{\mathbf{x}} \mathbf{x}'\Sigma^{-1}\mathbf{x} \text{ s.t. } \mathbf{w}'\mathbf{x} = -L \implies \mathbf{x}^* = -L \frac{\Sigma\mathbf{w}}{\mathbf{w}'\Sigma\mathbf{w}}, \quad d(\mathbf{x}^*) = \frac{L}{\sigma_P}$$

$\mathbf{x}^* \propto \Sigma \mathbf{w}$: the most plausible loss scenario moves each factor in the direction of its contribution to portfolio risk; volatile factors correlated with the portfolio take more of the shock. The distance needed equals L/σ_P , "how many portfolio standard deviations this loss is". This links reverse stress testing to VaR: under Normality, the loss at $d = 2.326$ is exactly the 99% VaR.

But a Normal Mahalanobis distance **understates** how plausible extreme scenarios are: fat tails and crisis correlation shifts make an "8-sigma" event far more common than the Normal suggests (§11.3). Use it to **rank** scenarios, not to convert them into probabilities.

$\mathbf{x}^* \propto \Sigma \mathbf{w}$: 最可能的致损情景，就是每个因子按"它对组合风险的贡献方向"移动。与组合正相关、波动大的因子承担更多冲击。所需的马氏距离恰好等于 L/σ_P ——即"这个损失相当于组合几个标准差"。这把反向压力测试与 VaR 联系了起来：在正态假设下， $d = 2.326$ 对应的损失就是 99% VaR。

但正态下的马氏距离会低估极端情景的可能性：厚尾与危机中相关性的变化会让"8个标准差"的事件远比正态分布暗示的更常见（见 §11.3）。因此马氏距离适合在情景之间比较合理性，不宜直接换算成概率。

10.6 Liquidity stress testing 流动性压力测试

$$\text{days to liquidate}_i = \frac{Q_i}{\theta \cdot \text{ADV}_i}, \quad \text{cost}_i \approx \frac{1}{2} \text{spread}_i + k \sigma_i \sqrt{\frac{Q_i}{\text{ADV}_i}}, \quad \text{coverage}(h) = \frac{\text{assets liquidable within } h}{\text{stressed redemptions within } h}$$

- **Asset side**: days to liquidate each position at a participation rate θ (typically 10–25% of average daily volume) and the cost incurred; the second term is the common square-root market-impact model. Under stress, widen spreads and cut ADV together.
- **Liability side**: redemption scenarios from the worst historical outflow, investor concentration (top 5 investors redeem), peer outflows in past crises, plus derivative margin calls and repo roll failure.
- **Coverage** ≥ 1 means assets liquidable within h cover stressed redemptions.
- **Vertical slicing vs waterfall**: selling the most liquid assets first (waterfall) is cheapest but leaves remaining investors with a less liquid book, a first-mover advantage that invites more redemptions. Pro-rata selling (slicing) preserves the profile at higher cost. Report both.
- **Liquidity management tools (LMTs)**: gates, extended notice, redemption fees, swing pricing, anti-dilution levies, redemptions in kind, suspension, side pockets. Swing pricing and anti-dilution levies push trading costs onto redeemers and structurally weaken the first-mover advantage.
- **资产端**: 每个头寸在给定参与率 θ (通常为日均成交量 ADV 的 10–25%) 下需要几天变现、付出多少成本。第二项是常见的平方根市场冲击模型。压力下应同时放大价差、缩小 ADV。
- **负债端**: 赎回情景可取历史最大赎回、投资者集中度 (前 5 大投资者全部赎回)、同类基金在危机中的赎回，以及衍生品追加保证金和回购续作失败。
- **覆盖率** ≥ 1 表示在期限 h 内可变现资产足以应付压力赎回。
- **纵向切片 vs 瀑布式**: 先卖最流动的资产 (瀑布式) 成本最低，但留下的投资者持有更不流动的组合，产生"先跑者优势"，进一步诱发赎回。按比例卖出 (纵向切片) 保持组合结构，但成本更高。流动性压力测试应同时报告两种方式的结果。
- **流动性管理工具 (LMT)**: 赎回闸门、延长通知期、赎回费、摆动定价、反稀释征费、实物赎回、暂停赎回、侧袋。摆动定价和反稀释征费把交易成本转给赎回者，从机制上削弱先跑者优势。

10.7 Limits of stress testing 压力测试的局限

Stress tests carry no probability and cannot "prove" safety; scenarios are human-designed and tend to test only the risks people already think of; severity calibration is subjective. Good practice: a scenario library mixing historical and hypothetical cases, periodic independent challenge, reverse stress tests to find blind spots, and a genuine "use test" where results drive decisions rather than just reports.

压力测试没有概率，无法"证明"组合安全；情景由人设计，容易只测自己想到风险；幅度校准主观。好的实践是：情景库兼顾历史与假设；定期由独立团队挑战；用反向压力测试寻找盲点；并且把结果真正用于决策 ("使用测试")，而不仅是报告。

11 Stress Testing: Practice and Worked Examples

压力测试：实操与算例

11.1 Design process 设计流程

1. **Identify vulnerabilities**: concentration, leverage, non-linearity, liquidity mismatch, counterparties, basis.
2. **Map factors**: map every position to core and satellite risk factors (§10.4).
3. **Scenario library**: sensitivity grid + 3–5 historical + 2–3 hypothetical + reverse + liquidity scenarios; review the library annually.
4. **Compute P&L**: sensitivities for linear positions, full revaluation for options and convexity (§10.3).
5. **Aggregate and limit**: compare with stress-loss limits, VaR limits and liquidity coverage; set amber warnings (e.g. at 80% of a limit).
6. **Escalate and act**: results go to the risk committee, with pre-agreed actions on breach (hedge, deleverage, activate LMTs).
7. **Document and validate**: record assumptions, data, code, results and decisions; independent periodic validation.
1. **识别弱点**: 集中度、杠杆、非线性、流动性错配、交易对手、基差。
2. **因子映射**: 把所有头寸映射到核心与卫星风险因子 (§10.4)。
3. **情景库**: 敏感性网格 + 3–5 个历史情景 + 2–3 个假设情景 + 反向压力测试 + 流动性情景；每年复核一次情景库。
4. **损益计算**: 线性头寸用敏感度，期权和凸性头寸全重估 (§10.3)。
5. **汇总与限额**: 与压力损失限额、VaR 限额、流动性覆盖率比较，设琥珀色预警 (如达到限额 80%)。
6. **上报与行动**: 结果进入风险委员会；预先写明触发后的行动 (对冲、降杠杆、启用 LMT)。
7. **留档与验证**: 假设、数据、代码、结果与决策全部留档，独立团队定期验证。

11.2 Worked example: stress testing a multi-asset UCITS 算例：多资产 UCITS 组合的压力测试

Take a €100m multi-asset UCITS: EUR equities €30m, US equities €20m (USD unhedged), EUR government bonds €30m (modified duration 7, convexity 60), IG credit €15m (duration 5, convexity 30), cash €5m. Shocks are instantaneous and **illustrative**, sized to the order of magnitude of the relevant historical episodes.

设一只 €1 亿的多资产 UCITS：欧元股票 €3,000 万、美国股票 €2,000 万（美元未对冲）、欧元政府债 €3,000 万（修正久期 7，凸性 60）、投资级信用债 €1,500 万（久期 5，凸性 30）、现金 €500 万。冲击为瞬时，数值为示意，幅度参照相应历史时期的量级。

情景 Scenario	股票 Equity	利率 Rates	利差 Spreads	美元 USD	股票损益	政府债	信用债	合计 €m / % NAV
Sensitivity: equity -10% 敏感性：股票 -10%	-10%	0	0	0	-5.00	0.00	0.00	-5.00
Sensitivity: rates +100bp 敏感性：利率 +100bp	0	+100	0	0	0.00	-2.01	-0.73	-2.74
Sensitivity: spreads +100bp 敏感性：利差 +100bp	0	0	+100	0	0.00	0.00	-0.73	-0.73
Sensitivity: USD -10% 敏感性：美元 -10%	0	0	0	-10%	-2.00	0.00	0.00	-2.00
Historical: 2008 GFC-type 历史型：2008 金融危机	-40%	-150	+300	+15%	-18.20	+3.35	-1.07	-15.92
Historical: 2020 COVID-type 历史型：2020 新冠冲击	-34%	-50	+150	+3%	-16.60	+1.07	-0.73	-16.26
Historical: 2022 inflation-type 历史型：2022 通胀加息	-20%	+250	+100	+12%	-8.08	-4.69	-2.35	-15.12
Hypothetical: stagflation 假设型：滞胀	-25%	+150	+200	0	-12.50	-2.95	-2.35	-17.80

Reading the results:

- Under the covariance assumed in §11.3, the portfolio's 1-month 99% VaR is about 5.4% of NAV, while the scenarios lose 15–18%, roughly three times VaR: the tail that VaR does not see.
- In the 2008 and 2020-type scenarios, government bond gains and USD appreciation partly offset equity losses; in the 2022-type and stagflation scenarios **bonds and equities fall together** and the hedge fails. Stagflation produces the largest loss and is absent from the historical library, which shows the value of hypothetical scenarios.
- Sensitivities add linearly, but historical-scenario losses are **not** the sum of sensitivities: they include convexity and the equity × FX cross term.

解读：

- 在 § 11.3 假设的协方差下，组合 1 个月 99% VaR 约为 NAV 的 5.4%，而各情景损失在 15–18%，约为 VaR 的 3 倍——这正是 VaR 看不到的尾部。
- 2008 与 2020 型情景中，政府债上涨和美元升值部分对冲了股票损失；2022 型与滞胀情景中，**股债同跌**，对冲失效。滞胀情景损失最大，而它恰恰不在历史情景库里——假设情景的价值由此可见。
- 敏感性结果可以线性加总，但历史情景的损失不等于各敏感性之和，因为包含了凸性和股票×汇率的交叉项。

11.3 Worked example: reverse stress test 算例：反向压力测试

Assume 1-month factor volatilities: equity 5%, rates 25bp, spreads 15bp, USD 3%; correlations: equity–rates +0.3, equity–spreads -0.6, equity–USD -0.3, rates–spreads -0.2, spreads–USD +0.2. Portfolio 1-month s.d. σ_P = €2.33m (2.33% of NAV).

假设 1 个月因子波动率：股票 5%、利率 25bp、利差 15bp、美元 3%；相关系数：股票–利率 +0.3、股票–利差 -0.6、股票–美元 -0.3、利率–利差 -0.2、利差–美元 +0.2。组合 1 个月标准差 σ_P = €233 万（2.33% NAV）。

Target loss $\bar{L}$ = €20m (20% of NAV, the UCITS absolute VaR cap). From §10.5 the most plausible scenario is **equity -40.0%, rates -5.5bp, spreads +74bp, USD +1.9%**, with Mahalanobis distance $d = 20/2.33 = 8.6$. For L = €10m the shocks halve and $d = 4.3$.

目标损失 L = €2,000 万（20% NAV，恰好是 UCITS 绝对 VaR 上限的水平）。由 § 10.5 的闭式解，最可能的情景是：**股票 -40.0%、利率 -5.5bp、利差 +74bp、美元 +1.9%**，马氏距离 $d = 20/2.33 = 8.6$ 。若 L = €1,000 万，则各冲击减半， $d = 4.3$ 。

Lesson: under Normality an 8.6-sigma monthly event has probability around 5×10^{-18} , practically impossible; yet the 2008-type historical scenario in §11.2 already loses 15.9%. The gap comes from fat tails, crisis correlation shifts and multi-period accumulation. Use Mahalanobis distance to **compare** scenarios; the final plausibility call must draw on history and expert judgement.

教训：按正态分布，8.6 个标准差的月度事件概率约为 5×10^{-18} ，几乎不可能；但 § 11.2 的 2008 型历史情景已造成 15.9% 的损失。差距来自厚尾、相关性在危机中的变化和多期累积。这说明：马氏距离用于在情景之间**比较**，而最终的"合理性"判断要结合历史与专家判断。

11.4 Worked example: liquidity stress test 算例：流动性压力测试

流动性分层 Bucket	占 NAV	正常变现天数 Days (normal)	压力变现天数 Days (stressed, ×2)	估计成本 Cost
现金 Cash	5%	1	2	0.0%
政府债 Govt bonds	30%	1	2	0.1%
大盘股 Large-cap equity	45%	3	6	0.2%
投资级信用 IG credit	15%	10	20	1.0%
小盘股 Small-cap equity	5%	20	40	3.0%

赎回情景 Redemption scenario	赎回	期限	期限内可变现	覆盖率 Coverage
Worst historical weekly outflow 历史最大周赎回	10%	5 日	80%	8.0×
Stress: 20% in a week, liquidation time doubled 压力：一周 20%，变现时间翻倍	20%	5 日	35%	1.75×
Stress: 40% in a month, liquidation time doubled 压力：一个月 40%，变现时间翻倍	40%	20 日	95%	2.38×

All three coverage ratios exceed 1, so daily dealing is appropriate. But in the "20% in a week" case, a waterfall sale of cash and government bonds lifts the share of illiquid assets (credit + small caps) held by remaining investors from 20% to 25%. Reverse liquidity test: with doubled liquidation times, about 35% of NAV can be met within a week; beyond that, gates or swing pricing are needed.

三个情景的覆盖率都大于 1，说明日度申赎的结构是合适的。但在"一周 20%"情景中，用瀑布式卖出现金和政府债后，剩余投资者持有的不流动资产（信用债 + 小盘股）占比从 20% 升到 25%。反向流动性测试：在变现时间翻倍的条件下，一周内最多可应付约 35% 的赎回；超过这一水平就需要启用闸门或摆动定价。

In practice • An example stress-limit framework / 实操要点 • 压力测试限额框架示例

- **Stress-loss limit:** worst historical/hypothetical loss ≤ 25% of NAV; amber at 20%, escalated to the risk committee.
 - **Single-factor limits:** e.g. loss at rates +100bp ≤ 5% of NAV, at equity −10% ≤ 7% of NAV.
 - **Liquidity coverage:** 5-day stressed coverage ≥ 1.2; below 1.0 requires a pre-agreed LMT plan.
 - **Action list:** hedge or de-risk proposal within 5 business days of an amber flag; immediate de-risking and board reporting on breach, notifying the regulator where required.
- **压力损失限额：**最差历史/假设情景损失 ≤ 25% NAV；达到 20% 为琥珀色预警，需提交风险委员会。
 - **单因子敏感性限额：**如利率 +100bp 损失 ≤ 5% NAV，股票 −10% 损失 ≤ 7% NAV。
 - **流动性覆盖率：**压力情景下 5 日覆盖率 ≥ 1.2；低于 1.0 需预先准备 LMT 方案。
 - **行动清单：**预警后 5 个工作日内给出对冲或降风险方案；突破后立即降风险并报告董事会，必要时通知监管。

12 Regulation of Medium-Term Risk Testing

中期风险的监管要求

12.1 UCITS UCITS

- **Stress tests for VaR funds** (CESR/10-788): at least monthly, more often on material portfolio or market change; must cover risks VaR misses (correlation breakdown, liquidity, jumps); must consider leverage scenarios producing negative NAV; results must feed investment and risk decisions.
 - **General risk management** (Directive 2010/43/EU Arts 38–40): the risk management process includes stress tests and scenario analyses "where appropriate", and liquidity stress tests under exceptional conditions. This applies to commitment-approach funds too.
 - **Liquidity stress testing** (ESMA guidelines ESMA34-39-897, applicable from 30 September 2020): at least annually, quarterly or more often recommended; assets and liabilities; historical and hypothetical scenarios plus reverse stress testing; documented policy and independent validation; material risks and actions notified to the regulator; the depositary checks the process exists.
 - **Liquidity management tools** (UCITS amendments from 16 April 2026; Ireland S.I. 182/2026): select at least two LMTs from the statutory list and disclose activation conditions in the prospectus.
- **VaR 基金的压力测试** (CESR/10-788)：至少每月一次，组合或市场显著变化时更频繁；须覆盖 VaR 未充分反映的风险（相关性失效、流动性、跳跃）；须考察杠杆导致 NAV 为负的情景；结果须用于投资决策和风险管理。
 - **一般风险管理** (Directive 2010/43/EU 第 38–40 条)：风险管理流程须"适当时"包括压力测试和情景分析，并对流动性风险在异常情况下进行压力测试。承诺法基金同样适用。
 - **流动性压力测试** (ESMA 指引 ESMA34-39-897，2020 年 9 月 30 日起适用)：至少每年一次，建议每季或更频繁；同时覆盖资产与负债；使用历史与假设情景，并做反向压力测试；政策须留档、独立验证；重大风险与应对措施须通知主管机关；存管人核实流程存在。
 - **流动性管理工具** (UCITS 修订，2026 年 4 月 16 日起；爱尔兰 S.I. 182/2026)：须从法定清单中选定至少两种 LMT，并在招股书中披露启用条件。

12.2 AIFMD AIFMD

- **Art. 15(3)(b):** risks of each position and their overall portfolio effect must be identified, measured, managed and monitored on an ongoing basis, **including through appropriate stress-testing procedures.**
 - **Art. 16(1):** regular stress tests under normal and exceptional liquidity conditions, with investment strategy, liquidity profile and redemption policy kept consistent.
 - **Level 2 Art. 45:** periodic stress tests and scenario analyses "where appropriate" covering adverse changes in market conditions; Art. 48 details liquidity management.
 - **Annex IV:** results of market and liquidity stress tests are reported.
 - The **ESMA LST guidelines** apply to AIFs too; from AIFMD II open-ended AIFs select at least two LMTs; leverage restrictions may follow under Art. 25.
- **第 15(3)(b) 条:** 须能持续识别、计量、管理和监控每个头寸的风险及其对组合的整体影响, **包括使用适当的压力测试程序。**
 - **第 16(1) 条:** 须定期在正常与异常流动性条件下进行压力测试, 评估 AIF 的流动性风险, 并确保投资策略、流动性与赎回政策一致。
 - **Level 2 第 45 条:** 适当时定期进行压力测试与情景分析, 覆盖可能不利影响 AIF 的市场条件变化; 第 48 条细化流动性管理要求。
 - **Annex IV:** 须报告市场风险与流动性压力测试的结果。
 - **ESMA 流动性压力测试指引**同样适用于 AIF; AIFMD II 起开放式 AIF 须选定至少两种 LMT; ESMA 可依第 25 条就杠杆发布限制建议。

12.3 MiFID II and investment-firm prudential rules MiFID II 与投资公司审慎规则

- **Organisation** (MiFID II Art. 16; Delegated Regulation 2017/565 Art. 23): policies and procedures to identify, manage and monitor firm risks, with an independent risk function.
 - **Internal capital and liquidity** (IFD Art. 24): ongoing assessment of internal capital and liquid assets; supervisory review expects stress testing and orderly wind-down planning.
 - **Trading book:** firms on CRR internal models currently compute stressed VaR; from FRTB in 2027 this is replaced by a 97.5% ES calibrated to a stress period.
 - **Product governance:** manufacturers run scenario analysis of poor client outcomes, including deteriorating markets (Delegated Directive 2017/593 Art. 9(10); see §15.3).
- **组织要求** (MiFID II 第 16 条; Delegated Regulation 2017/565 第 23 条): 须建立识别、管理和监控公司风险的政策与程序, 独立的风险管理职能。
 - **内部资本与流动性评估** (IFD 第 24 条): 须持续评估所需内部资本与流动资产; 监管审查 (SREP) 期望包括压力测试与有序清算规划。
 - **交易账簿:** 适用 CRR 内部模型的公司, 现行框架含压力 VaR; 2027 年 FRTB 起以压力期校准的 97.5% ES 取代。
 - **产品治理:** 制造商须对产品做情景分析, 评估"市场环境恶化"等情况下客户的不利结果 (Delegated Directive 2017/593 第 9(10) 条, 详见 § 15.3)。

要求 Requirement	UCITS	AIFMD	MiFID II / IFD
市场压力测试 Market stress	Monthly for VaR funds; "where appropriate" for all VaR 基金至少每月; 所有基金"适当时"	Art. 15(3)(b); L2 Art. 45 第 15(3)(b) 条; Level 2 第 45 条	Firm-level risk management; IFD assessment 公司层面风险管理与 IFD 内部评估
流动性压力测试 LST	ESMA: ≥ annual, quarterly advised ESMA 指引: ≥ 每年, 建议每季	Art. 16(1) + ESMA 第 16(1) 条 + ESMA 指引	IFD liquid-asset assessment IFD 流动资产评估
反向压力测试 Reverse	Required in LST guidelines LST 指引要求	Required in LST guidelines LST 指引要求	Supervisory expectation SREP 期望
流动性工具 LMTs	≥ 2 (from April 2026) ≥ 2 种 (2026 年 4 月起)	≥ 2 for open-ended 开放式 ≥ 2 种	—
报告 Reporting	RMP; notify material LST issues 风险管理流程; 重大 LST 结果通知	Annex IV stress results Annex IV 压力测试结果	SREP / supervisory reporting SREP / 监管报表

Long-Term Risk Testing · one year and beyond

第三部分 长期风险测试

Long-term testing asks "how deep can losses go, how long can they last, and how long will recovery take?". It is about the **path** rather than the distribution at one date: drawdown measures the worst loss investors actually live through, and scenario analysis turns macro narratives into multi-year portfolio paths.

长期测试回答"亏损会有多深、持续多久、多久能恢复"。它关注**路径**而非某一时点的分布：回撤度量投资者真实经历的最大损失，情景分析则把宏观叙事转化为多年的组合路径。

13 Drawdown: Theory and Practice

回撤：理论与实操

13.1 Definitions and common metrics 定义与常用指标

$$M_t = \max_{s \leq t} V_s, \quad DD_t = 1 - \frac{V_t}{M_t}, \quad MDD_T = \max_{t \leq T} DD_t, \quad \text{Calmar} = \frac{\text{CAGR}}{\text{MDD}}, \quad \text{Ulcer} = \sqrt{\frac{1}{n} \sum_t DD_t^2}$$

指标 Metric	含义 Meaning
回撤 DD_t	当前净值距历史最高点（高水位 M_t ）的跌幅 · Distance of NAV below its running peak (high-water mark)
最大回撤 MDD	观察期内最大的回撤 · Largest drawdown over the period
Decline / recovery / time under water 下跌期 / 恢复期 / 水下时间	峰值到谷底、谷底到收复峰值、两者之和 · Peak to trough, trough to new peak, and their sum
Calmar 比率	年化收益 ÷ 最大回撤，衡量“每单位最大痛苦的回报” · Annual return per unit of maximum pain
Ulcer 指数	回撤的均方根，同时惩罚深度与持续时间 · RMS of drawdowns, penalising depth and duration
条件回撤 $CDaR_\alpha$	最差 $(1 - \alpha)$ 比例回撤的平均，相当于回撤版本的 ES（Chekhlov 等 2005） · Average of the worst $(1 - \alpha)$ drawdowns, the drawdown analogue of ES

Deep dive · The asymmetry of drawdowns / 深度解析 · 回撤的不对称性

After a loss d , a gain of $d/(1 - d)$ is needed to break even: -10% needs +11.1%, -20% needs +25%, -33% needs +50%, -50% needs +100%. Required gain and recovery time grow non-linearly with depth. That is why long-term investors care more about drawdown than volatility: volatility treats ups and downs alike, drawdown records only the pain.

亏损 d 后需要上涨 $d/(1 - d)$ 才能回本：-10% 需 +11.1%，-20% 需 +25%，-33% 需 +50%，-50% 需 +100%。回撤越深，恢复所需的收益和时间呈非线性增长。这也是为什么长期投资者更在乎回撤而不是波动率：波动率对涨跌一视同仁，回撤只记录“痛苦”。

13.2 Theory: path dependence and horizon 理论：路径依赖与期限

Path versus end point. VaR looks at the distribution of the terminal return; drawdown looks at the lowest point along the way. For driftless Brownian motion the **reflection principle** gives:

路径与终点不同。 VaR 看的是期末收益的分布，回撤看的是途中最低点。对无漂移布朗运动，**反射原理**给出：

$$\Pr\left(\min_{0 \leq s \leq T} X_s \leq -a\right) = 2 \Pr\left(X_T \leq -a\right), \quad \mathbb{E}[MDD_T] \approx \sqrt{\frac{\pi}{2}} \sigma \sqrt{T} \approx 1.25 \sigma \sqrt{T} \quad (\text{log scale, zero drift})$$

Meaning: the probability of **touching** the 99% one-year VaR level at some point during the year is about **twice** the probability of ending below it (2% rather than 1%). Simulation check: with daily discrete paths, 1.01% end below it and 1.80% touch it (discrete sampling sits slightly below the theoretical 2%).

含义：在一年内**途中**触及“99% 年度 VaR”水平的概率约是期末跌破它的**两倍**（2% 而非 1%）。模拟验证：日度离散路径中，期末跌破的比例为 1.01%，途中触及的比例为 1.80%（离散采样略低于理论值 2%）。

Expected maximum drawdown is proportional to $\sigma \sqrt{T}$, the same root as VaR's $\sqrt{h}$ rule but with a larger constant (about 1.25, where VaR uses a quantile). With positive drift μ drawdown growth slows and eventually grows like $(\sigma^2/\mu) \ln T$ (Magdon-Ismail et al. 2004).

预期最大回撤与 $\sigma \sqrt{T}$ 成正比，和 VaR 的 $\sqrt{h}$ 法则同源，但常数更大（约 1.25，而 VaR 的因子是分位数）。有正漂移 μ 时，回撤增长逐渐放缓，长期约按 $(\sigma^2/\mu) \ln T$ 增长（Magdon-Ismail 等 2004）。

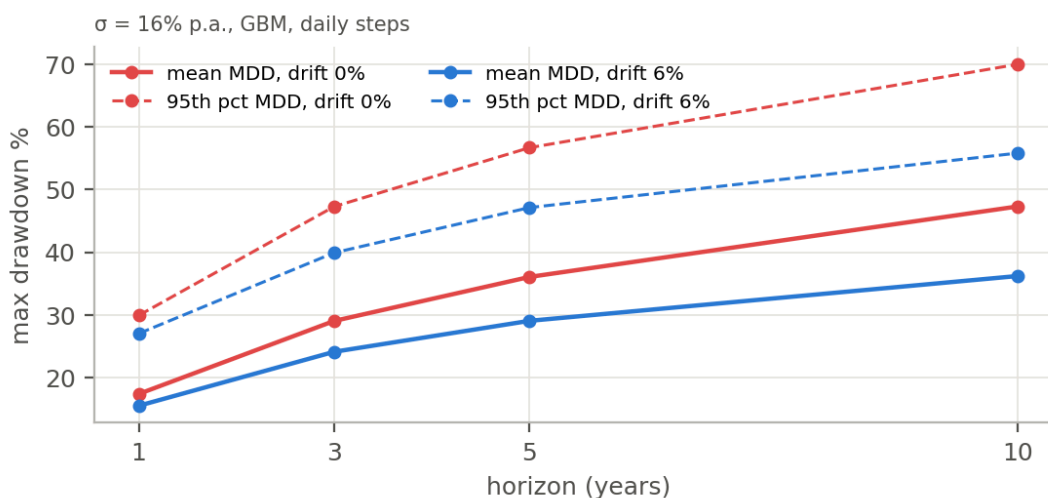


Figure 9. Monte Carlo: maximum drawdown (mean and 95th percentile) by horizon for an asset with 16% annual volatility.

图9 蒙特卡洛：年化波动率 16% 的资产在不同期限内的最大回撤（均值与第 95 百分位）。

期限 Horizon	Theory, zero drift 理论均值（无漂移）	Mean, 0% drift 模拟均值，漂移 0%	95th pct, 0% 第 95 百分位，漂移 0%	Mean, 6% drift 模拟均值，漂移 6%	95th pct, 6% 第 95 百分位，漂移 6%
1 年	18.2%	17.4%	29.9%	15.5%	27.0%
3 年	29.3%	29.0%	47.3%	24.1%	39.9%
5 年	36.1%	36.1%	56.7%	29.1%	47.1%
10 年	47.0%	47.3%	70.0%	36.2%	55.8%

In practice • "Is this drawdown unusual?" / 实操要点 • "这次回撤异常吗？"

Drawdowns are often read as evidence that "something is wrong" with a manager. The table gives a baseline: a strategy with 16% volatility and 6% expected return has a median 3-year maximum drawdown around 23% and a 5% chance of exceeding 40%. A 25% drawdown alone does not prove the strategy has failed; compare the realised drawdown with the simulated distribution for the same volatility and horizon, then look at its causes.

评价基金经理时，常把回撤当成“出问题”的证据。上表给出基准：一个年化波动率 16%、预期收益 6% 的策略，3 年内最大回撤的中位数约 23%，有 5% 的概率超过 40%。因此一次 25% 的回撤本身并不能说明策略失效；应把实际回撤放在同波动率、同期限的模拟分布中比较，再结合回撤原因判断。

13.3 Worked example: S&P 500 (2013–2018) 算例：标普 500（2013–2018）

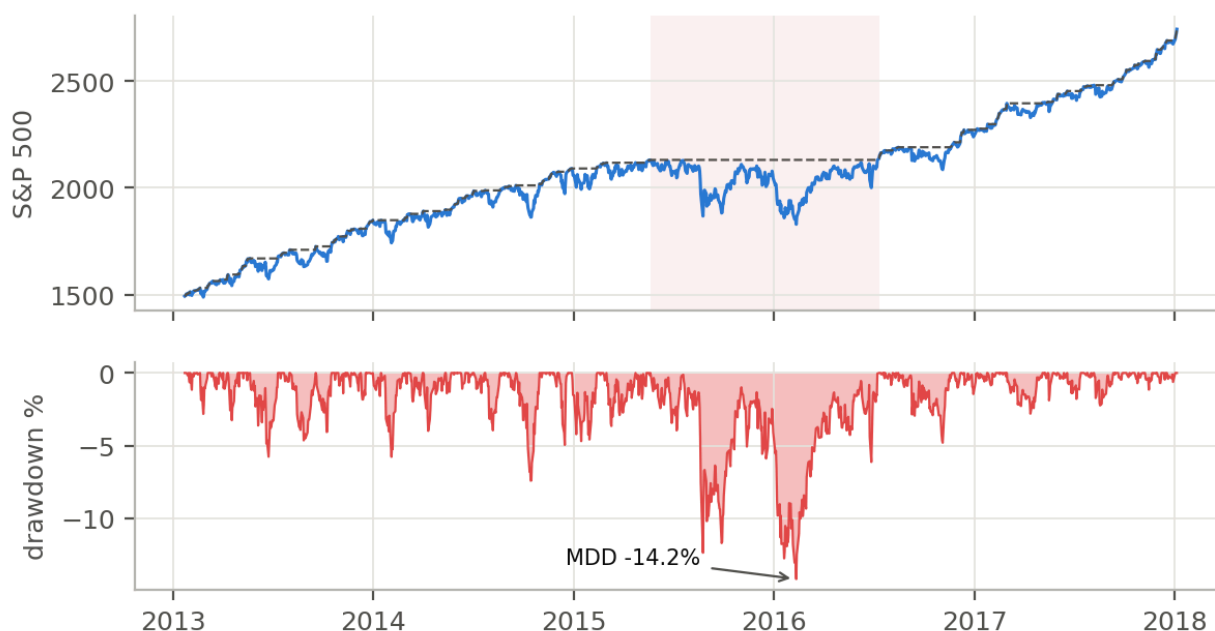


Figure 10. Top: index and high-water mark (dashed), with the maximum drawdown's time under water shaded; bottom: drawdown curve.

图 10 上：指数与高水位（虚线），阴影为最大回撤的水下期；下：回撤曲线。

-14.2% 最大回撤 · MDD (日度)	190 / 107 下跌 / 恢复 (营业日) · decline / recovery days	0.92 Calmar (价格指数 CAGR 13.1%)	-8.9% monthly-sampled MDD · 月度采样 MDD
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The maximum drawdown starts at the 2015-05-21 peak, runs through the August 2015 renminbi shock and the early-2016 oil slump, bottoms on 2016-02-11 (-14.2%) and regains the peak on 2016-07-11, about 297 business days under water. **Sampling frequency matters:** sampled at month-ends the same data show an MDD of only -8.9%, because month-end prices miss intra-month lows. Always state the frequency and use total-return NAV net of fees.

最大回撤从 2015-05-21 的高点开始, 经历 2015 年 8 月人民币贬值冲击与 2016 年初油价暴跌, 在 2016-02-11 见底 (-14.2%), 2016-07-11 收复高点, 水下时间约 297 个营业日。 **采样频率很重要:** 同一段数据按月末采样, MDD 只有 -8.9%, 因为月末价格错过了月中的低点。报告回撤时必须写明频率, 并使用含分红的总收益净值、扣除费用。

13.4 Drawdown limits and risk control 回撤限额与风险控制

- **Tiered triggers:** a common pattern is review at -8%, halve risk at -12%, stop adding risk and escalate to the board at -15%. Calibrate triggers against the simulated distribution in §13.2 so they do not fire on normal noise.
- **Drawdown-constrained allocation** (Grossman & Zhou 1993): if NAV must never fall below $(1 - \delta)$ of the high-water mark, the optimal risky exposure is proportional to the cushion $V_t - (1 - \delta)M_t$: less risk the closer to the floor. CPPI is a simplified version.
- **Costs:** stop-losses lock in losses, whipsaw in choppy markets, and sell together in broad declines, which is procyclical.
- **High-water mark and performance fees:** the HWM is the M_t in the drawdown definition; performance fees are only payable after past drawdowns are recovered (§15.1).
- **分级触发:** 常见做法如回撤 -8% 复核、-12% 风险敞口减半、-15% 停止新增风险并提交董事会。触发水平应参照 §13.2 的模拟分布设定, 避免在正常波动中频繁触发。
- **回撤约束配置** (Grossman & Zhou 1993): 若要求净值永不低于高水位的 $(1 - \delta)$, 最优风险敞口与缓冲 $V_t - (1 - \delta)M_t$ 成正比——越接近底线, 风险越小。CPPI 是其简化版本。
- **代价:** 止损会锁定亏损, 在震荡市中反复"止损-重新入场" (whipsaw), 且在市场整体下跌时同时卖出, 具有顺周期性。
- **高水位与业绩费:** 高水位就是回撤定义里的 M_t 。只有收复此前回撤后才能收取业绩费 (§15.1)。

VaR / ES		最大回撤 MDD
视角 Perspective	单期终点分布 · Single-period terminal distribution	整条路径 · The whole path
期限 Horizon	固定 (1 日、10 日、20 日) · Fixed	整个观察期, 随期限增长 · Grows with the observation period
可加性 Aggregation	可按资产分解 (ES 次可加) · Decomposable (ES subadditive)	不可简单分解 · Not simply decomposable
统计误差 Estimation	尾部样本少 · Few tail points	一个路径只有一个 MDD, 误差更大 · One MDD per path, noisier still
用途 Use	日常限额、资本、监管限额 · Daily limits, capital, regulatory caps	投资者体验、策略评估、止损框架 · Investor experience, strategy review, stop-loss

14 Scenario Analysis: Theory and Practice

情景分析: 理论与实操

14.1 Scenario analysis versus stress testing 情景分析与压力测试的区别

A stress test is usually an **instantaneous, single-period** shock: "what if X happens tomorrow?". Scenario analysis is a **multi-period, narrative-driven** path: "what if the world evolves like this over the next few years?". Growth, inflation, rates, equities, spreads and FX must be consistent year by year, with rebalancing, cash flows, fees and strategy responses included. The output is not one loss figure but a set of paths: terminal value, annualised return, maximum drawdown along the way, recovery time, and the chance of missing the investment objective.

压力测试通常是**瞬时、单期**的冲击, 问"如果明天发生 X"。情景分析是**多期、叙事驱动**的路径, 问"如果未来几年世界这样演变": 每年的增长、通胀、利率、股市、利差与汇率都要内部一致, 并考虑再平衡、现金流、费用与策略调整。它的产出不是一个损失数字, 而是一组路径: 期末价值、年化收益、途中最大回撤、恢复时间, 以及未达成投资目标的可能性。

14.2 Building a scenario 构建步骤

1. **Narrative:** one paragraph on cause, transmission and resolution (e.g. "a supply shock triggers stagflation, central banks hike, inflation fades after two years").
2. **Core variable paths:** year-by-year GDP, inflation, policy rate, 10-year yield, equity index, spreads, FX.
3. **Map to portfolio factors:** via regressions or the conditional expectation of §10.4.
4. **Project the portfolio:** period-by-period P&L with rebalancing, flows, fees and pre-agreed management actions.
5. **Outputs and decisions:** terminal value, annualised return, maximum drawdown, recovery time, probability of meeting objectives; discuss allocation or product-design changes.
1. **叙事:** 用一段话说清情景的起因、传导与结局 (如"供给冲击引发滞胀, 央行被迫加息, 两年后通胀回落")。
2. **核心变量路径:** 逐年设定 GDP、通胀、政策利率、10 年利率、股指、利差、汇率。
3. **映射到组合因子:** 用回归或 §10.4 的条件期望, 把核心变量映射到组合的全部风险因子。
4. **组合推演:** 逐期计算损益, 纳入再平衡、申赎、费用与预设的管理行动。
5. **输出与决策:** 期末价值、年化收益、最大回撤、恢复时间、目标达成概率; 讨论需要调整的配置或产品设计。

14.3 Types of scenario 情景的类型

类型 Type	说明 Description
Deterministic narrative 确定性叙事情景	3–5 个互相区分的情景（基准、不利、严重不利、结构性转变），便于沟通 · 3–5 distinct scenarios (base, adverse, severe, structural shift), easy to communicate
Historical analogue 历史类比	把 1970 年代滞胀、2000–2002 科技泡沫、2008–2009 危机的多年路径套用到当前组合 · Apply multi-year paths of past episodes to today's portfolio
Stochastic (Monte Carlo) 随机模拟	几何布朗运动、历史自助法或状态转换模型，生成数千条路径，得到终值与回撤的分布（如图 9）· GBM, bootstrap or regime-switching paths giving distributions of terminal value and drawdown
Climate 气候情景	参照 NGFS 的有序转型、无序转型、“温室世界”情景，区分转型风险（碳价、政策）与物理风险（灾害），期限常为 10–30 年 · NGFS orderly, disorderly and hot-house scenarios, separating transition risk (carbon prices, policy) from physical risk; horizons of 10–30 years

14.4 Worked example: five-year scenarios for a 60/40 portfolio 算例：60/40 组合的五年情景

Portfolio: 60% equities, 40% government bonds (duration 7, 3% carry), rebalanced annually. Four **illustrative** scenarios set yearly equity returns and yield changes; bond return $\approx 3\% - 7 \times \Delta y$.

组合：60% 股票、40% 政府债（久期 7，票息收益 3%），每年末再平衡。四个示意情景逐年设定股票收益与利率变化，债券收益 $\approx 3\% - 7 \times \Delta y$ 。

情景 Scenario	股票逐年收益 Equity by year	利率逐年变化 Yield change (bp)	5 年末 Terminal	年化 p.a.	最大回撤 MDD	收复年份 Recovered
基准 Base	+6% × 5	0	126.4	4.8%	0.0%	—
滞胀 Stagflation	-15, -5, +5, +8, +8	+150, +50, 0, -25, -25	101.1	0.2%	-14.8%	第 5 年
衰退后复苏 Recession & recovery	-30, +20, +15, +8, +7	-150, +50, +25, 0, 0	119.5	3.6%	-12.6%	第 3 年
无序气候转型 Disorderly transition	-10, -5, +4, +6, +6	+50, +25, 0, 0, 0	104.1	0.8%	-8.5%	第 5 年

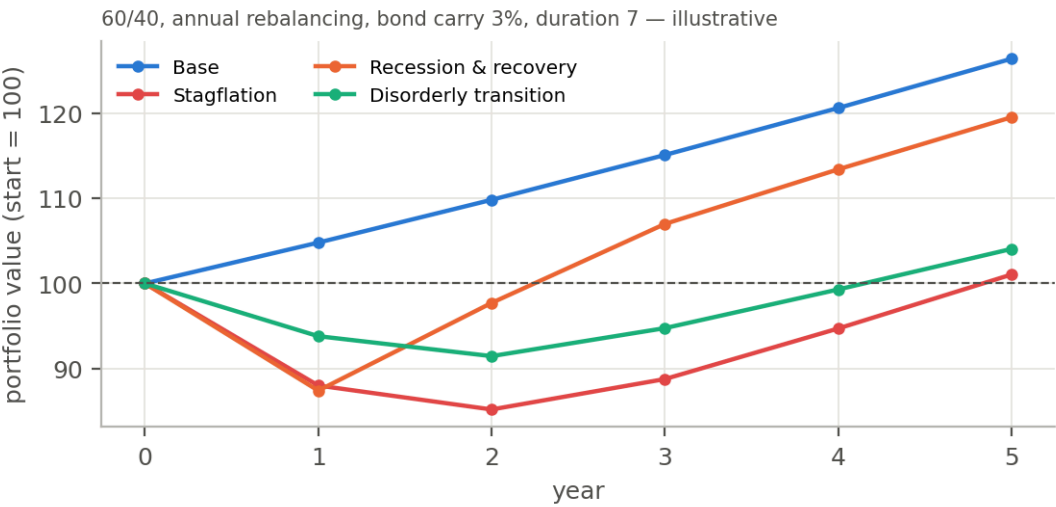


Figure 11. Five-year paths of a 60/40 portfolio under four scenarios (annual, illustrative).
图 11 四个情景下 60/40 组合的五年路径（年度，示意）。

Reading: the recession's first-year fall (equities -30%) is larger than stagflation's, but the bond rally cushions it and the portfolio recovers within three years; in stagflation bonds and equities fall together, the portfolio keeps sliding after year one and only breaks even in year five, at 0.2% a year. For a product targeting 4% a year, both stagflation and disorderly transition miss the objective: exactly the "poor outcomes" MiFID product governance asks firms to identify. Annual data also understate intra-year drawdowns (the sampling point in §13.3), so true maximum drawdowns would be deeper.

解读：衰退情景第一年的跌幅（股票 -30%）比滞胀大，但债券大涨缓冲了损失，组合三年即收复；滞胀情景股债同跌，第一年后继续下探，五年后才勉强回本，年化仅 0.2%。对一个目标“年化 4%”的产品，滞胀与无序转型都会导致目标落空——这正是 MiFID 产品治理要求识别的“不利结果”。另外，年度数据会低估年内回撤（§ 13.3 的采样问题），实际最大回撤应更深。

14.5 PRIIPs performance scenarios PRIIPs 业绩情景

A PRIIPs KID shows retail investors four performance scenarios, **stress**, **unfavourable**, **moderate** and **favourable**, at the recommended holding period (RHP) and at one year when the RHP is longer. Since 2023 (Delegated Regulation 2021/2268) the Category 2 method is:

- **Unfavourable / moderate / favourable**: over at least 10 years of history, take the outcomes of all rolling RHP-length windows and select the worst, median and best;
- **Stress**: compute rolling volatility over 21-day ($RHP \leq 1$ year) or 63-day ($RHP > 1$ year) sub-intervals, take the 99th or 90th percentile as stressed volatility, then a Cornish–Fisher 1% or 5% quantile.

This is "regulator-standardised scenario analysis", entirely historical, so it looks optimistic after long bull markets. Refer to the regulation for full computational detail.

14.6 Limits and governance 局限与治理

Scenarios are human-designed, prone to recency bias and hard to attach probabilities to; multi-period scenarios must also assume management actions and second-round effects (e.g. redemption-driven selling). Governance: review at least annually; build jointly across risk, investment and economics; record assumptions and sources; use results in allocation, product design and board discussion rather than an appendix.

PRIIPs KID 必须向零售投资者展示四个业绩情景：压力、不利、中性、有利，显示在建议持有期（RHP）末以及持有 1 年时（RHP 超过 1 年时）的结果。2023 年起（Delegated Regulation 2021/2268）第 2 类产品的方法：

- **不利 / 中性 / 有利**：在至少 10 年的历史中，取所有长度为 RHP 的滚动区间的结果，分别选最差、中位数与最好；
- **压力情景**：用 21 日（ $RHP \leq 1$ 年）或 63 日（ $RHP > 1$ 年）子区间计算滚动波动率，取第 99 或第 90 百分位作为压力波动率，再用 Cornish–Fisher 取 1% 或 5% 分位数。

这是一种"监管标准化的情景分析"，完全基于历史，因此在长期牛市后会显得乐观。具体计算细节以法规文本为准。

情景由人设定，容易受近期经验影响（近因偏差），也难以给出概率；多期情景还要假设管理层行动与二阶效应（如赎回潮引发的抛售）。治理上：情景至少每年复核；由风险、投资与经济研究共同制定；记录假设与来源；结果用于资产配置、产品设计与董事会讨论，而不只是附在报告里。

15 Regulation of Long-Term Risk Testing

长期风险的监管要求

15.1 UCITS UCITS

- **No drawdown limit**: UCITS sets none, but the risk management process must keep the risk profile consistent with objectives and strategy.
- **Performance fees and HWM** (ESMA guidelines, from January 2021): fees only on positive performance over the reference period; where an HWM model's reference period is shorter than the fund's life it must be at least a rolling 5 years; past losses or underperformance must be recovered first; crystallisation generally no more than once a year.
- **PRIIPs KID**: performance scenarios (§14.5); past-performance chart (up to 10 years); previous performance scenarios updated monthly and published.
- **Sustainability risk** (Delegated Directive 2021/1270, from August 2022): integrated into the risk management process; climate scenario analysis is common practice.
- **Marketing** (ESMA marketing guidelines): past performance must be fair and clear, with risk warnings.

- **无回撤上限**：UCITS 不设回撤限额，但风险管理流程须确保风险状况与投资目标、策略一致。
- **业绩费与高水位**（ESMA 业绩费指引，2021 年 1 月起）：业绩费只能对业绩参考期内的正收益收取；采用高水位模型且参考期短于基金存续期时，参考期至少为滚动 5 年；收取前须先弥补以往的亏损或跑输基准的部分；结算频率一般不超过每年一次。
- **PRIIPs KID**：业绩情景（§ 14.5）；过往业绩图表（最多 10 年）；以往业绩情景须每月更新并公开。
- **可持续风险**（Delegated Directive 2021/1270，2022 年 8 月起）：须把可持续风险纳入风险管理流程，气候情景分析是常见做法。
- **营销材料**（ESMA 营销指引）：展示过往业绩须公平、清晰，并附风险提示。

15.2 AIFMD AIFMD

- **Consistent risk profile** (Art. 15(3)(c)): the risk profile must match the AIF's size, portfolio structure, strategy and objectives; long-horizon scenarios help evidence this.
- **Annex IV**: monthly gross and net returns (from which regulators can compute drawdowns), plus stress-test results.
- **Investor disclosure** (Art. 23): historical performance where available; PRIIPs performance scenarios if sold to retail.
- **Sustainability risk** (Delegated Regulation 2021/1255, from August 2022): integrated into risk management and due diligence.
- **AIFMD II loan-originating funds**: leverage caps and 5% retention address long-term credit risk; closed-ended structures should match loan maturities.
- The ESMA performance-fee guidelines also apply to certain retail-facing AIFs.

- **风险状况一致性**（第 15(3)(c) 条）：风险状况须与 AIF 的规模、组合结构、策略和目标一致；长期情景有助于证明这一点。
- **Annex IV**：报告月度收益（毛、净），监管可据此计算回撤；另报告压力测试结果。
- **投资者披露**（第 23 条）：提供历史业绩（如有）；面向零售时适用 PRIIPs 业绩情景。
- **可持续风险**（Delegated Regulation 2021/1255，2022 年 8 月起）：纳入风险管理与尽职调查。
- **AIFMD II 贷款发放基金**：杠杆上限与 5% 风险保留等长期信用风险要求；封闭式结构须与贷款期限匹配。
- ESMA 业绩费指引同样适用于面向零售的某些 AIF。

15.3 MiFID II MiFID II

- **Product-governance scenario analysis** (Delegated Directive 2017/593 Art. 9(10)): manufacturers must run scenario analysis assessing the risk of **poor outcomes** for end clients and when they may arise, including deteriorating markets, distress at the manufacturer or a third party, the product ceasing to be commercially viable, and demand far above expectations. Art. 9(14) requires regular product review.
- **Suitability**: investment horizon and capacity for loss; periodic suitability reports; sustainability preferences (Delegated Regulation 2021/1253).
- **10% loss reporting** (2017/565 Art. 62): in effect a **drawdown trigger** measured from the last report.
- **Performance in marketing** (2017/565 Art. 44): past performance covers the preceding 5 years in complete 12-month periods; forward-looking figures must rest on reasonable, objectively supported assumptions and use **performance scenarios in different market conditions, both negative and positive**.
- **Firm level** (IFD Art. 24): the internal capital assessment typically spans a multi-year planning horizon, including orderly wind-down.

- **产品治理的情景分析** (Delegated Directive 2017/593 第 9(10) 条): 制造商须对金融工具进行情景分析, 评估产品给终端客户带来**不利结果**的风险及其发生情形, 包括市场环境恶化、制造商或第三方陷入困境、产品在商业上不再可行、需求远超预期。第 9(14) 条要求定期复核产品。
- **合适性**: 须考虑客户的投资期限与承受损失的能力; 定期合适性报告; 可持续偏好 (Delegated Regulation 2021/1253)。
- **10% 损失通知** (2017/565 第 62 条): 本质上是以上次报告为基准的**回撤**触发。
- **营销中的业绩** (2017/565 第 44 条): 过往业绩须覆盖最近 5 年 (按完整 12 个月区间); 对未来业绩的展示须基于有客观数据支持的合理假设, 并采用**不同市场条件下的业绩情景 (正面与负面)**。
- **公司层面** (IFD 第 24 条): 内部资本评估通常覆盖多年规划期, 包括有序清算情景。

要求 Requirement	UCITS	AIFMD	MiFID II
回撤相关 Drawdown-related	Performance-fee HWM (≥ 5-year reference) 业绩费高水位 (≥ 5 年参考期)	Annex IV monthly returns; retail AIF fee rules Annex IV 月度收益; 零售 AIF 业绩费规则	10% loss reporting 10% 损失通知
情景分析 Scenarios	PRIIPs performance scenarios PRIIPs 业绩情景	Risk-profile consistency; PRIIPs if retail 风险状况一致性; 零售时 PRIIPs	Product governance Art. 9(10); marketing scenarios 产品治理第 9(10) 条; 营销正负情景
可持续 / 气候 Sustainability	2021/1270	2021/1255	2021/1253 (合适性)
频率 Frequency	KID reviewed at least annually; monthly previous scenarios KID 至少每年复核; 以往情景每月公开	Per risk policy 按风险管理政策	Regular product review 产品定期复核

16 Integrated Framework, Checklist and Errata

三期限整合框架、实务清单与勘误

16.1 Linking the three horizons 把三个期限连成一个体系

The three toolsets do not replace one another; they layer: **VaR/ES** covers normal days, **stress testing** severe shocks, **drawdown and scenario analysis** multi-year bad paths. A complete framework shares one risk-factor mapping and dataset across all three, with limits calibrated to each other:

- high VaR-limit usage usually comes with high stress losses; joint warnings mean more than single ones;
- the worst stress loss should sit inside the drawdown triggers, otherwise one shock trips the stop-loss directly;
- poor outcomes found in long-term scenarios should feed back into product design (target market, SRI, choice of LMTs) and asset allocation.

三类工具不是互相替代，而是逐层覆盖：VaR/ES 管"正常日子"，**压力测试** 管"严重冲击"，**回撤与情景分析**管"持续多年的坏路径"。一个完整的风险框架应让三者共享同一套风险因子映射与数据，并在限额上相互校准：

- VaR 限额使用率高时，压力损失通常也高——两者同时预警比单独预警更有意义；
- 压力测试的最差情景损失应落在回撤触发线之内，否则一次冲击就会直接触发止损；
- 长期情景中出现的不利结果，应反馈到产品设计（目标市场、SRI、LMT 选择）与资产配置。

频率 Frequency	短期监测 Short-term	中期测试 Medium-term	长期测试 Long-term	主要监管依据 Main rule
每日 Daily	VaR/ES, leverage, limit usage VaR/ES、杠杆、限额使用率	Sensitivity snapshot (optional) 敏感性快照（可选）	Drawdown monitoring 回撤监控	UCITS daily VaR; MiFID 10% report UCITS 每日 VaR; MiFID 10% 通知
每月 Monthly	Backtesting 回溯	Market stress tests (VaR funds) 市场压力测试（VaR 基金）	—	CESR/10-788
每季 Quarterly	Parameter and data refresh 模型参数与数据更新	Liquidity stress tests (advised) 流动性压力测试（建议）	Scenario review 情景复核	ESMA LST; Annex IV (size-dependent) ESMA LST 指引; Annex IV（视规模）
每年 Annually	Model validation 模型验证	Scenario library and LMT review 情景库与 LMT 复核	Multi-year scenarios, product review, KID update 多年情景、产品复核、KID 更新	AIFMD L2 Art. 45; MiFID product governance; PRIIPs AIFMD L2 第 45 条; MiFID 产品治理; PRIIPs

16.2 Decision checklist 决策清单

#	决策 Decision	常见选择 Common choices	须写入文档 Document
1	收益定义 Return definition	Log (equity, commodity); absolute changes (rates, spreads, vols) 对数收益（股票、商品）；绝对变动（利率、信用利差、隐含波动率）	Shock type per factor class 每类风险因子的冲击类型
2	置信水平与期限 Confidence and horizon	99%/1D, 99%/10D, 97.5% ES; $\sqrt{10}$ or direct 10-day 99%/1D、99%/10D、97.5% ES; $\sqrt{10}$ 缩放或直接 10 日	Test of the scaling assumption 缩放假设的检验
3	估计窗口 Window	σ : 21–66 or EWMA; HS: 250–500 + stress; S,K: ≥ 250 σ : 21–66 obs 或 EWMA; HS: 250–500 + 压力期; S,K: ≥ 250	Data span, gaps, holidays 数据起止、缺失值与节假日处理
4	均值 Mean	通常设为 0 · usually zero	If kept, how it is estimated 若保留，说明估计方法
5	EWMA λ	0.94 (daily), 0.97; seed and burn-in 0.94（日）、0.97；初始化与热身期	Indexing convention $\sigma_{t t-1}$ 的索引约定
6	分位数约定 Quantile convention	Interpolation / [Np]-th worst / smoothing 插值 / 第 [Np] 差 / 多点平滑	Difference vs Excel/Python defaults 与 Excel、Python 默认值的差异
7	重估方式 Revaluation	Full revaluation; delta–gamma; sensitivities $\times$ shocks 全重估; Delta–Gamma 近似; 敏感度 $\times$ 冲击	Approximation error and PLA 近似误差与 PLA 结果
8	回溯 Backtesting	1-day 99% and 97.5%; APL and HPL; Kupiec + Christoffersen + traffic light 1 日 99% 与 97.5%; APL 与 HPL; Kupiec + Christoffersen + 交通灯	Eligible comparisons; sign convention 有效比较的定义、突破的符号约定
9	顺周期性 Procyclicality	Long-run floor, buffers, partial scaling 长期下限、缓冲、部分缩放	Peak-to-trough, n-day increase 峰谷比、n 日增幅
10	持续监控 Ongoing monitoring	Recalibration frequency, residual diagnostics, change control 参数重校准频率、残差诊断、模型变更管理	Triggers for model review 触发重审的阈值
11	压力情景库 Stress library	Sensitivity + historical + hypothetical + reverse + liquidity; annual review 敏感性 + 历史 + 假设 + 反向 + 流动性；每年复核	Sources, severity rationale, correlation assumptions 情景来源、幅度依据、相关性假设
12	回撤 Drawdown	Total-return NAV, daily frequency, tiered triggers 总收益净值、日度频率、分级触发线	Trigger calibration vs simulated distribution 触发线与模拟分布的校准
13	长期情景 Long-term scenarios	3–5 narratives + Monte Carlo; climate scenarios 3–5 个叙事情景 + 蒙特卡洛；气候情景	Assumptions, management actions, comparison with objectives 假设、管理行动、与产品目标的比较

16.3 Errata and clarifications on the source materials 原材料勘误与说明

位置 Location	原文 As written	更正 / 说明 Correction / note
课件：交通灯 Slides: traffic zones	$p_Y = 0.9599$	应为 0.9999（讲义正确）。Should be 0.9999 (the notes have it right).
课件：突破概率表 Slides: breach table	"Cum Binomial ... (T=252, 0.01)"	数值实为 T=252、p=5% 的概率质量（非累积）。99% VaR 的累积概率见图 5。Values are the PMF for T=252, p=5%, not cumulative. See Figure 5 for 99%.
讲义、白皮书：PRIIPs 公式 Notes, whitepaper: PRIIPs	$-0.146 S^2$	应为 $+0.146 S^2$ ，且 S、K 按 $1/\sqrt{N}$ 、 $1/N$ 缩放 (§6.2)。Should be $+0.146 S^2$, with S, K scaled by $1/\sqrt{N}$, $1/N$.
讲义：诊断比率 Notes: diagnostic ratio	Gaussian ρ "fixed at every c" 高斯 ρ "在所有 c 下固定"	正态 ρ 适中但随 $c \rightarrow 1$ 缓慢降向 1（ $1.254 \rightarrow 1.146 \rightarrow 1.090$ ）。Moderate but slowly decreasing to 1.
讲义：双资产例题 Solutions: two-asset	—	无误；相关系数为 1 时 VaR 为 £10,420，可用于说明分散化收益。Correct; with $\rho=1$ VaR is £10,420, illustrating diversification.
图 8 回测图 Figure 8 chart	Green ≤ 5 , Yellow ≤ 10	应为绿 0–4、黄 5–9、红 ≥ 10 。Should be 0–4 / 5–9 / ≥ 10 .

17 AIFMD Annex IV Reporting: A Step-by-Step Process

AIFMD Annex IV 报告：逐步准备流程

Annex IV is the periodic supervisory report an AIFM files with its regulator under AIFMD Articles 3(3)(d) and 24, with the template set by Annex IV of Delegated Regulation 231/2013 plus ESMA's reporting guidelines and XML schema (XSD). It has an **AIFM report** (manager level) and an **AIF report** (one per fund, and per compartment of an umbrella). The AIF report pulls together almost every output in this handbook: VaR and sensitivities (Part I), stress tests and liquidity (Part II), monthly returns (Part III). It therefore closes the book as the regulatory landing point of the three-horizon framework.

Annex IV 是 AIFMD 第 3(3)(d) 条与第 24 条下，AIFM 向主管机关定期报送的监管报告，模板由 Delegated Regulation 231/2013 附件 IV 与 ESMA 报告指引、XML 模式（XSD）规定。它分为 **AIFM 报告**（管理人层面）和 **AIF 报告**（每只基金、伞形基金的每个子基金各一份）。AIF 报告汇集了本手册三个部分的几乎所有产出：VaR 与敏感度（第一部分）、压力测试与流动性（第二部分）、月度收益（第三部分），因此放在全书最后，作为三期限框架在监管报送上的落地。

17.1 Process overview and timeline 流程总览与时间表

时点 When	步骤 Step	主要负责 Owner	产出 Output
Annually / pre-period 每年 / 期前	Scope and frequency 1 确定范围与频率	合规 · Compliance	Frequency map, NCA list, portal access 报告频率表、NCA 清单、门户账户
期前 Pre-period	Governance and master data 2 治理与主数据	合规、运营 · Compliance, operations	RACI, identifiers, strategy classification RACI、标识符、策略分类
T+0 – T+5	Positions and valuation 3 头寸与估值数据	基金行政管理人 · Administrator	Period-end positions, NAV, trades, flows 期末头寸、NAV、交易与申赎
T+5 – T+15	Exposures and risk metrics 4 敞口分类与风险指标	风险管理 · Risk	Market, counterparty, liquidity, leverage, stress 市场、交易对手、流动性、杠杆、压力测试
T+15 – T+22	Validation 5 数据验证	风险、合规 · Risk, compliance	Checks, variance explanations 核对清单、差异说明
T+22 – T+28	XML, approval, submission 6 生成 XML、审批、提交	合规 · Compliance	Signed-off filing 已签署提交的报告
From T+30 T+30 起	Post-submission 7 提交后管理	合规 · Compliance	Amendments, records, lessons 更正、留档、复盘

T = reference date (last business day of March, June, September, December); the deadline is one month later (last calendar day of the following month in Ireland), with 15 extra days for fund-of-funds AIF reports.
T = 报告参考日（3 月、6 月、9 月、12 月的最后一个营业日）；截止日为参考日后一个月（爱尔兰为次月最后一个日历日），组合基金的 AIF 报告可延长 15 天。

17.2 Step 1: Determine scope and frequency 第 1 步：确定报告范围与频率

(a) **Compute regulatory AuM** (Level 2 Arts 2–5): unlike NAV, convert derivatives to underlying equivalents under the commitment method, include assets acquired through leverage, aggregate all AIFs managed, and exclude UCITS the AIFM manages as a management company. Compute at least annually and monitor threshold crossings continuously.

(b) **Set the frequency** using the table in the right-hand column. For unleveraged AIFMs whose AIFs have 5-year lock-ups, the half-yearly band starts at €500m rather than €100m.

(c) **Set the content layer**: Art. 24(1) for all AIFs (main instruments, markets, principal exposures and concentrations); Art. 24(2) for authorised AIFMs (risk profile, liquidity, leverage, stress tests); Art. 24(4) for substantial leverage, commitment above 3× NAV (sources of leverage and more).

(d) **Special cases**: after authorisation, report from the next quarter; switch frequency on crossing a threshold; file a "nil return" (no-reporting flag set) even when there is nothing to report; non-EU AIFMs marketing under national private placement regimes file separately with each host regulator.

(a) **计算监管口径的管理资产规模（AuM）**（Level 2 第 2–5 条）：与 NAV 不同，须把衍生品按承诺法换算为标的等值、计入通过杠杆取得的资产，汇总管理人管理的全部 AIF，但不计其作为管理公司的 UCITS。至少每年计算一次，并持续监控是否越过门槛。

(b) **确定频率**：

情形 Condition	频率 Frequency
Registered (sub-threshold) AIFM 注册制（低于门槛）AIFM	Annual, Art. 24(1) content only 每年，仅报第 24(1) 条内容
AIFM AuM €100m–€1bn (€500m–€1bn if unleveraged with 5-year lock-up) 管理人 AuM €1 亿–€10 亿（无杠杆且 5 年锁定的管理人为 €5 亿–€10 亿）	每半年 · Half-yearly
AIFM AuM > €1bn 管理人 AuM > €10 亿	每季 · Quarterly
Single AIF > €500m 单只 AIF AuM > €5 亿	该 AIF 每季 · Quarterly for that AIF
Unleveraged AIF acquiring control of non-listed companies 无杠杆、以控股非上市公司为目的的 AIF	该 AIF 每年 · Annual for that AIF

(c) **确定报送内容层级**：第 24(1) 条（所有 AIF：主要工具、市场、主要敞口与集中度）；第 24(2) 条（授权 AIFM：风险概况、流动性、杠杆、压力测试）；第 24(4) 条（实质性杠杆，承诺法 > 3 倍 NAV：杠杆来源等额外信息）。

(d) **特殊情形**：新授权后从下一个季度开始报告；越过门槛后改用新频率；当期无资产也须提交"零报告"（设置无报告标志）；非欧盟 AIFM 通过私募制度（NPPR）销售的，须分别向每个销售国主管机关报送。

17.3 Step 2: Governance and master data 第 2 步：治理与主数据

- **RACI**: compliance owns the process and filing; the administrator provides positions, NAV and flows; risk computes metrics and stress tests; the investment team confirms strategy classification; the board or delegate approves. A written procedure sets cut-offs and escalation.
- **Identifiers**: AIFM and AIF national codes (the Central Bank C-code in Ireland), LEIs, share-class ISINs, ECB codes where relevant, master–feeder links, LEI/BIC of prime brokers and counterparties.
- **Static attributes**: base currency, inception date, domicile, marketing jurisdictions, predominant AIF type (hedge fund, private equity, real estate, fund of funds, other) and strategy breakdown as % of NAV (summing to 100%).
- **Portal readiness**: portal accounts, user rights and the "uploader ≠ signer" four-eyes rule; track the XSD version and validation-rule updates published by the regulator.

- **分工 (RACI)**：合规负责整体流程与提交；基金行政管理人提供头寸、NAV 与申赎；风险管理计算风险指标与压力测试；投资团队确认策略分类；董事会或授权人员审批。书面程序写明截止时间与升级路径。
- **标识符**：AIFM 与 AIF 的国家代码（爱尔兰为 Central Bank 的 C-code）、LEI、ISIN（份额类别）、ECB 代码（如适用）、主联基金关系、主经纪商与交易对手的 LEI/BIC。
- **静态属性**：基础货币、设立日期、注册地、销售地、主要 AIF 类型（对冲基金、私募股权、房地产、组合基金、其他）以及投资策略占 NAV 的百分比（合计须为 100%）。
- **门户准备**：确认门户账户、用户权限与"上传人与签署人不同"的双人规则；关注主管机关发布的 XSD 版本与验证规则更新。

17.4 Step 3: Positions and valuation data 第 3 步：头寸与估值数据

- Obtain final reference-date positions and NAV from the administrator and reconcile to the published NAV.
- Map each position to ESMA **sub-asset type codes**, market (MIC), region, currency and counterparty.
- Aggregate: top 10 principal exposures, top 5 portfolio concentrations, principal markets, turnover by sub-asset class over the period, long/short exposure by currency, holdings with dominant influence over non-listed companies (private equity).
- Collect the **monthly series** for the period: gross and net returns, NAV changes, subscriptions and redemptions, which are the inputs for the drawdown analysis in Part III.

- 从行政管理人获取参考日的最终头寸与 NAV，并与已公布 NAV 对账。
- 每个头寸映射到 ESMA 的**子资产类别代码**、交易市场（MIC）、地区、货币与交易对手。
- 由此汇总：前 10 大主要敞口、前 5 大组合集中度、主要交易市场、按子资产类别的期间成交额、各货币多空敞口、对非上市公司具有支配性影响的持仓（私募股权）。
- 收集报告期内的**月度数据**：毛收益与净收益、NAV 变化、申购与赎回金额——这正是第三部分回撤分析所需的输入。

17.5 Step 4: Risk metrics (Arts 24(2) and 24(4)) 第 4 步：风险指标（第 24(2) 与 24(4) 条）

Annex IV 部分 Section	要填什么 What is reported	本手册对应 In this handbook
市场风险 Market risk	Net equity/FX/commodity delta; DV01 and CS01 by maturity bucket; vega; VaR and its method 净股票 Delta、净汇率 Delta、净商品 Delta；按期限分段的 DV01 与 CS01；Vega；VaR 及计算方法	Chs 2–6; §10.3 sensitivities 第 2–6 章；§ 10.3 敏感度
交易对手 Counterparty	Top-5 counterparty exposures (both directions), collateral, share cleared via CCP vs bilateral 对前 5 大交易对手的敞口（双向）、抵押品、经 CCP 清算与双边交易的比例	§8.2
组合流动性 Portfolio liquidity	% of NAV liquidable within 1 day, 2–7, 8–30, 31–90, 91–180, 181–365, >365 days; unencumbered cash 可在 ≤1 日、2–7 日、8–30 日、31–90 日、91–180 日、181–365 日、>365 日内变现的 NAV 比例；未设押现金	§10.6 days-to-liquidate; §11.4 § 10.6 变现天数模型；§ 11.4
投资者流动性 Investor liquidity	Redeemable % in the same buckets; dealing frequency and notice; lock-ups; gates, side pockets; top-5 investor concentration 同样期限段的可赎回比例；赎回频率与通知期；锁定期；闸门、侧袋等特殊安排；前 5 大投资者持有比例	§10.6 liabilities; §12.1 LMTs § 10.6 负债端；§ 12.1 LMT
融资流动性 Financing liquidity	Borrowing by remaining term bucket 借款按剩余期限分段	§10.6
操作与历史风险 Operational / historical	Open positions; monthly gross/net returns, NAV changes, flows 未平仓头寸数量；月度毛/净收益、NAV 变化、申赎	Ch. 13 drawdown 第 13 章回撤
压力测试结果 Stress tests	Results of market and liquidity stress tests (free text, Arts 15(3)(b) and 16(1)) 市场风险与流动性风险压力测试的结果（文字描述，依据第 15(3)(b) 与 16(1) 条）	Chs 10–11 第 10–11 章
杠杆 Leverage	Gross and commitment leverage; cash and securities borrowing; derivative exposure; re-hypothecation; (24(4)) top-5 sources of leverage 总额法与承诺法杠杆；现金与证券借款；衍生品敞口；再质押；（24(4)）前 5 大杠杆来源	§8.2

In practice • Writing the stress-test section / 实操要点 • 压力测试结果怎么写

The stress-test field is free text, and regulators most often criticise boilerplate. Use a fixed structure: (1) scenario and shock sizes; (2) NAV impact (%); (3) liquidity coverage; (4) comparison with internal limits; (5) actions taken or planned. For example: "2022-type inflation scenario (equity -20%, rates +250bp, spreads +100bp, USD +12%); NAV -15.1%, within the 25% stress limit; 5-day stressed coverage 1.75x; no action required." Keep it consistent with the ESMA LST policy and the risk management policy.

压力测试栏是自由文本，监管最常批评的是"样板化"。建议固定结构：(1) 情景名称与冲击幅度；(2) 对 NAV 的影响 (%)；(3) 流动性覆盖率；(4) 与内部限额的比较；(5) 采取或计划采取的行动。例如："2022 型通胀情景（股票 -20%、利率 +250bp、利差 +100bp、美元 +12%）：NAV -15.1%，低于 25% 压力限额；5 日压力覆盖率 1.75x；无需行动。"内容应与 ESMA 流动性压力测试政策和风险管理政策保持一致。

17.6 Step 5: Validation 第 5 步：数据验证

1. **Internal reasonableness**: reconcile NAV with the administrator; strategy shares and liquidity buckets each sum to 100%; monthly returns agree with NAV changes and flows; portfolio liquidity agrees with dealing terms and LST conclusions; leverage methods match the prospectus; explain large period-on-period moves (e.g. $\pm 20\%$).
 2. **ESMA XSD**: structure and format.
 3. **ESMA business rules**: conditional fields, enumerations, cross-field consistency (separate rule sets for AIF and AIFM files).
 4. **Regulator-specific rules**: e.g. Ireland cross-checks against its fund register, and an unreported AIF blocks completion.
 5. **Four-eyes review and sign-off**: separate risk and compliance review, with records kept.
1. **内部合理性检查**：NAV 与行政管理人对账；策略占比、流动性分段各自合计 100%；月度收益与 NAV 变化、申赎一致；组合流动性与赎回条款、流动性压力测试结论一致；杠杆方法与招股书披露一致；与上期相比的大幅变动（如 $\pm 20\%$ ）逐项说明。
 2. **ESMA XSD 模式校验**：结构与格式。
 3. **ESMA 业务规则**：条件字段、枚举值、字段间一致性（AIF 层面与 AIFM 层面各有一套规则）。
 4. **主管机关附加规则**：如爱尔兰会与其内部的基金登记记录交叉核对，未报告的 AIF 会阻止提交完成。
 5. **四眼复核与签署**：风险与合规分别复核，留存复核记录。

17.7 Step 6: Generate XML and submit (Ireland as example) 第 6 步：生成 XML 与提交（以爱尔兰为例）

1. Generate AIFM and AIF files under the ESMA XSD (Ireland allows several AIFs in one file). The AIFM report is in EUR at the reference-date ECB rate; the AIF report is in the fund's base currency.
 2. File naming: C999100_YYYYMMDD_AIFM.xml (AIFM) or C999100_YYYYMMDD_AIF.xml (AIF), where C999100 is the institution's C-code.
 3. Log in to the Central Bank of Ireland Portal: Returns → View/Edit Returns → Add Return → choose the reference date (and C-code for AIF returns).
 4. Load a File; the portal validates automatically and emails the result; fix errors and re-upload.
 5. A **different user** finalises and signs off. **An unsigned return is not filed**; once signed off it is locked.
 6. Use INIT for the first filing and AMND for corrections, resending the whole record rather than changed fields only.
1. 按 ESMA XSD 生成 AIFM 文件与 AIF 文件（爱尔兰允许一个文件包含多只 AIF）。AIFM 报告以欧元报送，按参考日 ECB 汇率换算；AIF 报告以基金基础货币报送。
 2. 文件命名：C999100_YYYYMMDD_AIFM.xml (AIFM) 或 C999100_YYYYMMDD_AIF.xml (AIF)，C999100 为机构 C-code。
 3. 登录 Central Bank of Ireland Portal: Returns → View/Edit Returns → Add Return → 选择参考日 (AIF 报告再选择 C-code)。
 4. Load a File 上传，系统自动校验并发送邮件告知结果；有错误则修正后重新上传。
 5. 由另一名用户完成定稿与签署 (sign-off)。**未签署的报告不算已提交**；签署后报告被锁定。
 6. 首次提交用 INIT，更正时整份重报并用 AMND，不能只报变动字段。

17.8 Step 7: After submission 第 7 步：提交后管理

- Keep the acknowledgement, XML, source data, workpapers and approvals for regulatory queries and audit.
 - On finding an error, ask the regulator to unlock and file a correction; late or incorrect returns may lead to supervisory action.
 - If a fund of funds uses the 15-day extension, the AIF report follows the AIFM report; confirm consistency and amend the AIFM report if needed.
 - Run a post-mortem each period: delays, validation failures, data-quality issues; update procedures and controls.
 - Changes in manager-fund relationships (launch, liquidation, change of AIFM) must be notified separately as the regulator requires.
- 保存提交回执、XML 文件、源数据、计算底稿与审批记录，以便监管问询与审计。
 - 发现错误时联系主管机关解锁并提交更正；晚报或错报可能招致监管行动。
 - 组合基金若使用 15 天延期，AIF 报告晚于 AIFM 报告，需确认两者一致，必要时更正 AIFM 报告。
 - 每期结束后复盘：延误原因、验证失败类型、数据质量问题，并更新程序与控制。
 - 管理人与基金的关系变化（新设、清盘、更换管理人）须按主管机关要求另行通知。

Pitfall • Frequent Annex IV errors / 常见陷阱 • Annex IV 高频错误

- Using NAV instead of regulatory AuM, so the frequency is wrong;
 - AIFM report not converted to EUR, or the wrong FX date;
 - Enumerations outside ESMA's list, or confusing "registration status" with the "AIF reporting code";
 - Conditional fields out of step: a flag set to "no" with sub-fields populated (or the reverse);
 - Reusing INIT, or sending only changed fields on correction;
 - Portfolio liquidity contradicting dealing terms or LST conclusions (e.g. a daily-dealing fund reporting most assets as needing 90+ days);
 - A stress-test field that says only "stress tests performed, results acceptable";
 - Missing nil returns, or omitting a compartment of an umbrella.
- 把 NAV 当成监管口径 AuM，导致频率判断错误；
 - AIFM 报告未换算成欧元，或汇率日期不对；
 - 枚举值不在 ESMA 允许列表内，或把"注册状态"与"AIF 报告代码"混淆；
 - 条件字段不匹配：标志为"否"却填了下级字段（或反之）；
 - 重复使用 INIT，或更正时只发送变动字段；
 - 组合流动性与赎回条款、LST 结论相互矛盾（例如日度赎回基金却报告大部分资产需 90 天以上变现）；
 - 压力测试栏只写"已进行压力测试，结果可接受"；
 - 遗漏零报告，或伞形基金漏报某个子基金。

17.9 Outlook: the new reporting framework under AIFMD II 展望：AIFMD II 下的新报告框架

AIFMD II mandates ESMA to redesign supervisory reporting and brings UCITS into comparable periodic reporting for the first time. ESMA's direction: one **modular template** integrating AIFMD, UCITS, MMF and ECB statistical reporting; **ISO 20022 XML** with a common data dictionary (LEI, ISIN, CFI); **security-by-security** reporting where useful; a single collection point that "reports once, uses many times". Technical standards are still being drafted, and Level 2 measures are expected only after October 2027. Preparation now: map data overlaps across reports, assess the IT impact of security-level reporting, confirm the national collection authority, and build the risk data from all three parts of this handbook into a traceable data store.

AIFMD II 授权 ESMA 重新设计监管报告，并首次把 UCITS 纳入类似的定期报告。ESMA 的方向是：以一个**模块化模板**整合 AIFMD、UCITS、MMF 与 ECB 统计报告；采用 **ISO 20022 XML** 与统一数据字典（LEI、ISIN、CFI）；在有价值处改为**逐只证券**报送；实行"一次报送、多方使用"的单一收集点。相关技术标准仍在制定中，Level 2 细则预计 2027 年 10 月以后才会落地。现在可以做的准备：盘点各类报告之间的数据重叠、评估逐只证券报送的 IT 影响、确认本国的收集机关，并把风险数据（本手册三个部分的产出）建成可追溯的数据仓库。

18 Risk Management in Practice for Derivative-Heavy Funds

衍生品密集型基金的风险管理实操

Alternative UCITS, macro and CTA strategies, volatility strategies, option overlays and hedge-fund AIFs rely mainly on derivatives: futures, interest rate swaps, CDS, total return swaps, options, variance swaps and FX forwards. Notionals can run to several times NAV, risk is highly non-linear, and the fund depends on daily margin and collateral liquidity. This chapter strings the earlier methods into an executable daily process, mapped item by item to UCITS, AIFMD and MiFID II requirements.

另类 UCITS、宏观与 CTA 策略、波动率策略、期权叠加策略、对冲基金 AIF 往往以衍生品为主要工具：期货、利率互换、CDS、总收益互换、期权、方差互换、外汇远期。它们的名义本金可达 NAV 的数倍，风险高度非线性，并且依赖保证金与抵押品的日常流动性。本章把前面各章的方法串成一套日常可执行的流程，并逐项对照 UCITS、AIFMD 与 MiFID II 的要求。

18.1 New-instrument approval: before the first trade 新工具准入：用第一笔交易之前

1. **Eligibility** (UCITS): the underlying must be eligible assets, financial indices, interest rates, FX rates or currencies (no direct commodities, only via eligible indices); OTC counterparties must be prudentially supervised institutions in categories approved by the regulator; OTC derivatives need reliable, verifiable daily valuation and must be closable at fair value at any time (UCITS Directive Art. 50(1)(g), Directive 2007/16/EC Art. 8).

2. **Risk management process (RMP)**: new derivative types must be written into the RMP; in Ireland a UCITS files its RMP with the Central Bank before using derivatives, and types not covered require an update first. AIFs update the risk policy and maximum-leverage disclosure.

3. **Legal documentation**: ISDA master and CSA, clearing agreements, prime brokerage agreements; confirm netting is enforceable in the counterparty's jurisdiction (legal opinions).

4. **Systems and models**: booking, pricing model, risk-factor mapping (§6.5), whether VaR captures gamma, vega and basis (§10.3), commitment conversion (§8.6), Annex IV coding (Ch. 17).

5. **Valuation source**: an independent price source or model, not counterparty quotes alone.

6. **Operations**: margin and collateral processes, settlement, EMIR trade reporting.

7. **Dry run and sign-off**: test the full chain with small or simulated trades; risk, compliance, operations and valuation sign off together.
1. **合格性** (UCITS)：标的须为合格资产、金融指数、利率、汇率或货币（不能直接持有商品，只能通过合格指数）；场外对手须为受审慎监管且属主管机关认可类别的机构；场外衍生品须每日有可靠、可验证的估值，并能随时按公允价值平仓（UCITS 指令第 50(1)(g) 条、Directive 2007/16/EC 第 8 条）。

2. **风险管理流程** (RMP)：新类型衍生品须写入 RMP；在爱尔兰，UCITS 使用衍生品前须向 Central Bank 提交 RMP，未列入 RMP 的衍生品类型须先更新。AIF 须更新风险管理政策与最高杠杆披露。

3. **法律文件**：ISDA 主协议与 CSA、清算协议、主经纪协议；确认净额结算在对手所在法域可执行（法律意见书）。

4. **系统与模型**：交易录入、定价模型、风险因子映射（§ 6.5）、VaR 能否捕捉其 Gamma、Vega 与基差风险（§ 10.3）、承诺法换算（§ 8.6）、Annex IV 编码（第 17 章）。

5. **估值来源**：独立定价源或独立模型，不能只依赖对手报价。

6. **运营**：保证金与抵押品流程、结算、EMIR 交易报告。

7. **试运行与签署**：用小规模或模拟交易验证全流程，风险、合规、运营、估值共同签署。

18.2 The daily risk process 日常风险流程

时点 When	步骤 Step	要点 Key points
T+0 morning T+0 早晨	Position reconciliation 头寸对账	Reconcile fund system, prime brokers, counterparties and administrator; clear unconfirmed trades and breaks the same day 基金系统、主经纪商、对手与行政管理人三方对账；未确认交易与差异当日解决
T+0	Market data and valuation 市场数据与估值	Closing prices, curves, vol surfaces; independent price verification; start CSA dispute resolution when valuation gaps exceed thresholds 收盘价、曲线、波动率曲面；独立价格验证（IPV）；与对手的抵押品估值差异超过阈值即启动争议流程
T+0	Risk calculation 风险计算	VaR/ES, Greeks ladders, leverage (sum of notionals, commitment, gross), counterparty exposure, margin forecast, liquidity coverage VaR/ES、Greeks 阶梯、杠杆（名义本金之和、承诺法、总额法）、交易对手敞口、保证金预测、流动性覆盖率
T+0	Limit checks and escalation 限额检查与上报	Warning and hard limits; same-day escalation to PM and CRO, with actions and deadlines logged 预警线、硬限额；突破当日上报投资经理与风险负责人，记录处理与期限
By T+1 T+1 前	Margin and collateral 保证金与抵押品	Issue and meet margin calls, move collateral, update cash forecast 发出和应付追加保证金、抵押品调拨、现金预测更新

18.3 Leverage on several bases 杠杆的多种口径

The same fund shows very different leverage numbers under different rules, so always state the basis (table in the right-hand column).

Example: a macro UCITS holds €3bn of interest-rate futures and swaps, €0.5bn of equity index futures and €0.5bn of FX forwards on €0.5bn NAV. Sum of notionals is 800% of NAV; because much of it is short-dated rates, duration-adjusted risk is modest and the 20-day 99% VaR may be only 8–10% of NAV. An 800% notional is not a breach, but it must match the expected leverage in the prospectus; persistently exceeding it calls for a disclosure review.

同一只基金在不同规则下会得到截然不同的杠杆数字，报告时必须写明口径：

口径	规则	用途
名义本金之和	UCITS VaR 基金	招股书预期杠杆、年报实际杠杆
承诺法（含轧差、对冲）	UCITS 承诺法基金；AIFMD	≤ 100% NAV 限额（UCITS）；最高杠杆披露、> 3 倍为实质性杠杆（AIFMD）
总额法（不含轧差、对冲）	AIFMD	最高杠杆披露、Annex IV
风险口径（VaR）	UCITS VaR 基金	绝对 20% / 相对 2 倍限额

例：一只宏观 UCITS 持有利率期货与互换名义本金 €30 亿、股指期货 €5 亿、外汇远期 €5 亿，NAV €5 亿。名义本金之和为 800% NAV；因大量短久期利率合约，按久期调整后风险其实不大，20 日 99% VaR 可能只有 NAV 的 8–10%。名义本金 800% 并不违规，但须与招股书披露的预期杠杆一致；持续高于预期应评估是否更新披露。

18.4 Non-linear risk and Greek limits 非线性风险与 Greeks 限额

- **Greek limits:** set delta, gamma and vega limits by underlying and expiry bucket; bucket vega by tenor since short and long vols move differently.
- **Spot-vol ladder:** show full-revaluation P&L daily for spot ±5%, ±10%, ±20%, ±30% combined with vol −5 to +20 points (§10.3.2); it exposes short-option tails far better than a single VaR.
- **Spotting negative gamma:** short options, autocallable hedges and short variance swaps earn steady returns with low VaR in calm markets but lose at an accelerating rate in large moves. Give negative-gamma books a dedicated stress-loss limit.
- **Expiry concentration and pin risk:** many options on one expiry with spot near the strike make delta jump before expiry and hedging costs unpredictable.
- **Jump and gap risk:** overnight or weekend gaps cannot be hedged dynamically; include gaps in scenarios.

- **Greeks 限额：**按标的与到期分段设定 Delta、Gamma、Vega 限额；Vega 按期限分段，因为短期与长期波动率走势不同。
- **现货-波动率阶梯：**每日展示现货 ±5%、±10%、±20%、±30% 与波动率 −5 至 +20 个点组合下的全重估损益（§ 10.3.2），比单一 VaR 更能暴露卖出期权的尾部。
- **负 Gamma 识别：**卖出期权、自动赎回票据对冲、方差互换空头在平静期赚取稳定收益、VaR 很低，但在大波动时亏损加速。负 Gamma 头寸应设专门的压力损失限额。
- **到期集中与钉住风险：**大量期权集中在同一到期日、标的价格贴近行权价时，Delta 在到期前剧烈跳动，对冲成本难以预测。
- **跳跃与缺口风险：**隔夜或周末的价格跳空无法通过动态对冲规避，情景中应包含跳空。

18.5 Margin and collateral liquidity 保证金与抵押品流动性

Derivative-heavy funds most often fail not from losses as such but from **being unable to post margin in time:** variation margin (VM) is paid in cash daily, and initial margin (IM) rises with volatility at CCPs or under SIMM. In March 2020 and in the 2022 UK LDI crisis, many funds were solvent yet forced into fire sales because they could not raise cash fast enough.

Margin coverage = (cash + assets immediately usable as collateral) ÷ stressed VM and IM calls over 5 days. An internal floor of 1.2 is common; below 1, actions must be pre-arranged.

衍生品密集型基金最常见的失败方式不是亏损本身，而是**来不及交保证金**：变动保证金（VM）须每日以现金支付，初始保证金（IM）在波动上升时会被中央对手方或 SIMM 模型同步调高。2020 年 3 月与 2022 年英国 LDI 危机中，许多基金资产并未资不抵债，却因短期内无法筹集足现金而被迫低价抛售。

保证金覆盖率 = (现金 + 可即时用作抵押品的资产) ÷ 压力情景下 5 日内的 VM 与 IM 追加额。内部通常要求 ≥ 1.2，低于 1 须预先安排行动。

Worked example 13 • Margin stress test / 例题 13 • 保证金压力测试

A €500m NAV fund: interest rate swaps with total DV01 of €400k per bp (receiving fixed), long equity index futures of €1bn notional, current initial margin €80m; cash and T-bills €180m. Stress: rates +100bp, equities −15%, IM up 30%. Cash needs in the table in the right-hand column.

Coverage = 1.8 ÷ 2.14 = **0.84 < 1**. The fund is not insolvent (VM losses cut NAV by about 38% but it stays positive), yet faces a cash gap of about €34m and must sell other assets under stress. Pre-emptive measures: a larger cash buffer, options instead of some futures to cap VM, several clearing brokers, pre-arranged repo lines, and a margin-coverage limit.

NAV €5 亿的基金：利率互换合计 DV01 为 €40 万/bp（收固定），股指期货多头名义 €10 亿，当前初始保证金 €8,000 万；现金与短期国债共 €1.8 亿。压力情景：利率 +100bp、股票 −15%、初始保证金上调 30%。

项目	现金需求
利率互换 VM: €40 万 × 100	€4,000 万
股指期货 VM: €10 亿 × 15%	€1.5 亿
IM 上调: €8,000 万 × 30%	€2,400 万
合计	€2.14 亿

覆盖率 = 1.8 ÷ 2.14 = **0.84 < 1**。基金在经济上没有资不抵债（VM 造成的 NAV 损失约 38%，仍为正），但会出现约 €3,400 万的现金缺口，必须在压力下出售其他资产。事前可采取的措施：提高现金缓冲、以期权替代部分期货以限制 VM、分散清算经纪商、预先安排回购额度、设定保证金覆盖率限额。

18.6 Counterparties, collateral and legal terms 交易对手、抵押品与法律条款

- **Limits:** UCITS ≤ 5% of assets per OTC counterparty (10% for credit institutions), net of eligible collateral (§8.7); AIFs set internal limits. Diversify prime brokers and counterparties so one default cannot trap positions.
 - **CSA terms:** thresholds, minimum transfer amounts, eligible collateral and haircuts, rating triggers, dispute timelines; term differences directly change exposure and liquidity (see §18.9).
 - **Counterparty-default scenario:** assume the largest counterparty defaults; compute replacement cost (re-establishing the same positions) plus losses on collateral liquidation.
 - **Collateral reuse:** rehypothecation adds leverage and counterparty risk; include it in leverage and report under SFTR.
- **限额:** UCITS 对单一 OTC 对手 ≤ 5% 资产（信用机构 10%），扣除合格抵押品后计算（§ 8.7）；AIF 自设内部限额。分散主经纪商与对手，避免单一对手违约导致无法平仓。
 - **CSA 条款:** 阈值（threshold）、最低转账金额、合格抵押品与折扣率、评级触发条款、争议解决时限；条款差异直接影响风险敞口与流动性（详见 § 18.9）。
 - **对手违约情景:** 假设最大对手违约，计算替换成本（重新建立相同头寸的成本）加抵押品变现损失。
 - **抵押品再利用:** 再质押会增加杠杆与对手风险，须纳入杠杆计算并按 SFTR 报告。

18.7 Stress-scenario checklist for derivative funds 衍生品基金的压力情景清单

情景 Scenario	考察什么 What it tests
Vol spike (+20 points) with spot fall 波动率骤升（+20 个点）叠加现货下跌	Short gamma and vega; IM increases 负 Gamma、负 Vega 头寸；IM 上调
Overnight gap (~10%) 跳空（隔夜 ~10%）	Failure of dynamic hedging 动态对冲失效
Correlation flip (bonds and equities fall together) 相关性翻转（股债同跌）	Relative-value and hedged books 相对价值与对冲组合
Basis widening (futures–cash, CDS–bond, swap spreads) 基差扩大（期货对现货、CDS 对债券、掉期利差）	Arbitrage and hedge assumptions 套利与对冲假设
Margin spiral (§18.5) 保证金螺旋（§ 18.5）	Cash and collateral liquidity 现金与抵押品流动性
Largest counterparty default 最大对手违约	Replacement cost and collateral 替换成本与抵押品
Hedge-instrument illiquidity 对冲工具流动性枯竭	Cost of being unable to close or rebalance 无法平仓或调整对冲的成本
Reverse: what takes NAV down 20% or margin coverage below 1 反向压力：何种情景使 NAV 跌 20% 或保证金覆盖率跌破 1	Hidden concentrations (§10.5) 隐藏集中度（§ 10.5）

18.8 Regulatory mapping and control frequency 监管对照与控制频率

控制 Control	频率 Frequency	UCITS	AIFMD	MiFID II
VaR/ES and limit usage VaR/ES 与限额使用率	Daily 每日	绝对 20% / 相对 2 倍（§8.1）	内部限额（第 15 条）	Mandate limits for discretionary accounts 全权委托账户按合同限额
Leverage bases 杠杆（名义本金、承诺法、总额法）	Daily 每日	Commitment ≤ 100% or disclose notionals 承诺法 ≤ 100% 或披露名义本金之和	Both methods + maximum; > 3× extra reporting 两种方法 + 最高杠杆；> 3 倍须额外报送	10% loss report for leveraged accounts 含杠杆账户 10% 损失通知
Counterparty and collateral 交易对手与抵押品	Daily 每日	5% / 10%; ESMA collateral criteria 5% / 10%; ESMA 抵押品标准	Due diligence, Annex IV top 5 尽职调查、Annex IV 前 5 大对手	Client asset safeguarding 客户资产保护
Margin liquidity 保证金流动性	Daily forecast, weekly stress 每日预测，每周压力	LST includes margin calls LST 须含保证金追加（§ 12.1）	Art. 16 liquidity 第 16 条流动性管理	IFD liquid assets IFD 流动资产评估
Stress tests and Greek ladders 压力测试与 Greeks 阶梯	Daily ladder, monthly full 每日阶梯，每月全面	At least monthly for VaR funds VaR 基金至少每月（§ 12.1）	Art. 15(3)(b); Annex IV results 第 15(3)(b) 条；Annex IV 报告结果	Product-governance scenarios 产品治理情景分析
Backtesting 回测	Monthly 每月	> 4 次须上报（§7.3）	Level 2 第 45 条	—
Model and valuation validation 模型与估值验证	Annually and on material change 每年及重大变更时	Independent validation; RMP update 独立验证；RMP 更新	Art. 19 valuation; L2 Art. 45 第 19 条估值；Level 2 第 45 条	Appropriateness (derivatives are complex) 适当性（衍生品为复杂工具）
Disclosure and reporting 披露与报告	Annual / periodic 年度 / 定期	Prospectus expected leverage; annual VaR usage and leverage 招股书预期杠杆；年报 VaR 使用率与杠杆	Annex IV (Ch. 17) Annex IV（第 17 章）	EMIR reporting; cost disclosure EMIR 报告；成本披露

Pitfall • Derivative funds / 常见陷阱 • 衍生品基金

- Watching VaR but not notionals and Greeks: a low VaR can hide the tail of heavy short optionality;
 - Treating "not insolvent" as "safe": the binding constraint is often cash, not NAV;
 - Relying on counterparty valuations: in a dispute you cannot independently judge exposure or margin;
 - Hedge ratios failing under stress: basis and correlation shifts make "hedged" positions lose together;
 - Trading new instruments before the process is ready: derivatives outside the RMP or not yet mapped leave VaR and leverage understated.
- 只看 VaR 不看名义本金与 Greeks: 低 VaR 可能掩盖大量卖出期权的尾部;
 - 把"未资不抵债"当成"安全": 真正的约束常常是现金而不是 NAV;
 - 依赖对手估值: 估值争议时无法独立判断敞口与保证金;
 - 对冲比例在压力下失效: 基差与相关性变化让"已对冲"头寸同时亏损;
 - 新工具先交易后补流程: 未列入 RMP 或未完成系统映射的衍生品会造成 VaR 与杠杆漏算。

18.9 CSA requirements and collateral management CSA 要求与抵押品管理

The **CSA** (Credit Support Annex) is an annex to the ISDA master agreement setting when, in what assets and at what valuation the parties exchange collateral. It directly drives counterparty exposure, margin cash needs and even derivative valuation (CSA discounting, §6.9).

CSA（Credit Support Annex，信用支持附件）是 ISDA 主协议的附件，约定双方何时、以什么资产、按什么估值交换抵押品。它直接决定交易对手敞口、保证金现金需求，甚至衍生品的估值（CSA 贴现，§ 6.9）。

文件 Document	特点 Features
1994 New York law CSA 1994 年纽约法 CSA	Security interest: the poster keeps ownership, the taker holds a security interest 担保权益方式：抵押品所有权仍属交出方，收取方享有担保权益
1995 English law CSA 1995 年英国法 CSA	Title transfer: ownership passes to the taker and is netted on default 所有权转移方式：抵押品所有权转给收取方，违约时并入净额结算
2016 VM CSA 2016 年 VM CSA	Built for regulatory variation margin: zero threshold, daily exchange 为满足监管变动保证金要求而设计：零阈值、每日交换
2018 IM documents (CSA / CSD) with custodian control agreements 2018 年 IM 文件（CSA / CSD）及托管账户控制协议	Regulatory initial margin, segregated at a third-party custodian, no reuse 监管初始保证金：由第三方托管人隔离持有，不得再使用

Key terms 关键条款

条款 Term	含义与风险影响 Meaning and risk impact
阈值 Threshold	Exposure below it is uncollateralised 低于该金额的敞口不要求抵押；阈值以内即为无抵押敞口
最低转账金额 MTA	Calls below it are not made, avoiding small transfers 追加额低于该值不转账，避免小额频繁转账
Independent amount / IM 独立金额 / 初始保证金	An extra buffer independent of MtM 与市值无关、额外交付的缓冲
Eligible collateral and haircut 合格抵押品与折扣率	Which assets and at what haircut; sets usable liquidity 可交哪些资产、各打几折；决定可动用的流动性
Valuation agent, dates, settlement 估值代理、估值日与交付时间	Who calculates, how often, settlement lag (usually T+1) 谁算、多久算一次、几天内交付（通常 T+1）
Interest on cash collateral 现金抵押品利率	Usually the overnight rate (e.g. €STR) ± spread, with negative-rate handling; sets the discount curve 通常为隔夜利率（如 €STR）加减点差，并约定负利率处理；决定贴现曲线
Dispute resolution 争议解决	Timelines, recalculation and escalation when valuations differ 估值分歧时的时限、重新估值与升级机制
Substitution and reuse 替换与再使用	Whether collateral can be swapped and whether the taker may rehypothecate 能否替换抵押品、收取方能否再质押
Additional termination events 附加终止事件	Rating downgrades, NAV declines (common for funds, e.g. -15% in 1 month, -25% in 3 months, -35% in 12 months), key-person events triggering early termination 评级下调、NAV 下跌（基金常见：例如 1 个月跌 15%、3 个月跌 25%、12 个月跌 35%）、关键人员离职等触发提前终止

Regulatory requirements 监管要求

框架 Framework	要求 Requirement
EMIR margin RTS EMIR 保证金规则 (Delegated Regulation 2016/2251)	Uncleared OTC derivatives: VM calculated daily with same-day calls and zero VM threshold; combined VM and IM minimum transfer amount $\leq$ €500k; IM exchanged when both parties exceed €8bn average aggregate notional (final phase from September 2022), with a group IM threshold of up to €50m; IM segregated and not reused; eligible-collateral list, standard haircuts and concentration limits; physically settled FX forwards exempt from IM. 非清算 OTC 衍生品: 变动保证金每日计算、当日发出追加, VM 阈值为零; VM 与 IM 合计的最低转账金额不超过 €50 万; 双方未清算衍生品平均名义本金均超过 €80 亿 (2022 年 9 月起的最后阶段) 须交换初始保证金, 集团层面可设最高 €5,000 万的 IM 阈值; IM 须隔离持有、不得再使用; 合格抵押品清单、标准折扣率与集中度限制; 实物交割外汇远期豁免 IM。
EMIR risk mitigation EMIR 风险缓释	Portfolio reconciliation: daily for 500+ outstanding trades with a counterparty, weekly for 51–499, quarterly for up to 50; disputes above €15m outstanding for over 15 business days reported monthly to the regulator. 组合对账频率: 与同一对手未平仓合约 $\geq$ 500 笔须每个营业日、51–499 笔每周、 $\leq$ 50 笔每季度; 超过 €1,500 万且持续 15 个营业日以上的争议须按月报告主管机关。
UCITS	Collateral received must meet ESMA criteria (liquidity, daily valuation, credit quality, low correlation, diversification, third-party custody, enforceability, §8.7); non-cash collateral may not be sold, reinvested or pledged; unprotected collateral posted (e.g. non-segregated IM) counts as counterparty exposure; uncollateralised exposure within the CSA threshold counts towards the 5% / 10% limit. 收到的抵押品须满足 ESMA 标准 (流动性、每日估值、信用质量、低相关、分散、第三方托管、可随时执行, § 8.7); 非现金抵押品不得出售、再投资或质押; 交给对手且未受保护的抵押品 (如未隔离的 IM) 计入对手敞口; CSA 阈值以内的无抵押敞口计入 5% / 10% 对手限额。
AIFMD	Collateral reuse counts in leverage; the depositary oversees assets and cash flows; Annex IV reports collateral; reuse is reported under SFTR. 抵押品再使用计入杠杆; 存管人监督资产与现金流; Annex IV 报告抵押品; 再使用须按 SFTR 报告。
MiFID II	Investment firms may not conclude title-transfer collateral arrangements with retail clients (Art. 16(10)); client assets must be safeguarded and segregated. 投资公司不得与零售客户订立所有权转移式抵押安排 (第 16(10) 条); 客户资产须妥善保管与隔离。

Worked example 15 • CSA threshold and the UCITS counterparty limit / 例题 15 • CSA 阈值与 UCITS 对手限额

A €150m NAV UCITS has a legacy CSA with a **non-credit-institution** counterparty: threshold €10m, MTA €0.5m. Current MtM with that counterparty is +€12.3m and collateral held is €1.5m. Required collateral = $12.3 - 10 = \text{€}2.3\text{m}$, so a €0.8m call is made ($\geq$ MTA). Uncollateralised exposure afterwards = $12.3 - 2.3 = \text{€}10\text{m} = \mathbf{6.7\% \text{ of NAV}}$, above the 5% limit for non-credit institutions. Conclusion: UCITS CSAs should use zero thresholds (2016 VM CSA), otherwise the threshold alone can cause a breach.

NAV €1.5 亿的 UCITS 与一家**非信用机构**对手签有旧式 CSA, 阈值 €1,000 万、MTA €50 万。当前对该对手的衍生品市值为 +€1,230 万, 已收抵押品 €150 万。按 CSA, 需抵押额 = $1,230 - 1,000 = \text{€}230$ 万, 追加 €80 万 ($\geq$ MTA)。追加后无抵押敞口 = $1,230 - 230 = \text{€}1,000$ 万 = **6.7% NAV**, 超过对非信用机构 5% 的上限。结论: UCITS 与对手的 CSA 应采用零阈值 (2016 年 VM CSA), 否则阈值本身就可能导致违规。

In practice • The daily margin process / 实操要点 • 每日保证金流程

1. Morning: compute collateral receivable or payable per CSA from prior-day closing valuations;

2. Issue or receive calls and agree valuations; start disputes when gaps exceed internal thresholds;

3. Choose collateral to deliver, weighing haircuts, reuse rights and cheapest-to-deliver while keeping §18.5 margin coverage;

4. Settle and reconcile: confirm receipt, log pending and failed transfers;

5. Monitor: ageing of open disputes, uncollateralised exposure per counterparty as % of NAV, distance to NAV triggers.
1. 早晨: 按前一日收盘估值计算每个 CSA 的应收 / 应付抵押品;

2. 发出或接收追加通知, 与对手核对估值; 差异超过内部阈值即启动争议流程;

3. 选择交付的抵押品: 考虑折扣率、再使用权与"最便宜交付", 同时保证剩余现金满足 § 18.5 的保证金覆盖率;

4. 结算与对账: 确认到账, 记录在途与失败交接;

5. 监控: 未解决争议账龄、对手无抵押敞口占 NAV 比例、NAV 触发条款的距离。

18.9.1 Applying and calculating the threshold 阈值的应用与计算

$$CSA = \max(E + IA_{\text{cpty}} - IA_{\text{own}} - TH, 0), \quad \text{Value} = MV \times (1 - h), \quad \text{Delivery} = CSA - \text{Value} \text{ if } \geq \text{MTA}$$

The threshold (TH) is the amount of exposure the counterparty may leave uncollateralised, in effect an agreed unsecured credit line. Three steps:

1. **Credit Support Amount:** exposure E plus independent amounts owed by the counterparty, minus those owed by us, minus the counterparty's threshold, floored at zero; without independent amounts it is $\max(\text{MtM} - \text{threshold}, 0)$.
2. **Compare with collateral held:** collateral value = market value $\times (1 - \text{haircut } h)$. Call if the shortfall $\geq$ MTA; return if the excess $\geq$ MTA; otherwise do nothing.
3. **Round:** calls usually round up, returns down (e.g. to €100k); once above the MTA the **whole** difference moves, not just the excess.

阈值 (TH) 是在该金额以内对手可以不交抵押品——本质上是双方约定的"无抵押授信额度"。计算分三步:

1. **应抵押额** (Credit Support Amount): 敞口 E 加对手应交的独立金额, 减己方应交的独立金额, 再减对手阈值, 负数取 0; 无独立金额时即 $\max(\text{市值} - \text{阈值}, 0)$ 。
2. **与已持有抵押品价值比较:** 抵押品价值 = 市价 $\times (1 - \text{折扣率 } h)$ 。差额 $\geq$ MTA 则追加, 已持有价值超出应抵押额 $\geq$ MTA 则退还, 否则不动。
3. **取整:** 追加通常向上、退还向下取整 (如到 €10 万); 超过 MTA 后转移全部差额, 而非仅超出部分。

Worked example 16 • Daily margining under a threshold / 例题 16 • 阈值下的逐日保证金

The fund is in the money; counterparty threshold €5m, MTA €0.5m, rounding to €0.1m; the counterparty delivers government bonds with a 2% haircut. Figures in € millions in the table in the right-hand column.

Whenever MtM exceeds the threshold, uncollateralised exposure is pinned near the threshold (plus at most just under one MTA). The threshold is thus a standing unsecured exposure.

基金为有利方；对手阈值 €500 万、MTA €50 万、按 €10 万取整；对手以政府债交付，折扣率 2%。单位：百万欧元。

日	市值	应抵押额	差额	动作	已持有	无抵押敞口
1	4.0	0	0	无（阈值内）	0	4.0
2	6.2	1.2	+1.2	追加 1.2（债券市值约 1.224）	1.2	5.0
3	6.5	1.5	+0.3	无（< MTA）	1.2	5.3
4	8.0	3.0	+1.8	追加 1.8	3.0	5.0
5	5.3	0.3	-2.7	退还 2.7	0.3	5.0

只要市值超过阈值，无抵押敞口就被"钉"在阈值附近（最多再加不到一个 MTA）。阈值因此是一笔持续存在的无抵押风险。

影响 Effect	说明 Explanation
Exposure ceiling 敞口上限	Worst exposure $\approx$ threshold + MTA + MtM move over the margin period of risk (usually 10 business days bilaterally) 最坏敞口 $\approx$ 阈值 + MTA + 保证金风险期（双边通常 10 个营业日）内的市值变动
UCITS counterparty limit UCITS 对手限额	Uncollateralised exposure within the threshold counts towards 5% / 10%; keep thresholds well inside the headroom, e.g. $\leq 1-2\%$ of NAV (Example 15) 阈值以内的无抵押敞口计入 5% / 10% 限额；实务上阈值应远低于限额余量，如 $\leq 1-2\%$ NAV（例题 15）
CVA	Expected exposure is roughly capped near threshold + MTA + MPOR move; higher thresholds mean higher CVA (§8.7) 预期敞口大致被截在"阈值 + MTA + 风险期变动"附近，阈值越大 CVA 越高（§ 8.7）
Bank capital (SA-CCR) 银行资本（SA-CCR）	Replacement cost for margined netting sets $RC = \max(V - C, TH + MTA - NICA, 0)$; the threshold enters capital directly, so banks price high-threshold trades higher 有保证金净额结算集合的重置成本 $RC = \max(V - C, TH + MTA - NICA, 0)$ ；阈值直接进入资本，银行对高阈值交易定价更高

场景 Context	阈值规定 Threshold rule
EMIR variation margin EMIR 变动保证金	Must be zero between in-scope parties (2016 VM CSA) 范围内双方之间必须为零（2016 年 VM CSA）
EMIR initial margin EMIR 初始保证金	Up to €50m at group level, allocated across counterparties; IM due = $\max(\text{SIMM IM} - \text{allocated threshold}, 0)$ 集团层面最高 €5,000 万，跨对手合计分配；应交 IM = $\max(\text{SIMM IM} - \text{分配阈值}, 0)$
Minimum transfer amount 最低转账金额	Combined VM and IM $\leq$ €500k VM 与 IM 合计 $\leq$ €50 万
Legacy CSAs / out-of-scope trades 旧 CSA / 不在范围内交易	Positive thresholds possible, but UCITS remain bound by 5% / 10% 可约定正阈值，但 UCITS 仍受 5% / 10% 限额约束
Rating-linked thresholds 评级挂钩阈值	Step down or to zero on downgrade; mitigates deterioration but can trigger concentrated calls under stress 评级下调时自动降低或归零；缓解信用恶化，但可能在压力时触发集中追加

In practice • Managing thresholds / 实操要点 • 阈值管理

1. Monitor "uncollateralised exposure $\div$ NAV" per CSA daily against counterparty limits;
2. Negotiate: UCITS should seek zero thresholds; AIFs accepting positive thresholds include them in internal counterparty limits;
3. Check both directions: a threshold granted to the fund helps liquidity (less collateral posted), but counterparties may charge more or shorten its validity;
4. Reconcile systems: threshold, MTA, rounding and haircuts in the collateral system must match the CSA text; mis-keyed parameters are a common source of margin disputes.

1. 每日监控每个 CSA 的"无抵押敞口 $\div$ NAV"，与对手限额比较；
2. 谈判：UCITS 争取零阈值；AIF 接受正阈值时计入内部对手限额；
3. 双向检查：己方获得的阈值有利于流动性（少交抵押品），但对手可能要求更高价格或更短有效期；
4. 系统核对：抵押品系统中的阈值、MTA、取整、折扣率须与 CSA 文本一致——参数录错是保证金争议的常见原因。

18.9.2 How much of the exposure does collateral cover? 抵押品能覆盖多少敞口

$$\text{Coverage} = \frac{\sum_j MV_j (1 - h_j)}{E}, \quad \text{with a threshold: Coverage} \approx \frac{\max(E - TH, 0)}{E}$$

There is no fixed ratio; it depends mainly on the CSA terms (table in the right-hand column):

- **With a threshold:** coverage $\approx (E - TH) \div E$, zero when $E \leq TH$; the gap is the threshold itself (day 4 of Example 16: $3.0 \div 8.0 = 37.5\%$).

没有固定比例，主要由 CSA 条款决定：

- **Zero-threshold VM CSA:** about 90–100%; the gap comes from amounts below the MTA, the T+1 settlement lag and intraday MtM moves.
- **VM + IM:** above 100% (over-collateralised), as IM covers potential losses between default and close-out.
- **Haircuts:** coverage uses post-haircut value; €1.224m of bonds at a 2% haircut counts as €1.2m.
- **Value under stress:** collateral itself falls in a crisis; if it is correlated with the counterparty's credit (wrong-way risk, e.g. its home sovereign or its own securities), effective coverage is lower.
- **UCITS:** only collateral meeting ESMA criteria, at post-haircut value, reduces the 5% / 10% counterparty limit; collateral the fund **posts** without segregation (e.g. non-segregated IM) **adds** to exposure to that counterparty.
- **Monitoring:** report coverage per CSA daily and recompute it under stress (collateral value falls, haircuts rise).

安排	典型覆盖率	缺口来源
有阈值	$(E - TH) \div E$; $E \leq TH$ 时为 0	阈值本身（例题 16 第 4 天： $3.0 \div 8.0 = 37.5\%$ ）
零阈值 VM CSA	约 90–100%	低于 MTA 的未追加部分、T+1 交付时差、当日市值变动
VM + IM	> 100%（超额抵押）	IM 覆盖违约到平仓期间的潜在损失

- **折扣率：**覆盖率按折扣后价值计算——市价 €122.4 万、折扣率 2% 的政府债只算 €120 万。
- **压力下贬值：**抵押品本身会在危机中下跌；若抵押品与对手信用高度相关（错向风险，如对手交付本国政府债或自身发行的证券），实际覆盖更低。
- **UCITS：**收到的抵押品只有满足 ESMA 标准、按折扣后价值计算，才能冲减 5% / 10% 对手限额；基金**交给**对手且未隔离保护的抵押品（如未隔离的 IM）反而**增加**对该对手的敞口。
- **监控：**按 CSA 每日报告覆盖率，并在压力情景下（抵押品贬值、折扣率上调）重算。

Appendix A Python Reference Implementation

Python 参考实现

The code below runs as is (numpy, pandas, scipy) and reproduces every result of §9.2 (T=998 due to a one-day alignment offset; breach counts are identical), plus the reverse stress test of §11.3, the liquidity coverage of §11.4 and the drawdown statistics of §13. Design rules: the forecast for day t uses data up to $t - 1$ only; zone boundaries use "strictly below ppf".

以下代码可直接运行（依赖 numpy、pandas、scipy），复现第 9.2 节的全部结果（因起点对齐差 1 日，T=998，突破次数不变），以及 §11.3 的反向压力测试、§11.4 的流动性覆盖率和 §13 的回撤统计。设计原则：第 t 日的预测只使用 $t - 1$ 及以前的数据；区域边界用"严格小于 ppf"判断。

```
import numpy as np, pandas as pd
from scipy.stats import norm, binom, chi2

def log_returns(px: pd.Series) -> pd.Series:
    return np.log(px).diff().dropna()

def ewma_sigma(r: pd.Series, lam=0.94, seed_n=250) -> pd.Series:
    """sigma_{t|t-1}: forecast for day t using returns up to t-1."""
    v = np.empty(len(r)); v[0] = np.mean(r.values[:seed_n] ** 2)
    for t in range(1, len(r)):
        v[t] = lam * v[t-1] + (1 - lam) * r.values[t-1] ** 2
    return pd.Series(np.sqrt(v), index=r.index)

def cf_factor(p, S, K):
    z = norm.ppf(p)
    return z + (z*z-1)*S/6 + (z**3-3*z)*K/24 - (2*z**3-5*z)*S*S/36

def normal_es_factor(c):
    return -norm.pdf(norm.ppf(1-c)) / (1-c) # -2.6652 at c = 0.99

def var_forecasts(r, c=0.99, window=250, lam=0.94):
    p, z = 1 - c, norm.ppf(1 - c)
    sig = ewma_sigma(r, lam)
    out = pd.DataFrame(index=r.index)
    out["roll21"] = z * r.rolling(21).std().shift(1)
    out["ewma"] = z * sig
    out["hs"] = r.rolling(window).quantile(p).shift(1)
    resid = r / sig # devolatilise
    out["fhs"] = resid.rolling(window).quantile(p).shift(1) * sig # revolatilise
    return out

def backtest(r, var, p=0.01):
    m = var.notna(); hit = (r[m] < var[m]).astype(int).values
    T, x = len(hit), hit.sum(); ph = x / T
    ll = lambda q, k0, k1: (k0*np.log(1-q) if k0 else 0) + (k1*np.log(q) if k1 else 0)
    pof = -2 * (ll(p, T-x, x) - ll(ph, T-x, x))
    a, b = hit[:-1], hit[1:]
    n00, n01 = ((a==0)&(b==0)).sum(), ((a==0)&(b==1)).sum()
    n10, n11 = ((a==1)&(b==0)).sum(), ((a==1)&(b==1)).sum()
    pi0, pi1 = n01/(n00+n01), (n11/(n10+n11) if n10+n11 else 0.0)
    pi = (n01+n11)/(n00+n01+n10+n11)
    ind = -2 * (ll(pi, n00+n10, n01+n11) - ll(pi0, n00, n01) - ll(pi1, n10, n11))
    k250 = hit[-250:].sum()
    g, y = binom.ppf(0.95, 250, p), binom.ppf(0.9999, 250, p) # 5, 10
    zone = "green" if k250 < g else ("yellow" if k250 < y else "red")
    return dict(T=T, x=int(x), rate=ph, LR_pof=pof, p_pof=chi2.sf(pof, 1),
                LR_ind=ind, p_ind=chi2.sf(ind, 1), cc=pof+ind, p_cc=chi2.sf(pof+ind, 2),
                pil=pi1, last250=int(k250), zone=zone)

# ----- Part II: stress testing -----
def reverse_stress(w, cov, loss):
    """Most plausible scenario x* (min Mahalanobis) with w'x = -loss."""
    w, cov = np.asarray(w, float), np.asarray(cov, float)
    var_p = w @ cov @ w
    x = -loss * cov @ w / var_p
    return x, loss / np.sqrt(var_p) # scenario, Mahalanobis distance

def coverage(buckets, redemption, horizon, stress_mult=1.0):
    """buckets: list of (share_of_nav, days_to_liquidate)."""
    avail = sum(a for a, d in buckets if d * stress_mult <= horizon)
    return avail / redemption

# ----- Part III: drawdown -----
def drawdown_stats(nav: pd.Series) -> dict:
    peak = nav.cummax(); dd = nav / peak - 1
    trough = dd.idxmin(); start = nav.loc[:trough].idxmax()
    after = nav.loc[trough:]; rec = after[after >= nav.loc[start]]
    yrs = (nav.index[-1] - nav.index[0]).days / 365.25
    cagr = (nav.iloc[-1] / nav.iloc[0]) ** (1 / yrs) - 1
    return dict(mdd=dd.min(), peak=start, trough=trough,
                recovery=rec.index[0] if len(rec) else None,
                calmar=cagr / -dd.min(), ulcer=np.sqrt((dd ** 2).mean()))

def mc_mdd(sigma=0.16, mu=0.0, years=3, n_paths=10000, steps=252, seed=1):
    rng = np.random.default_rng(seed)
    lr = (mu - 0.5 * sigma**2) / steps + sigma / np.sqrt(steps) * rng.standard_normal((n_paths, years * steps))
```

```
lv = np.concatenate([np.zeros((n_paths, 1)), np.cumsum(lr, axis=1)], axis=1)
mdd = 1 - np.exp(lv - np.maximum.accumulate(lv, axis=1))
return mdd.max(axis=1) # distribution of MDD

if __name__ == "__main__":
    px = pd.read_csv("sp500.csv", index_col=0, parse_dates=True)["Px"]
    r = log_returns(px)
    f = var_forecasts(r).iloc[251:]
    for col in f:
        print(col, {k: (round(v, 4) if isinstance(v, float) else v)
                    for k, v in backtest(r.loc[f.index], f[col]).items()})
    print(drawdown_stats(px))
    m = mc_mdd(years=3, mu=0.06); print("3y MDD median/95%:", np.median(m).round(3), np.quantile(m, .95).round(3))
```

Appendix B Excel Formula Cheat Sheet

Excel 公式速查

用途 Purpose	公式 Formula	说明 Note
对数收益 Log-return	=LN(B8/B7)	
正态因子 Normal factor	=NORM.S.INV(0.01)	-2.32635
正态 ES 因子 Normal ES factor	=-NORM.S.DIST(NORM.S.INV(0.01),FALSE)/0.01	-2.6652
滚动 σ (21 obs) Rolling σ	=STDEV.S(C8:C28)	下一行窗口下移 1 行 · shift one row
10 日 VaR	=STDEV.S(C8:C28)*SQRT(10)*Factor_VaR	命名单元格 Factor_VaR
10 日前瞻收益 Forward 10D return	=LN(B39/B29)	与 VaR 同一行对齐 · align with VaR row
突破标记 Breach flag	=IF(E28<D28,1,0)	负数口径 · negative convention
EWMA 方差 EWMA variance	=E5*\$I\$2+D5*(1-\$I\$2)	D 列为 u^2 , I2 为 λ
HS VaR	=PERCENTILE.INC(C8:C257,0.01)	或 SMALL(range,k)
二项累积 Binomial CDF	=BINOM.DIST(k,250,0.01,TRUE)	交通灯 · traffic light
Kupiec p 值	=CHISQ.DIST.RT(LR,1)	
偏度 / 超额峰度 Skew / excess kurtosis	=SKEW(range),=KURT(range)	KURT 即超额峰度 · KURT is excess

Appendix C Glossary

中英术语表

English	中文	一句话解释 / One-line meaning
Value at Risk (VaR)	风险价值	Loss percentile at a given confidence and horizon · 给定置信水平与期限下的损失分位数
Expected Shortfall (ES), CVaR	预期损失 / 预期亏空	Average loss beyond VaR · 超过 VaR 的平均损失
Quantile / percentile	分位数 / 百分位数	Point where cumulative probability reaches a level · 累积概率达到某水平的点
Inverse CDF (quantile function)	逆累积分布函数	Maps probability to value · 由概率求数值
Supremum / infimum	上确界 / 下确界	Least upper / greatest lower bound · 集合的最小上界 / 最大下界
Subadditivity	次可加性	Merging does not add risk · 合并不增加风险
Coherent risk measure	一致性风险度量	Satisfies the four axioms · 满足四条公理的风险度量
EWMA	指数加权移动平均	Exponentially decaying weights · 按入指数衰减加权
Plateau effect / ghost effect	平台效应 / 幽灵效应	Step-like VaR from equal-weighted windows · 等权窗口导致 VaR 阶梯状变化
Parametric / historical / Monte Carlo VaR	参数法 / 历史模拟 / 蒙特卡洛 VaR	Distribution assumed, historical, or simulated · 分布分别来自假设、历史、随机模型
Marginal / component / incremental VaR	边际 / 成分 / 增量 VaR	Derivative, contribution and change of VaR by position · VaR 对头寸的导数、贡献与变化
Absolute / relative VaR (UCITS)	绝对 / 相对 VaR	$VaR \leq 20\%$ of NAV / fund-to-reference VaR ratio ≤ 2 · $VaR \leq 20\%$ NAV / 基金与参考组合 VaR 之比 ≤ 2
Tracking-error VaR	跟踪误差 VaR	VaR of the active return · 主动收益（基金减基准）的 VaR
Reference portfolio (UCITS)	参考组合	Unleveraged benchmark for relative VaR · 相对 VaR 的无杠杆比较基准
Stressed VaR (sVaR)	压力 VaR	VaR calibrated on a stress period · 用压力期数据校准的 VaR
Unsmoothing (Geltner)	去平滑	Restores volatility hidden by appraisal smoothing · 恢复评估价格被平滑的波动
Netting / hedging arrangements	轧差 / 对冲安排	The two ways to offset exposure under the commitment approach · 承诺法下可抵消敞口的两类安排
Duration netting	久期轧差	Partial netting of rate derivatives across maturity buckets · 利率衍生品按期限区间部分轧差
CE / PFE / EE	当前 / 潜在未来 / 预期敞口	Three measures of counterparty exposure · 交易对手敞口的三种度量
Credit valuation adjustment (CVA)	信用估值调整	Valuation charge for counterparty default · 对手违约风险的估值扣减
Wrong-way risk	错向风险	Default probability rising with exposure · 违约概率与敞口同向上升
Variation / initial margin (VM / IM)	变动 / 初始保证金	Daily MtM settlement / cover for close-out risk · 按市值每日结算 / 覆盖平仓期风险
Taylor expansion	泰勒展开	Polynomial approximation built from derivatives · 用各阶导数构成的多项式近似函数
Full revaluation	全重估	Repricing with the valuation model under shocked data · 用定价模型在冲击后的市场数据下重新定价
Sticky strike / sticky moneyness	黏性行权价 / 黏性价值状态	Two rules for moving the vol surface with spot · 波动率曲面随现货移动的两种规则
Barrier option / autocallable	障碍期权 / 自动赎回票据	Typical path-dependent products · 典型的路径依赖产品
Common random numbers	公共随机数	Same draws for base and shocked Monte Carlo · 基准与冲击估值共用随机数以降低噪声
Multi-factor model / factor exposure	多因子模型 / 因子暴露	Explains returns with a few common factors · 用少数共同因子解释资产收益
Specific (idiosyncratic) risk	特质风险	Residual risk not explained by factors · 因子无法解释的残差风险
Sensitivities-based method (SBM) / ISDA SIMM	敏感度法 / SIMM	Prescribed factor-sensitivity models · 监管或行业规定的因子敏感度模型
Risk factor eligibility test (RFET)	可建模性测试	Whether a factor can enter the internal model · 判断风险因子能否纳入内部模型
Sum of notionals	名义本金之和	Leverage basis disclosed by UCITS VaR funds · UCITS VaR 基金披露的杠杆口径
Margin coverage ratio	保证金覆盖率	Available cash and collateral ÷ stressed margin calls · 可用现金与抵押品 ÷ 压力保证金追加
Pin risk / gap risk	钉住风险 / 跳空风险	Spot near strike at expiry / discontinuous price jumps · 临近到期贴近行权价 / 价格不连续跳动
Independent price verification (IPV)	独立价格验证	Valuation check independent of trading · 独立于交易的估值核对
Credit Support Annex (CSA)	信用支持附件	ISDA annex governing collateral exchange · 约定抵押品交换条件的 ISDA 附件
Threshold / minimum transfer amount (MTA)	阈值 / 最低转账金额	Uncollateralised allowance / smallest call · 免于抵押的敞口 / 最小追加额
Additional termination event (ATE)	附加终止事件	E.g. NAV-decline early termination · 如 NAV 下跌触发提前终止
Discount / projection curve	贴现曲线 / 预测曲线	The two curve roles in multi-curve pricing · 多曲线框架下的两类曲线

Bootstrapping	剥离法	Sequentially solving discount factors · 逐段求解贴现因子构建曲线
Key-rate DV01	关键期限 DV01	P&L for a 1bp bump at one tenor · 单个期限点上移 1bp 的损益
Principal component analysis (PCA)	主成分分析	Level, slope, curvature · 水平、斜率、曲率
Credit Support Amount	应抵押额	$\max(\text{exposure} - \text{threshold} \pm \text{independent amounts}, 0) \cdot \max(\text{敞口} - \text{阈值} \pm \text{独立金额}, 0)$
Margin period of risk (MPOR)	保证金风险期	Time from default to close-out · 违约到完成平仓的时间
Collateral coverage	抵押品覆盖率	$\text{Post-haircut collateral} \div \text{exposure} \cdot \text{折扣后抵押品价值} \div \text{风险敞口}$
Half-life	半衰期	Days for weight to halve · 权重衰减一半所需天数
Historical simulation (HS)	历史模拟	Revalue under historical scenarios · 用历史情景重估组合
Filtered historical simulation (FHS)	过滤历史模拟	HS on devol/revol returns · 去波动化再波动化的 HS
Devolatilise / revolatilise	去波动化 / 再波动化	Divide by own-day σ / multiply by today's σ · 除以当日 σ / 乘以今日 σ
Procyclicality	顺周期性	Risk estimates amplify the cycle · 风险度量随市场周期同向放大
Skewness / excess kurtosis	偏度 / 超额峰度	三阶、四阶标准化矩 (峰度减 3) · Third and fourth standardised moments (kurtosis - 3)
Generalised Pareto distribution (GPD)	广义帕累托分布	Limit law of threshold exceedances · 超阈值损失的极限分布
Backtesting / breach / exception	回测 / 突破 / 例外	Realised loss exceeds VaR · 实现损失超过 VaR
Eligible comparisons	有效比较次数	Days with both VaR and realised return · 同时有 VaR 与实现收益的天数
Unconditional / conditional coverage	无条件覆盖 / 条件覆盖	Right frequency / frequency and independence · 频率正确 / 频率与独立性都正确
Traffic-light approach	交通灯方法	Green/yellow/red by breach count · 按突破次数分绿黄红区
Fundamental Review of the Trading Book (FRTB)	交易账簿根本性审查	Basel III market-risk framework · Basel III 市场风险框架
Internal models approach (IMA) / standardised approach (SA)	内部模型法 / 标准法	
HPL / APL / RTPL	假设损益 / 实际损益 / 风险理论损益	see §8.5 · 见 § 8.5
P&L attribution (PLA) test	损益归因测试	Consistency of RTPL with HPL · RTPL 与 HPL 的一致性检验
VaR-equivalent volatility (VEV)	VaR 等效波动率	Input to the PRIIPs risk class · PRIIPs 风险等级的输入
Initial margin (IM)	初始保证金	Collateral for close-out risk · 覆盖平仓期潜在损失的抵押品
Stress test / sensitivity analysis	压力测试 / 敏感性分析	P&L under a given shock · 给定冲击下的损益
Reverse stress test	反向压力测试	Solve for scenarios causing an unacceptable loss · 由不可接受的损失反推情景
Mahalanobis distance	马氏距离	Standard-deviation distance accounting for correlation · 考虑相关性的"几个标准差"
Liquidity stress test (LST)	流动性压力测试	Stress on liquidation and redemptions · 资产变现与赎回的压力测试
Coverage ratio	覆盖率	$\text{Liquidable assets} \div \text{stressed redemptions} \cdot \text{可变现资产} \div \text{压力赎回}$
Liquidity management tool (LMT)	流动性管理工具	Gates, swing pricing, anti-dilution levies, etc. · 闸门、摆动定价、反稀释征费等
Swing pricing	摆动定价	NAV adjusted for net flows · 按净申赎方向调整 NAV
Drawdown / maximum drawdown (MDD)	回撤 / 最大回撤	Fall from the running peak · 距历史高点的跌幅
Time under water	水下时间	Time from peak to new peak · 从高点到收复高点的时长
High-water mark (HWM)	高水位	Highest NAV to date · 历史最高净值
Calmar ratio / Ulcer index	Calmar 比率 / Ulcer 指数	Drawdown-adjusted return and pain measure · 回撤调整后的收益与痛苦度量
Reflection principle	反射原理	Touching probability is twice the terminal one · 途中触及概率为期末概率的两倍
Scenario analysis	情景分析	Multi-period, narrative-driven path analysis · 多期、叙事驱动的路径分析
Performance scenarios (PRIIPs)	业绩情景	Stress, unfavourable, moderate, favourable · 压力、不利、中性、有利
Recommended holding period (RHP)	建议持有期	Horizon for PRIIPs scenarios and SRI · PRIIPs 情景与风险指标的期限
Global exposure (UCITS)	全球风险暴露	Derivative risk under commitment or VaR · 承诺法或 VaR 法计量的衍生品风险
Gross / commitment method (AIFMD)	总额法 / 承诺法	Two leverage calculations · 两种杠杆计算方法
Summary risk indicator (SRI)	综合风险指标	PRIIPs 1-7 risk class · PRIIPs 1-7 级风险等级